

BitZ: Proofs in arbitrary domains with commitments in binary fields

immediate

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Abstract

Contents

1 Introduction

A Succinct Non-interactive Argument of Knowledge (SNARK) for a relation $\mathcal{R}$ is a non-interactive protocol that enables a prover to convince a verifier that, for a given public input $\mathbf{x}$, they know a witness $\mathbf{w}$ such that $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$, and where communication and verifier runtime are sublinear (i.e. the scheme is succinct).

In the vast majority of scenarios, $\mathcal{R}$ is defined over one (or more) algebraic object R , usually a finite field, and $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$ if and only if $(\mathbf{x}, \mathbf{w})$ consists of elements that satisfy some polynomial equations. The choice of R has a crucial impact on the design and end-to-end performance of the proof system, for two reasons:

1. Usually the computations one wants to prove are not native to R , so one has to perform a more or less complex emulation of the computation into polynomial equations over R . This can affect the size of the input-witness pair $\mathbf{x}, \mathbf{w}$ significantly [?]. Additionally, the emulation step can be technically involved and prone to security errors [?].
2. The prover and verifier algorithms efficiency and design is affected by the choice of R : for example, modern CPUs are more performant for operations in certain small fields (e.g. M31 [?, ?], Babybear [?], Goldilocks [?]) than in arbitrary fields, especially if these are larger. Importantly, another class of fields where CPUs operate exceptionally fast are *binary fields*, i.e. fields of order a power of two.

Recently, two independent lines of research have received increased attention:

- Designing proof systems for relations expressed over $R = \mathbb{Z}$ the ring of integers, and over other rings that are not fields [?, ?, ?, ?, ?, ?]. The motivation behind this is to address Item ?? above: in many practical scenarios, the computation to be proved is most succinctly and easily expressed over such rings, see ?? [Albert: more refs?](#).

- Designing proof systems for $R = \mathbb{F}_{2^\nu}$ a binary field, for example [?, ?, ?, ?, ?]. The reason for this is twofold: first, some computations like classic hashes (SHA-256, Blake, etc.) are easier to express over $\mathbb{F}_{2^\nu}$ than over other structures such as prime fields (however, as we discuss in ??, these hash functions remain relatively un-native to $\mathbb{F}_{2^\nu}$ due to the hardness of emulating operations modulo 2^{32} or 2^{64}). Second, binary fields are highly performant on modern CPU’s, and further enable using a boon of protocol techniques that further benefit performance, most importantly *packed* polynomial commitment schemes [?, ?].

1.1 Our contributions

Albert: It should be very clear throughout the paper: The PIOP machinery is for the most part Zinc+’s (which builds on Zinc and Zaratan). The dual field PCS primitive is novel; its instantiation is pretty much novel (though ideas can be traced back to [?, ?]), and its enabling optimizations are novel. The PCS in binary fields is completely known.

In this work we bring these two lines of research together. We present BitZ, a hash-based proof system that:

1. BitZ can prove polynomial constraints over essentially arbitrary rings R , including $\mathbb{Z}$, prime fields, lattices, binary fields, arbitrary field extensions, polynomial rings, etc., addressing Item ?? above (BitZ can handle constraints over different rings at once).
2. In its initial commitment step, BitZ’s decomposes the witness w into a vector of bits $\text{bits}(w)$, and commits to $\text{bits}(w)$ over a binary field $\mathbb{F}_{2^\nu}$ with an error correcting code (leveraging the packing technique from [?, ?]).
3. The initial polynomial constraints over the ring(s) R are reduced to an MLE evaluation claim (or more generally, an inner product claim) over a finite field $\mathbb{F}$, of the form $\langle \mathbf{v}, \pi(w) \rangle = \mu \in \mathbb{F}$, where π is a projection from R to $\mathbb{F}$ and $\mathbf{v}$ is a vector of elements from $\mathbb{F}$. The field $\mathbb{F}$ can be a fixed field extension, a random prime field (following [?]), or a fixed prime field, depending on what the ring(s) R is (are).
4. BitZ’s Polynomial Commitment Scheme (PCS) enables proving such linear claim efficiently. It operates, for the most part, over $\mathbb{F}_{2^\nu}$. Thus, BitZ benefits of many of the advantages of running proof systems over binary fields (Item 2 discussed earlier one). See below for a brief description of BitZ’s PCS.

BitZ provides, among others, a compiler that takes any PCS over $\mathbb{F}_{2^\nu}$ and transforms it into a PCS over any of the above rings R , following the “compiler” frameworks of [?, ?].

We implement and open-source BitZ [?], and use it to prove a number of computations of practical interest. In ?? we provide a summary of the results. In all computations but one, BitZ outperforms the current state-of-the-art approaches, sometimes by orders of magnitudes (for example: ***). BitZ is even able to prove Babybear multiplications more efficiently than Plonky3, a mature proof system that operates natively on the Babybear field, and which is used in industry within a variety of projects [?]. Albert: Mention what is the security parameter

We manage to instantiate our protocol using $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$ in GHASH representation, attain 128 bits of security, and have no limitation on the size of the ring R we start with. See ???? for more comments and details on this.

Computation	BitZ	Other schemes
SHA-256 (7KB) + ECDSA		Albert: Zinc+,Vega,Longfewllow
2^{17} 32 bit integer mults		Albert: Binius,Limber,Zinc+
2^{20} Babybear mults		Albert: Plonk3
Poseidon hash + RSA [?]		Albert: Limber
Falcon		
SHA-256 in 8KB message (2^7 compressions)		Albert: flock/binius
SHA-256 in 2MB message (2^{15} compressions)		Albert: flock/binius

Table 1: Albert: Include memory. Albert: Revisit: should we include the experiemnts on hashing big messages here? Performance of BitZ on a suite of various computations of interest. For each such computation, we also benchmark the performance of prior state of the art protocols. All experiments were run on a MacBook Air M4 16GB, multi-threaded. Albert: Mention what is the security parameter Albert: Maybe report numbers for both the AI and human implementation

BitZ’s PCS in a nutshell. BitZ proves the aforementioned claim $\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \mu$ over a finite field $\mathbb{F}$ as follows. We describe first an initial, mostly insufficient, attempt.

1. **Bitification.** First, it transforms the claim into a claim of the form $\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu$ over $\mathbb{F}$, for some $\mathbf{u}$.
2. **Lift to integers.** Assuming $\mathbb{F}$ is a prime field of size q and $\pi_q : \mathbb{Z} \rightarrow \mathbb{F}$ is reduction modulo $|\mathbb{F}|$, the prover sends $\mu' \in \mathbb{Z}$ such that $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$ in $\mathbb{Z}$.
3. **Proof in the exponent of $\mathbb{F}_{2^\nu}$.** Now take a generator g of the multiplicative subgroup $\mathbb{F}_{2^\nu}^*$ of $\mathbb{F}_{2^\nu}$, and prove that $g^{\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle} = g^{\mu'}$ in $\mathbb{F}_{2^\nu}$. Concretely, we have

$$g^{\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle} = \prod_i \left((g^{\pi_q^{-1}(\mathbf{u}_i)} - 1) \cdot \text{bits}(\mathbf{w})_i + 1 \right) = g^{\mu'}, \quad (1)$$

which is a polynomial expression over $\mathbb{F}_{2^\nu}$ on the vector $\text{bits}(\mathbf{w})$ that we committed over $\mathbb{F}_{2^\nu}$.

This is a standard situation where there is no “wrong-field” arithmetic, hence in principle one can use any proof system over $\mathbb{F}_{2^\nu}$ to prove (??).

There are several issues with this attempt:

- First, $\mathbb{F}$ may not be a prime field, in which case there is no natural lift to integers in Step 2; rather, there is a lift to integer polynomials $\mathbb{Z}[X]$.
- Second, even if $\mathbb{F}$ is a prime field, $q = |\mathbb{F}|$ may be large and the linear expression $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle$ may overflow the order of $g \in \mathbb{F}_{2^\nu}^*$, which is $2^\nu - 1$. In this case, proving $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$ “in the exponent of g ” is unsound.
- Third, even if neither of these problems hold, (??) is, in principle, an expensive constraint to prove: indeed, it is a length $|\text{bits}(\mathbf{w})|$ grand product of, essentially, arbitrary factors from $\mathbb{F}_{2^\nu}$ (typically ≈ 128 bit elements). Grand products can be expensive to prove, and furthermore the fact that the factors are large prevents us from using known sumcheck optimizations such as univariate skips [?] and Albert: what else?.
- Four, when proving (??), typically one ends up with an MLE evaluation claim on the vector $(g^{\pi_q^{-1}(\mathbf{u}_i)})_i$ (and another on $\text{bits}(\mathbf{w})$, which is proved with any PCS over $\mathbb{F}_{2^\nu}$). If the verifier checks the claim itself, then it pays a linear cost. If the prover proves this claim, then it is essentially back to square one.

We solve these issues as follows: 1) if $\mathbb{F}$ is a proper extension, at Step 2 we lift to a claim

$\langle \pi_{\text{canon}}^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu' \in \mathbb{Z}[X]$ over $\mathbb{Z}[X]$, where $\pi_{\text{canon}} : \mathbb{Z}[X] \rightarrow \mathbb{F}$ is a canonical projection of $\mathbb{Z}[X]$ onto $\mathbb{F}$. Then the verifier sends a random prime q' of $\approx |\mathbb{F}_{2^\nu}^*|/|\text{bits}(\mathbf{w})|$ size (small enough to avoid the issue of overflowing in the exponent), and a random element $\alpha \in \mathbb{F}_{q'}$; then the previous claim is replaced by $\langle \pi_{q',\alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}), \pi_{q',\alpha}(\text{bits}(\mathbf{w})) \rangle = \pi_{q',\alpha}(\mu') \in \mathbb{F}_{q'}$ where $\pi_{q',\alpha}(f(X)) = \pi_{q'}(f)(\alpha) \in \mathbb{F}_{q'}$. Now we re-enter Step 2 with this new claim.

2) If $\mathbb{F}$ is too large of a prime field, we follow a similar approach as above: once we have the integer claim $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$, the verifier samples a random “small” prime q' , reduce the claim modulo q' , and then re-enter Step 2 with this new claim.

Albert: remove? We emphasize again that we manage to instantiate BitZ with 128 bits of security, using $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$, while having no limitation on the size of the ring R we start with.

3) The main idea is to split (??) into $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ grand products of the same form as (??), so that, even though in total there are $|\text{bits}(\mathbf{w})|$ factors among all of them, these factors can only take $\sqrt{|\text{bits}(\mathbf{w})|}$ values. More concretely, we assume $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ tensor-decomposes into $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ sized factors (this is typically the case since $\mathbf{v}$ is usually an “eq” vector), and then, assuming the notation of Step 3 above, we aim to prove $\langle \pi_q^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle = \mu'_j$ in $\mathbb{Z}$ for all $j \in [|\mathbf{u}^{(2)}|]$, such that $\sum_j \pi_q(\mu'_j) \cdot \mathbf{u}_j^{(2)} = \mu$, where we index $\text{bits}(\mathbf{w})$ with indices $i \in [|\mathbf{u}^{(1)}|], j \in [|\mathbf{u}^{(2)}|]$. We then set up a grand product for each of these claims, obtaining

$$g^{\langle \pi_q^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle} = \prod_i (g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} - 1) \cdot \text{bits}(\mathbf{w})_{ij} + 1 = g^{\mu'_j} \quad \text{for all } j. \quad (2)$$

There are $|\text{bits}(\mathbf{w})|$ factors in total, but only $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ different values these factors can take.

We say (??) is a *low entropy batched grand product*. To prove (??) we use a batched GKR protocol with the following optimization, among others: because of the low-entropy phenomenon, we build several small lookup tables of size $O(\sqrt{|\text{bits}(\mathbf{w})|})$ (of a few MB at most) which are used during the most expensive sumcheck rounds of the GKR protocol. Concretely, the last sumcheck (the most expensive one), uses 3 lookup tables to compute the first three messages of the prover; the second last sumcheck does the same for the first two prover messages (with two tables); and the third last sumcheck does the same for the first prover message (with one table).

The above solution also takes care of Issue 4: the batched GKR protocol outputs MLE evaluation claims on $(g^{\pi_q^{-1}(\mathbf{u}^{(1)})_i})_i$ (and $\text{bits}(\mathbf{w})$, which are dealt with the PCS over $\mathbb{F}_{2^\nu}$). This is an $O(\sqrt{|\text{bits}(\mathbf{w})|})$ vector, and the verifier can check this claim by itself cheaply.

Virtual $\mathbb{F}_2$ -linear transforms, NP-complete dually linear relations, and arithmetization of SHA-256 Observe that, once we have a standard linear claim on $\text{bits}(\mathbf{w})$ in $\mathbb{F}_{2^\nu}$, we could replace $\text{bits}(\mathbf{w})$ by a linear expression on some other committed vector $\mathbf{f}$ over $\mathbb{F}_{2^\nu}$, so that $\text{bits}(\mathbf{w}) = M \cdot \mathbf{f}$ for some matrix M . The final linear claim can then be replaced by a linear claim for $\mathbf{f}$. Hence, we can, instead of committing to $\text{bits}(\mathbf{w})$, commit to $\mathbf{f}$ and treat $\mathbf{w}$ and $\text{bits}(\mathbf{w})$ as virtual vectors. See ?? for details.

This allows BitZ to prove polynomial constraints over R on vectors $\mathbf{w}$ that are a virtual $\mathbb{F}_2$ -linear combination of a committed vector $\mathbf{f} \in \mathbb{F}_{2^\nu}^{\ell'}$. For example, BitZ can efficiently prove the following R1CS-like relation:

$$\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z})$$

$$:= \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, A, B, C, M) \in \mathbb{F}_2^k \times (\mathbb{Z}^{m \times \ell})^3 \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \mid \begin{array}{l} (A \cdot \mathbf{w}) \circ (B \cdot \mathbf{w}) = C \cdot \mathbf{w} \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{z}, \mathbf{f})) \end{array} \right\},$$

where π_2 denotes reduction modulo 2. A particularly interesting subclass of this relation is that obtained by setting $A = B = \mathbf{0}$, in which case the resulting relation, which we call the *CM relation*, degenerates to linear constraints over $\mathbb{Z}$ and over $\mathbb{F}_2$, i.e. a “dually linear” constraint.

In ?? we show that the CM relation is NP-complete [Albert: Or NP-hard?](#). In ?? we arithmetize SHA-256 as an instance of the CM relation, with *** smaller witnesses than Binius64 [?] and Flock [?] [Albert: add numbers. Mention that we don't prove hashes faster though \(but we have better proof size and memory so...\)](#)

1.2 Related work

Proof systems over integers and other rings. In [?], Campanelli and Hall-Andersen propose proving polynomial constraints over integers, by projecting them onto a random prime field (sampled by the verifier), a technique which the authors call *fingerprinting*. Subsequent works have adopted this idea [?, ?, ?, ?]. [?] also proposes a hidden-order group based PCS suitable for their PIOP, similar to the ones in [?, ?]. The PCS is not implemented and so we do not compare against it.

The Zinc [?] work adopts *fingerprinting* and designs a hash-based PCS over the rationals $\mathbb{Q}$, suitable for the fingerprinting approach, providing the first relatively practical proof system over the integers. [?] addresses subtle technical issues that arise when aiming to commit to *integer* vectors with error correcting codes (namely, malicious provers can potentially use rational numbers that are very close to being integers of the target magnitude).

The Zinc+ [?] work extends [?] in two directions: 1) It proposes a constraint system that includes polynomial constraints simultaneously over many types of rings¹ and fields, and provides a method for constructing PIOP's for those. This method extends the fingerprinting approach by not only projecting $\mathbb{Z}$ onto a random prime field, but projecting all sorts of rings and fields onto a random prime field. 2) It improves the soundness analysis of the PCS in [?], plus it proposes a new linear code over $\mathbb{Z}$ or $\mathbb{Q}$, called Integer Pseudo Reed-Solomon (IPRS) code, obtained by “lifting” (in a suitable way) an FFT encoding algorithm over a small finite field, to the integers. [Albert: compare BitZ to Zinc+?](#)

In [?] the authors propose Qaren, a compiler that takes any PCS over a finite field of sufficiently large characteristic and creates a PCS for integer polynomials (or, more precisely, rational polynomials close to the target integer polynomials, see the overview of [?] above). The authors instantiate their compiler to obtain the first transparent and fully succinct PCS over integers/rationals. The prior work [?, ?] is not succinct on the bit-size of the integers being committed. No implementation or performance reports is previded in [?].

The concurrent work Limber [?] proposes a new compiler that transforms any PCS over fields of sufficiently large characteristic q (in this case, q is $\Omega(\lambda^2 \log(N))$, N being the witness size, while in [?] it is $\Omega(2^\lambda)$). Additionally Limber compiles into strictly integer PCS, and does not present the issue that a malicious prover could use rationals that are “almost integers”. Limber is instantiated in practice with the Hyrax PCS (in contrast to [?, ?] which use a hash based PCS). [Albert: Wait](#)

¹In the terminology of [?], the constraints are over *polynomial rings with ideal membership constraints*, for certain types of ideals (generated by a monic and non-constant polynomial, or zero). However this is equivalent to saying the constraints are over different types of rings (within a certain class)

for the implementation to come out so we can discuss its performance against prior work better. In [?] we compare BitZ against Limber, obtaining *** **Albert: fill in.**

Proof systems over binary fields. Hash-based SNARKs over binary fields were pioneered by the line of work of Ben-Sasson et al. [?, ?, ?, ?], but did not see widespread adoption until the introduction of *packed* polynomial commitment schemes by Diamond and Posen [?, ?]; see also [?]. Such schemes allow one to commit to a bit-valued witness over an extension field without incurring embedding overhead, while still providing evaluation proofs for its original “uncompressed” representation. The *ring-switching* technique of Diamond and Posen [?] is generic, enabling the use of codes over larger alphabets without introducing representational redundancy.² A particularly attractive choice is Reed–Solomon codes over $GF(2^k)$ [?, ?], which offer an excellent trade-off between distance and encoding cost, although several works [?, ?, ?] explore this design space using more efficient codes.

At the time of writing, two further works push the performance envelope of binary proof systems. Bünz, Rothblum, and Wang [?] combine [?] with the batched R1CS strategy inspired by [?], achieving a striking proving throughput of several hundred thousand hashes per second on an M4 Max CPU. BinarySpartan [?], in turn, incorporates the optimizations of [?]-Flock into Spartan, resulting in a proof system with again high hash-proving throughput.

Commitments in binary fields with openings in rings R of characteristic different than 2. To our knowledge this primitive is studied only in Soukhanov’s note [?] and in the present work. The note [?] proposes a scheme for committing over the binary field $\mathbb{F}_{2^{127}}$ and then executing a PCS opening over $\mathbb{F}_{2^{127}-1}$ (a prime field), by running the PCS in the exponent of a generator of $\mathbb{F}_{2^{127}}^*$. The approach, substantially different to ours, is limited to $R = \mathbb{F}_{2^{127}-1}$ (or analogous configurations where $2^n - 1$ is prime) and does not seem to be able to leverage our aforementioned “low-entropy” based optimizations. The note does not provide an implementation nor performance reports. We discuss this further in ??.

Qaren and Limber [?, ?] provide a related primitive when $R = \mathbb{Z}$, but can only support commitments over fields of relatively large characteristic (compared to 2).

Albert: more? Swirl, Frepack?

1.3 Acknowledgements

We thank Lev Soukhanov and Giacomo Fenzi for insightful and helpful discussions. **Albert: More if not authors: Marcin**

Albert: World Foundation

Usage and non-usage of AI This document was essentially written without the assistance of AI. It was only used to identify typos, make refactorings, and for the cost accounting in ??.

We used AI in our implementation and experiments, as explained in ??.

²Alternative approaches based on Galois conjugates have been proposed thereafter [?, ?]. Remarkably, the Galois approach is present in prior work [?, Appendix F], although not explicitly framed in full generality.

2 Technical overview

Albert: Need to discuss somewhere that range checks are free

Albert: discuss somewhere (technical overview) the succinctness/non-succinctness of verifier

2.1 A simple version of BitZ

We explain how BitZ works. Throughout the exposition we make several simplifying assumptions, for the sake of clarity. Then we address the general cases in ??.

Let R be a ring, for example R may be the integers $\mathbb{Z}$, a prime field, a lattice ring, a finite field extension, etc. For simplicity, for now let R be either $\mathbb{Z}$ or a finite field $\mathbb{K}$ (possibly an extension field). In ?? we treat the general case. If $R = \mathbb{Z}$, then we assume a public parameter B specifying the maximum bit-size of a witness element.

In BitZ, given a public vector $\mathbf{x} = \mathbf{x} \in R^k$, the prover wants to prove it knows $\mathbf{w} = \mathbf{w} \in R^n$ such that

$$\mathbf{p}_i(\mathbf{x}, \mathbf{w}) = 0 \quad \text{in } R \quad \text{for all } i \in [m], \quad (3)$$

where $\mathbf{p} = (\mathbf{p}_i)_{i \in [m]}$ is a collection of polynomials over R . For example, in an *R1CS constraint*,

$$\mathbf{p}_i(\mathbf{x}, \mathbf{w}) = (A \cdot (\mathbf{x}, \mathbf{w}))_i \cdot (B \cdot (\mathbf{x}, \mathbf{w}))_i - (C \cdot (\mathbf{x}, \mathbf{w}))_i$$

where A, B, C are $m \times (k + n)$ matrices over R .

Step 1: Commitment in $\mathbb{F}_{2^\nu}$. The prover creates a vector $\text{bits}(\mathbf{w}) \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ of integer bits, according to some public specification of how to represent elements $r \in R$ as a linear combination of bits, i.e. $r = L(\text{bits}(r))$ for $L : \{0, 1\}^{\ell/n} \rightarrow R$ a linear map. For example, if $R = \mathbb{Z}$, then $L(b_1, \dots, b_B) = \sum_{i \in [B]} 2^{i-1} \cdot b_i$.

Then the prover looks at $\text{bits}(\mathbf{w})$ as a vector from $\mathbb{F}_{2^\nu}^\ell$, encodes it with a Reed-Solomon code (or any other code) over $\mathbb{F}_{2^\nu}$, and sends a Merkle root of the resulting codeword to the verifier.

When encoding $\text{bits}(\mathbf{w})$, every ν bits $b_1, \dots, b_\nu$ are packed as a single field element $\sum_{i \in [\mathbb{F}_{2^\nu}]} X^{i-1} \cdot b_i$ from $\mathbb{F}_{2^\nu}$, following [?] (one can use a different $\mathbb{F}_2$ -base than the monomial base $(X^i)_i$).

Step 2: Projection to a finite field $\mathbb{F}$. Albert: If necessary, the verifier samples a random map π from R to a finite field $\mathbb{F}$ of $\approx \lambda$ bit size (not necessarily a prime field). Then the claim (??) is replaced by

$$\pi(\mathbf{p}_i)(\pi(\mathbf{x}), \pi(\mathbf{w})) = 0 \quad \text{in } \mathbb{F} \quad \text{for all } i. \quad (4)$$

The form of the samplable maps π depends on the type of ring R we started with. For example, if $R = \mathbb{Z}$, then π is reduction modulo a random prime q , following [?]. If $R = \mathbb{K}$ is a finite field (not necessarily a prime field), then π is always the identity. In ?? we discuss other rings and other projections.

This step can be skipped if one already has an efficient PIOP for polynomial constraints over R ; for example, if R is a finite field of reasonable size.

If R is a finite field of very large size, one may prefer to lift the initial polynomial constraints to $\mathbb{Z}$ first (or $\mathbb{Z}[X]$ if R is field extension), and then project onto a random prime field $\mathbb{F}$ of bit-size $\approx \lambda$ during this step. This can compress the bit-size of the witness in the next steps by a very large factor.

Step 3: PIOP over $\mathbb{F}$. As in [?, ?, ?], a PIOP is run over $\mathbb{F}$, reducing (??) to a linear claim over $\mathbb{F}$ or a field extension $\mathbb{F}_{\text{ext}(e)}$ of $\mathbb{F}$. The claim is of the form

$$\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \mu \quad \text{in } \mathbb{F}_{\text{ext}(e)}, \quad (5)$$

for some $\mu \in \mathbb{F}_{\text{ext}(e)}$, $\mathbf{v} \in (\mathbb{F}_{\text{ext}(e)})^\ell$.

In general, if $\mathbb{F}$ is a prime field of $\approx \lambda$ bits (e.g. if we started with $R = \mathbb{Z}$), then $\mathbb{F}_{\text{ext}} = \mathbb{F}$. The field $\mathbb{F}_{\text{ext}}$ may be a proper extension if, for example, we started with $\mathbb{R} = \mathbb{K}$ a small finite field (in which case $\pi = \text{Id}$ and $\mathbb{K} = \mathbb{F}$), and the PIOP had to use challenges in an extension of $\mathbb{F}$ to guarantee soundness.

We assume $\mathbf{v}$ admits a tensor decomposition $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)} \in \mathbb{F}_q^{n_1 \cdot n_2}$. This is typically the case since $\mathbf{v}$ is usually an “eq” vector. This will be leveraged crucially, for efficiency purposes, at the later Step 5.2.

For simplicity, we assume $\mathbb{F}_{\text{ext}} = \mathbb{F}$ is a prime field. We discuss the general case in ??.

Step 4: Bitification of the linear claim. The claim (??) is replaced by

$$\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu \quad \text{in } \mathbb{F}, \quad (6)$$

for a new $\mathbf{u} \in \mathbb{F}^\ell$. This transformation is done using the linear bit representation L mentioned in Step 1. Concretely, representing L as matrix multiplication and indexing vectors appropriately, $\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \langle \mathbf{v}, \pi(L \cdot \text{bits}(\mathbf{w})) \rangle = \langle \pi(L)^\top \cdot \mathbf{v}, \pi(\text{bits}(\mathbf{w})) \rangle$.

The new vector $\mathbf{u}$ still admits tensor decompositions $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ with, say $\mathbf{u}^{(1)} \in \mathbb{F}^{\ell_1}$ and $\mathbf{u}^{(2)} \in \mathbb{F}^{\ell_2}$. Indeed, one can see that $L = \text{Id}_{n \times n} \otimes L'$ for some vector $L' \in R^{\ell/n}$ and Id the identity matrix of dimension n . Then for example $\mathbf{u}^{(1)} = \mathbf{v}^{(1)}$ and $\mathbf{u}^{(2)} = \mathbf{v}^{(2)} \otimes \pi(L')$, see ?? for details [Albert: Review. Maybe I could prove it here](#). Other choices of decompositions may lead to better efficiency later.

We assume that $|\mathbb{F}| < |\mathbb{F}_{2^\nu}|/\ell_1$, a condition needed for soundness later in Step 5.2. Below in in ?? we explain how to handle larger $\mathbb{F}$. We note that this is a very constraining assumption because, when for example aiming to use $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$ (a highly performant binary field) and 128 bits of security, one can easily have $\mathbb{F}$ with $|\mathbb{F}| \geq |\mathbb{F}_{2^\nu}|/\ell_1$.

Step 5: BitZ Polynomial Commitment Scheme (PCS). Our BitZ PCS enables efficiently proving (??) with $\text{bits}(\mathbf{w})$ being committed over $\mathbb{F}_{2^\nu}$ (in packed form). This is done as follows:

Step 5.0: reducing the prime modulus if necessary. This step triggers if the condition $|\mathbb{F}| < |\mathbb{F}_{2^\nu}|/\ell_1$ is not met. We explain this step below in ?? and assume for now that the condition holds.

When $\mathbb{F}_{\text{ext}}$ is a proper field extension, this step always triggers.

Step 5.1: Transformation of (??) into a batch of low-magnitude integer claim. The prover organizes $\text{bits}(\mathbf{w})$ into a $\ell_1 \times \ell_2$ matrix with columns $\text{bits}(\mathbf{w})_{*j} \in \{0, 1\}^{\ell_1}$ ($j \in [\ell_2]$). Then it publishes $\eta_1, \dots, \eta_{\ell_2} \in \mathbb{Z}$ defined as $\eta_j = \langle \pi^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle$ in $\mathbb{Z}$, for all $j \in [\ell_2]$. Then the verifier checks that [Albert: Add a remark about the possibility of proving the below with a sumcheck](#)

$$\sum_{j \in [\ell_2]} \pi(\eta_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F},$$

and (??) is replaced by the integer claim

$$\langle \pi^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle = \eta_j \quad \text{in } \mathbb{Z} \quad \text{for all } j \in [\ell_2], \quad (7)$$

where π^{-1} is the canonical representation of elements from $\mathbb{F}$ as integers in the range $[0..|\mathbb{F}| - 1]$.

Step 5.2: Exponentiation in $\mathbb{F}_{2^\nu}$ and grand product in $\mathbb{F}_{2^\nu}$. Since we assumed $|\mathbb{F}| < |\mathbb{F}_{2^\nu}|/\ell_1$, (??) is true if and only if

$$\begin{aligned} g^{\eta_j} &= g^{\langle \pi^{-1}(\mathbf{u}), \text{bits}(\mathbf{w})_{*j} \rangle} \\ &= g^{\sum_{i \in [\ell_1]} \pi^{-1}(\mathbf{u}_i) \cdot \text{bits}(\mathbf{w})_{ij}} = \prod_{i \in [\ell_1]} g^{\pi^{-1}(\mathbf{u}_i) \cdot \text{bits}(\mathbf{w})_{ij}} \\ &= \prod_{i \in [\ell_1]} \left((g^{\pi^{-1}(\mathbf{u}_i)} - 1) \cdot \text{bits}(\mathbf{w})_{ij} + 1 \right) \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2], \end{aligned} \quad (8)$$

where g is a generator of the multiplicative subgroup of $\mathbb{F}_{2^\nu}$, and in the middle expression, $\text{bits}(\mathbf{w})_{ij}$ denotes the i, j -th entry (a bit) in $\text{bits}(\mathbf{w})$ seen as an element of $\mathbb{F}_{2^\nu}$.

The prover and verifier run a batched grand product PIOP over the binary field $\mathbb{F}_{2^\nu}$ for the claim (??), obtaining as a result a new claim of the form

$$\langle \mathbf{z}, \text{bits}(\mathbf{w})_{*j} \rangle = c_j \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2], \quad (9)$$

for some $\mathbf{z} \in \mathbb{F}_{2^\nu}^{\ell_1}$ and $c_j \in \mathbb{F}_{2^\nu}$. In general, these claims can be batched to

$$\langle \mathbf{z}', \text{bits}(\mathbf{w}) \rangle = c \quad \text{in } \mathbb{F}_{2^\nu} \quad (10)$$

for $\mathbf{z}' \in \mathbb{F}_{2^\nu}^\ell, c \in \mathbb{F}_{2^\nu}$ with, for example, $\mathbf{z}' = (\alpha_1, \dots, \alpha_{\ell_2}) \otimes \mathbf{z}$ for $\alpha_1, \dots, \alpha_{\ell_2}$ verifier's randomness, and $c = \sum_{j \in [\ell_2]} \alpha_j \cdot c_j$. However, in most instantiations this step may not be necessary, especially if using an interleaved code to commit to $\text{bits}(\mathbf{w})$ [Albert: In the GKR grand product section of the main body I may need to mention batching?](#).

This exponentiation trick is inspired by [?, ?], where the same idea is used in different contexts, see ??.

This is by far the most expensive step of BitZ. Our grand product PIOP is a simple GKR protocol for grand products, for which we design novel [Albert: to our knowledge](#) prover optimizations, see ?? below and ??. These optimizations exploit the “low entropy” of the factors $(g^{\pi^{-1}} - 1) \cdot \text{bits}(\mathbf{w})_{ij} + 1$ across $(i, j) \in [\ell_1] \times [\ell_2]$, in the sense that across $\ell = \ell_1 \cdot \ell_2$ factors, there are only $\ell_1 + 1 = O(\sqrt{\ell})$ possible values taken by these factors. This is why it is important to have the vectors $\mathbf{v}$ and $\mathbf{u}$ admit a tensor split into $\approx$ square-root sized factors.

Step 5.3: Proving linear claims over $\mathbb{F}_{2^\nu}$. At this point, since $\text{bits}(\mathbf{w})$ was committed over $\mathbb{F}_{2^\nu}$ (in packed form), one can prove (??) (or the batched version of it) via standard methods such as in [?, ?, ?]. Concretely, we use the so-called ring switch from [?, ?] to turn (??) into linear claims about the packed binary field elements we encoded, and then use WHIR [?] over $\mathbb{F}_{2^\nu}$ to prove these claims. See ???? for details on ring-switching.

In ?? we discuss how to efficiently instantiate BitZ with $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$, and attain 128 bits of security.

Remark 2.1 (Description of Step 5.0). **Albert:** In the actual protocol this step is done after Step 5.1. I think they are equivalent but should unify Assume that $\mathbb{F}_{\text{ext}} = \mathbb{F}$ is a prime field (the general case is described in ??), in Step 5.0, the condition $|\mathbb{F}| < |\mathbb{F}_{2^\nu}|/\ell_1$ does not hold. Then the prover sends an integer $\mu' \in \mathbb{Z}$ such that

$$\langle \pi^{-1}(\mathbf{u}^{(1)}) \otimes \pi^{-1}(\mathbf{u}^{(2)}), \text{bits}(\mathbf{u}) \rangle = \mu' \quad \text{in } \mathbb{Z},$$

The verifier checks that $\pi(\mu') = \mu$, and responds with a random prime q such that $q < |\mathbb{F}_{2^\nu}|/\ell_1$. Then prover and verifier proceed to Step 5.1 after updating $\mathbb{F} \leftarrow \mathbb{F}_q$ the prime field with q' elements, $\pi \leftarrow \pi_{q'}$, $\mu \leftarrow \pi_q(\mu')$, and $\mathbf{u} \leftarrow \pi_q \circ \pi^{-1}(\mathbf{u}^{(1)}) \otimes \pi_q \circ \pi^{-1}(\mathbf{u}^{(2)})$.

Remark 2.2 (Other usages of the “prove-in-the-exponent” approach). There are other projects (e.g. Binius64 [?] and Shoukhanov’s note [?]) that use exponentiation to prove integer (or mod $|\mathbb{F}_{2^\nu}| - 1$) claims in binary fields, none of these (to our knowledge) is used or is usable in a context that results in the low entropy grand products of ours, where our optimizations apply. We discuss this further in ??

2.1.1 Intuition of soundness proofs

We prove that the above protocol (and the more general version in ??) is round-by-round knowledge sound. We next provide the main ideas of why this is the case.

Albert: todo

2.2 Virtual $\mathbb{F}_2$ -linear transforms in $\mathbb{Bit}\mathbb{Z}$ and NP-complete dually linear relations

Observe that, at the end of the day, Step 5.2 outputs a linear claim of the form³

$$\langle \mathbf{z}, \text{bits}(\mathbf{w}) \rangle = c \quad \text{in } \mathbb{F}_{2^\nu}, \quad (11)$$

for some $\mathbf{z} \in \mathbb{F}_{2^\nu}^\ell$ and $c \in \mathbb{F}_{2^\nu}$. This is a standard linear claim over $\mathbb{F}_{2^\nu}$ for the vector $\text{bits}(\mathbf{w})$ which we committed to previously.

Now, say that we know that $\text{bits}(\mathbf{w})$ is a linear transformation of a vector $\mathbf{f} \in \mathbb{F}_2^{\ell'} \subseteq \mathbb{F}_{2^\nu}^{\ell'}$, so that

$$\text{bits}(\mathbf{w}) = M \cdot \mathbf{f} \quad \text{in } \mathbb{F}_{2^\nu},$$

for some matrix $M \in \mathbb{F}_2^{\ell \times \ell'}$. Then

$$\langle \mathbf{z}, \text{bits}(\mathbf{w}) \rangle = \langle M^\top \cdot \mathbf{z}, \mathbf{f} \rangle = c \quad \text{in } \mathbb{F}_{2^\nu}. \quad (12)$$

Hence, in this scenario, one can commit to $\mathbf{f}$ instead of $\text{bits}(\mathbf{w})$ (again, in packed formed if necessary), replace the claim (??), and prove (??) in Step 5.3 of $\mathbb{Bit}\mathbb{Z}$.

Thus, $\mathbb{Bit}\mathbb{Z}$ can be expanded so as to prove claims where we are given a public vector $\mathbf{x} = \mathbf{x} \in R^k$, and the prover claims it knows $\mathbf{w} = \mathbf{f} \in \mathbb{F}_{2^\nu}^{\ell'}$ such that

$$\begin{cases} \mathbf{p}_i(\mathbf{x}, \mathbf{w}) = 0 & \text{in } R & \text{for all } i \in [m], \\ \text{bits}(\mathbf{w}) = \pi_2^{-1}(M \cdot \mathbf{f}) \end{cases} \quad (13)$$

³We start with the batched version of the claim, but the ideas in this section can be adapted to (??) as well.

where $\mathbf{p} = (\mathbf{p}_i)_{i \in [m]}$ is a collection of polynomials over R ; and π_2 is reduction modulo 2 [Albert: I may have been loose about when \$\text{bits}\(\mathbf{w}\)\$ is in \$\mathbb{F}_2\$ and when in integers during the technical overview, review.](#)

A particularly interesting instantiation of this relation is the following R1CS-like relation over the integers, where $\mathbf{w}$ is a vector of bits (hence $\mathbf{w} = \text{bits}(\mathbf{w})$) and $\mathbf{w} = \pi_2^{-1}(M \cdot \mathbf{f})$:

$$\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z}) := \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, A, B, C, M) \in \mathbb{F}_2^k \times (\mathbb{Z}^{m \times \ell})^3 \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \left| \begin{array}{l} (A \cdot \mathbf{w}) \circ (B \cdot \mathbf{w}) = C \cdot \mathbf{w} \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{z}, \mathbf{f})) \end{array} \right. \right\}$$

An interesting subset of $\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z})$ is that where A and B are zero matrices. We call the resulting relation the CM relation because of the matrices C and M :

$$\mathcal{R}_{\text{CM}}(\mathbb{Z}) := \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, C, M) \in \mathbb{F}_2^k \times \mathbb{Z}^{m \times \ell} \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \left| \begin{array}{l} C \cdot \mathbf{w} = 0 \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{z}, \mathbf{f})) \end{array} \right. \right\}$$

In a sense, the CM relation places only linear constraints over its witness (with some constraints happening in $\mathbb{F}_{2^\nu}$ and other in $\mathbb{Z}$). Yet, the CM relation is NP-complete, as shown below in ??.

Indeed, in ?? **we arithmetize SHA-256 as a CM instance**. Our arithmetization leads to witnesses of size (in bits) **** [Albert: report comparison with Flock/Binius in witness size. Mention as well the performance comparison.](#)

Remark 2.3 (The $\mathcal{R}_{\text{CM}}$ relation is NP-complete). Any Boolean circuit can be arithmetized as an instance of $\mathcal{R}_{\text{CM}}$. Indeed, since any boolean circuit can be written as a system of equations using the operations XOR and AND, it suffices to prove that, given $x, y, z \in \{0, 1\}$, we can express both of the following constraints as an instance of $\mathcal{R}_{\text{CM}}$:

$$x \text{ XOR } y = z, \quad x \text{ AND } y = z.$$

Indeed, $x \text{ XOR } y = z$ if and only if $\pi_2^{-1}(\pi_2(x) + \pi_2(y)) = z$, where here $+$ means addition modulo 2. Further, $x \text{ AND } y = z$ if and only if

$$2 \cdot z = x + y - \pi_2^{-1}(\pi_2(x) + \pi_2(y)).$$

One can verify this last equivalence for example by directly inspecting that it holds for the four possible combinations of $x, y \in \{0, 1\}$.

[Albert: Todo: Expressiveness result of Remco](#)

2.3 GKR for low-entropy batched grand products

Recall that, in Step 5.2 of $\mathbb{B}\text{it}\mathbb{Z}$ (??) the prover must prove that

$$g^{\langle \pi^{-1}(\mathbf{u}), \text{bits}(\mathbf{w})_{*j} \rangle} = \prod_{i \in [\ell_1]} \left(g^{\pi^{-1}(\mathbf{u}_i)} - 1 \right) \cdot \text{bits}(\mathbf{w})_{ij} + 1 = g^{\eta_j} \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2]. \quad (14)$$

For efficiency purposes, we assume both ℓ_1 and ℓ_2 are $O(\sqrt{\ell})$ (see ?? for exact parameter instantiations). At a protocol level, we simply prove (??) with a standard GKR protocol for grand products

[?]. Concretely, let $\ell_1 = 2^{r_1}$ and $\ell_2 = 2^{r_2}$. We set up multilinear polynomials $L_0, L_1, \dots, L_{r_1}$ on $r_2 + i$ variables for $i \in [0..r_1]$, respectively, such that

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = \left(g^{\pi^{-1}(\mathbf{u}_i)} - 1 \right) \cdot \text{bits}(\mathbf{w})_{ij} + 1 \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2},$$

and, for $i = 0, \dots, r_1 - 1$

$$L_i(\mathbf{b}, \mathbf{c}) = L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \quad \text{for all } \mathbf{b} \in \{0, 1\}^i \times \{0, 1\}^{r_2}. \quad (15)$$

We then use a series of sumchecks to certify ?? for all $i = 0, \dots, r_1 - 1$, and $L_0(\mathbf{c}) = g^{\eta_{\text{int}}(\mathbf{c})}$ for all $\mathbf{c} \in \{0, 1\}^{r_2}$.

We emphasize again that, at a protocol level, this GKR scheme is completely standard. See ?? for all the details.

At the end of the GKR grand product, the prover and verifier are left with MLE evaluation claims of the form:

$$\widetilde{g^{\pi^{-1}(\mathbf{u})}}(\mathbf{r}) = a, \quad \widetilde{\text{bits}(\mathbf{w})_{*j}}(\mathbf{r}) = a_j \quad \text{for all } j \in [\ell_2] \quad (16)$$

for some $\mathbf{r} \in \mathbb{F}_{2^\nu}^{r_1}$, where $\widetilde{*}$ denotes MLE extension, and $g^{\pi^{-1}(\mathbf{u})}$ is the vector with entries $g^{\pi^{-1}(\mathbf{u}_i)}$.

Since $\mathbf{u}$ is public, the verifiers checks the first claim of (??) on its own in time $O(\sqrt{\ell})$. This is one, among several, reasons why the tensor split of $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ and the splitting the original claim into a batch of ℓ_2 claims (see Step 5.1 of ??) pays off.

We design novel (to our knowledge [Albert: check](#)) prover optimizations for the GKR grand product (without altering the protocol itself), see ?? for details. Namely, we exploit the following property, which is enabled, again, by the tensor split of $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ and the splitting the original claim into a batch of ℓ_2 claims:

Low entropy grand products: There are at most $\ell_1 = O(\sqrt{\ell})$ values that $L_{r_1}(\mathbf{b}, \mathbf{c})$ can take on the hypercube of size $\ell_1 \cdot \ell_2 = \ell$.

This property trickles down to different stages of the GKR protocol. For example, the above property implies that $L_{r_1-1}(\mathbf{b}', \mathbf{c})$ as well can take only an $O(\sqrt{\ell})$ amount of values (with worse constants now) across all $(\mathbf{b}', \mathbf{c}) \in \{0, 1\}^{r_1-1} \times \{0, 1\}^{r_2}$. Similarly, the coefficients of the first messages sent by the prover at the last stages of the GKR protocol (which involve the most expensive sumchecks), also benefit from a low entropy property.

This allows the prover to compute small lookup tables that are kept on cache and reused across all values of $\mathbf{c} \in [\ell_2]$. We build different lookup tables used at different stages of the GKR protocol. For example, we use 4 lookup tables in the first round of the last sumcheck of GKR (the most expensive sumcheck, being on $r_1 + r_2$ variables). These tables are used to compute the first prover message of such sumcheck. We then construct 3 more tables to compute the second's prover message, and finally we use one last table to compute the third prover message. Similarly, we build lookup tables for the first rounds of the second and third last sumchecks of the GKR protocol.

2.3.1 Comparison to other “prove-in-the-exponent” usages

[Albert: Missing refs?](#) [Longfellow?](#)

Binius64 For example, [?] multiplies pairs of 64-bit integer witness elements in the exponent of g . This is substantially different from our setting where, when binary fields come into play, we are only interested in proving an integer claim about a linear combination of elements of the form “(public values) · (witness elements)”, i.e. $\langle \mathbf{v}, \mathbf{w} \rangle = \mu$ in $\mathbb{Z}$, with the public values $\mathbf{v}$ admitting a tensor split $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}$ (also BitZ can prove multiplication of all sorts of ring elements, not just 64-bit integers). This simpler setting is what allows us to use the low entropy optimizations designed in this work. Crucially, **the reason why we only need to address this setting is because we start off with constraints and a PIOP over integers/fields/rings (Steps 2 to 4 inclusive), and only ever move to work on binary fields when we need to prove an MLE evaluation of the witness (a linear claim) over a “small” prime field (Step 5).**

[?] on the other hand multiplies 64-bit integer witness in the exponent directly. More precisely, [?] proves the identity $g^{\mathbf{w}_{1i} \cdot \mathbf{w}_{2i}} = g^{2^{64} \cdot \mathbf{w}_{3i}} \cdot g^{\mathbf{w}_{4i}}$ for $\mathbf{w}_1, \dots, \mathbf{w}_4$ vectors of 64-bit integers (together with a small additional constraint to address possible overflow). This is done through a combination of LogUp* [?], a GKR grand product (for relatively small products), and the Frobenius automorphism trick. It does not seem possible to follow a similar optimization strategy to ours.

We emphasize again that BitZ supports proving operations of all kinds of ring elements (small and large integers, field elements, lattice ring elements, extension field elements), as opposed to ?? where multiplication is limited to 64-bit integers.

In ?? we compare the performance of BitZ and Binius64 when multiplying integers. BitZ outperforms Binius64 by [Albert: fill in](#).

The note [?]. In the note [?], Soukhanov proposes a scheme for committing to a vector of bits $\mathbf{w}$ over the binary field $\mathbb{F}_{2^{127}}$, and proving MLE evaluation claims over the prime field $\mathbb{F}_{2^{127}-1}$, for a vector of elements from $\mathbb{F}_{2^{127}-1}$ obtained by packing bits from $\mathbf{w}$ together. To do so, at a high level, the MLE evaluation claims are proved “in the exponent” in the binary field $\mathbb{F}_{2^{127}}$, leveraging that the multiplicative subgroup of $\mathbb{F}_{2^{127}}$ has prime order $|\mathbb{F}_{2^{127}}^*| = 2^{127} - 1$. The proposed approach is substantially different from ours: [?] essentially takes a polylogarithmic PCS over the prime field $\mathbb{F}_{2^{127}-1}$, such as Basefold [?], WHIR [?], etc., and executes it “in the exponent of $\mathbb{F}_{2^{127}}$ ”. To do so, it relies on the lookup argument LogUp* [?], grand product arguments (for small products), the Frobenius trick, and additional commitments, for every of the $\log \ell$ rounds.

In its execution, [?] and Binius64 [?] seem to us to be more similar than any of these two to BitZ. Indeed, they both use, mostly, the same collection of subprotocols, in a similar manner. Because of this we expect both approaches to have comparable performance profiles, though we do not validate this hypothesis due to [?] not reporting performance nor providing an implementation.

Conceptually, [?] shares with BitZ the aim of committing to bits in a binary field, while proving MLE claims of those bits in some other structure R . Note that [?] is limited to $R = \mathbb{F}_{2^{127}-1}$, while BitZ can handle essentially all rings and fields R of practical interest. Additionally it is not clear whether [?] can leverage any of our low-entropy oriented optimizations without substantially modifying the approach, incorporating some of our ideas. **Albert: Ask Lev if he has implemented his thing so we can compare. Also ask him if Binius64 is essentially following his method (Logup*, grand product, Frobenius)**

Albert: The comparison of Lev and BitZ right now has several unbacked claims (which I believe are true, but can’t really prove them, especially without an implementation). So probably it’s better to not push it, remove the unbacked claims, and leave it as it is

2.4 BitZ in full generality

In general, BitZ can prove polynomial constraints over multiple rings R_0 (in the same instance) of the form $R[X]/(h(X))$, where R is either the ring of integers $\mathbb{Z}$ or a finite field (not necessarily prime), and $h(X) \in R[X]$ is either the zero polynomial, or a monic non-constant polynomial. Examples of such rings R_0 include: the integers $\mathbb{Z}$, arbitrary finite fields, and lattice rings of the form $\mathbb{F}_q[X]/(\Phi(X))$ for $\Phi(X)$ a cyclotomic polynomial and q a prime power. More generally, R_0 can be any two-generated ring of zero or prime characteristic.

Concretely, we define a target relation, which we call *Universal Polynomial Satisfiability (UPESAT)*, as follows. Let ℓ be a witness length and k a public input length. Let R be either $\mathbb{Z}$ or a finite field. Let $\mathbf{R} = (R_i)_{i \in [\iota]}$ be a collection of rings of the form $R_i = R[X]/(h_i(X))$ for $h_i(X) \in R[X]$ either zero or a monic non-constant polynomial. For each $i \in [\iota]$, let $\mathbf{p}^{(i)}$ be a vector of n polynomials

$$\mathbf{p}^{(i)} = \left(\mathbf{p}_j^{(i)} \mid j \in [n] \right) \in ((R[X])[Y_1, \dots, Y_{k+\ell}])^n$$

with coefficients in $R[X]$ and on $k+\ell$ variables $Y_1, \dots, Y_{k+\ell}$. Given $a \in R[X]$ and $i \in [\iota]$, we say that $a = 0$ holds in R_i if the image of a under the canonical projection $R[X] \rightarrow R_i$ is zero. Finally, given a degree bound d and a bit-size bound, we let $R_{\leq B}^{\leq d}[X]$ be the subset of $R[X]$ formed by polynomials of degree $< d$ and coefficients in R of bit-size $\leq B$ (see Step 1 of ?? for a formalization of how we bit-decompose elements from R).

We define our target relation as

$$\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}[X], \mathbf{p}, \mathbf{R}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \left(R_{\leq B}^{\leq d}[X] \right)^k \\ \mathbf{w} = \mathbf{w} \in \left(R_{\leq B}^{\leq d}[X] \right)^\ell \end{array} \mid \mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{w}) = 0 \text{ in } R_i \text{ for all } j \in [n], i \in [\iota] \right\}$$

Remark 2.4 (Non prime characteristics or arithmetic modulo arbitrary integers). UPESAT does not allow to work with non-prime and non-zero characteristics m . I.e. it does not support, for example, $\mathbb{Z}/2^n\mathbb{Z}$ (i.e. arithmetic modulo $m = 2^n$). Following [?, ?], we arithmetize such characteristics in $\mathbb{Z}$ or $\mathbb{Z}[X]$ via the standard trick of adding a term of the form $+m \cdot w$ to the constraints, for w new witness elements. The correctness and soundness of this step follows from the fact that $a + b = c$ in $\mathbb{Z}/m\mathbb{Z}$ if and only if $a + b = c + m \cdot w$ in $\mathbb{Z}$ for some $w \in \mathbb{Z}$. Similar identities hold for rings of non-prime and non-zero characteristic other than $\mathbb{Z}/m\mathbb{Z}$. See the introduction and technical overviews of [?, ?] for more details.

We note that UPESAT is a subset of the more general UCS relation from [?]. See ?? for a comparison. [Albert: Mention we could allow for different characteristics in the same relation?](#)

Similarly as in ??, BitZ can also handle a variation of $\mathcal{R}_{\text{UPESAT}}$ where $\text{bits}(\mathbf{w})$ is a $\mathbb{F}_2$ -linear transformation of a vector of bits $\mathbf{f} \in \mathbb{F}_2^{\ell'}$.

Example 2.5. [Albert: I could briefly explain here how to arithmetize the computations in our experiments.](#)

We now explain how to modify the basic version of BitZ in ?? for the relation $\mathcal{R}_{\text{UPESAT}}$. There are two aspects that need modification:

1. In Step 2 we need to design a method for transforming constraints in R_i to constraints in a finite field $\mathbb{F}$.
2. In Step 3, $\mathbb{F}_{\text{ext}}$ could be a proper extension of a prime field $\mathbb{F}_p$. This occurs if for example $R_i = \mathbb{F}_p$ for all $i \in [\iota]$ (i.e. all our constraints are over the prime field $\mathbb{F}_p$), and p is small (say, 32-64 bits), and so the PIOP in Step 3 needs to work on a field extension of $\mathbb{F}_p$.

Generalized Step 2. Regarding Step 2, we discuss first the case that $\iota = 1$ and $h_1 = (0)$, so that all constraints are over $R[X]$, for R the integers or a finite field, with witnesses belonging to $R_{\leq B}^{<d}[X]$. In this case the verifier samples a map π of one of the following forms: If $R = \mathbb{Z}$ and $d > 0$,

$$\pi : \mathbb{Z}[X] \rightarrow \mathbb{F}_q, \quad g(X) \mapsto \pi_q(g)(\alpha)$$

for a random prime q and a random $\alpha \in \mathbb{F}_q$, where π_q denotes reduction modulo q .

If $R = \mathbb{F}$ is a finite field of characteristic q , then

$$\pi : \mathbb{F}[X] \rightarrow \mathbb{F}', \quad g(X) \mapsto g(\alpha)$$

for $\mathbb{F}'$ possibly a field extension of $\mathbb{F}$ (so that $|\mathbb{F}'| = O(2^\lambda)$) and $\alpha \in \mathbb{F}'$. [Albert: Say a few words of why this is sound \(once I've written the soundness intuition for the basic protocol\)](#)

The general case where not all ideals $(h_i(X))$ are zero is dealt in the same way as in Zinc+ [?]. We omit the details here for the sake of brevity. We refer to ?? (Items 3 and 4 of Round 1, and Round 2, of ??) or to the technical overview of [?] [Albert: Should I describe this here?](#)

The case when $\mathbb{F}_{\text{ext}}$ is a proper extension Next we discuss the scenario where $\mathbb{F}_{\text{ext}}$ is a proper field extension at the end of Step 3 of ??. In this case, Step 4 proceeds in the same way as in the normal case. At the end of Step 4 prover and verifier are left with the claim

$$\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu \quad \text{in } \mathbb{F}_{\text{ext}},$$

for some $\mathbf{u} \in \mathbb{F}_{\text{ext}}^\ell$ and $\mu \in \mathbb{F}_{\text{ext}}$.

After Step 4, Step 5.0 triggers [Albert: Again, in the main body this step triggers after step 5.1 and is slightly different](#). Since $\mathbb{F}_{\text{ext}}$ is a finite field, it is of the form $\mathbb{F}_{\text{ext}} = \mathbb{Z}[X]/(q, f(X))$ for an irreducible polynomial $f(X)$ over $\mathbb{F}_q[X]$. Let $\pi_{\text{canon}} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{\text{ext}}$ be the canonical homomorphism sending $g(X) \in \mathbb{Z}[X]$ to its reduction modulo q and modulo $f(X)$.

The prover sends an integer polynomial $\mu' \in \mathbb{Z}[X]$ such that

$$\langle \pi_{\text{canon}}^{-1}(\mathbf{u}^{(1)}) \otimes \pi_{\text{canon}}^{-1}(\mathbf{u}^{(2)}), \text{bits}(\mathbf{u}) \rangle = \mu' \quad \text{in } \mathbb{Z}[X],$$

The verifier checks that $\pi_{\text{canon}}(\mu') = \mu$, and responds with a random prime q' such that $q' < |\mathbb{F}_{2^\nu}|/\ell_1$ and with a random prime field element $\alpha \in \mathbb{F}_{q'}$. Then prover and verifier proceed to Step 5.1 after updating $\mathbb{F}_{\text{ext}} \leftarrow \mathbb{F}_{q'}$, $\pi \leftarrow \pi_{q', \alpha}$, $\mu \leftarrow \pi_{q', \alpha}(\mu') \in \mathbb{F}_{q'}$, and

$$\mathbf{u} \leftarrow \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}^{(1)}) \otimes \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}^{(2)}),$$

where $\pi_{q', \alpha} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{q'}$ is defined as

$$\pi_{q', \alpha}(g(X)) = \pi_{q'}(g)(\alpha) \in \mathbb{F}_{q'}.$$

Albert: Again, say a few words of why this is sound (once I’ve written the soundness intuition for the basic protocol)

One can, instead of sampling morphisms of the form $\pi_{q',\alpha}$, sample other types of $\mathbb{Z}$ -module homomorphisms $\pi' : \mathbb{Z}^{\deg(\mathbb{F}_{\text{ext}})}[X]$ to a prime field. See ???. This can modestly improve the soundness errors at the expense of increasing the communication complexity. Albert: It may be a good idea to present the protocols as parametrized by a family of module homomorphisms where we sample from. Right now the protocols have the fixed family above. Let’s see if there’s time

The rest of BitZ proceeds exactly as in ???.

Albert: To discuss somewhere: this technique allows us to do things such as: 1) start with a constraint in Babybear, do the PIOP in an extension, and then open with BitZ, 2) run PIOP’s with primes of size, say ≥ 128 bits, and then change the prime to a 108 bit prime to fit in the exponent of $\mathbb{F}_{2^\nu}^*$ without overflowing

Albert: Thoughts: We can handle arbitrary linear weights in our claims, of arbitrary bit-size and in arbitrary extension degrees, and still use a smallish binary field like GHASH during our protocol (not obvious at all because “proving in the exponent in principle only allows for using $\lesssim 90$ -ish integer bits. Albert: This is a good story for the technical overview

Scratch from another section. Delete? Albert: These two examples would be illustrative in the intro and TO To fix intuition, $\deg(\mathbb{K})$ and q are determined by the field or ring a PIOP runs on. For example, in our full proving system in ???, constraints can be (among others) over $\mathbb{Z}$ or $\mathbb{Z}[X]$, then they are projected to a random prime field $\mathbb{F}_q$, a PIOP is ran on them, and an MLE claim over $\mathbb{F}_p$ is produced. Then ??? is used to prove those claims. The size of q is set so that the PIOP achieves λ bits of security. A very conservative choice in this setting is $q \leq 2^{256}$, but also $q \leq 2^{128}$ is feasible if $\lambda = 100$ or one is willing to use grinding during the PIOP.

Another setting is to have constraints over a finite field $\mathbb{K}$ such as a degree 4 or 5 extension of the Babybear field, or a degree 2 or 3 extension of the Goldilocks field

3 Preliminaries

General notation Given an integer $n \geq 1$, we write $[n]$ to denote the set $\{1, \dots, n\}$. Any map $f : S \rightarrow T$ is extended entrywise to vectors and matrices with entries in S , and coefficient-wise to polynomials with coefficients in S .

We use $\circ$ to denote composition of maps, so that for maps $f : S_1 \rightarrow S_2$ and $g : S_2 \rightarrow S_3$, we have $f \circ g : S_1 \rightarrow S_3$ defined as $f \circ g(x) = f(g(x))$.

3.1 Algebra

We write $\mathbb{F}_q$ for the finite field with q elements (where q is a prime or a prime power), $\mathbb{Q}$ for the field of rational numbers, and $\mathbb{Z}$ for the ring of integers.

The *characteristic* of a ring R is the largest $c \geq 1$, if it exists, such that $1 + \dots + 1 = 0$ in R . Otherwise, if $1 + \dots + 1 \neq 0$ for all c , the characteristic is zero. By $[0..n]_R$ we denote all elements in R of the form $1 + \dots + 1$ for $i = 0, \dots, n$. For us, typically $R = \mathbb{Z}$ or R is a finite field $\mathbb{F}$.

We let $R[X]$ be the ring of univariate polynomials with coefficients in R , on the variable X . For a tuple of variables $\mathbf{Y} = (Y_1, \dots, Y_\mu)$, $R[\mathbf{Y}]$ denotes multivariate polynomials with coefficients in R .

Given a degree bound $d \geq 1$, a bit-size bound $B \geq 1$, we define

$$R_{\leq B}^{\leq d}[X] := \left\{ a(X) \in R[X] \mid a(X) = \sum_{i=0}^d a_i \cdot X^i, \text{ for some } a_i \in [0..2^B - 1]_R, \text{ for all } i \right\},$$

When we place no degree or bit-size bound, we use the notation $R_{\leq B}[X]$ and $R^{<d}[X]$, respectively. Note that $R^{\leq d}[X]$ is an R -module of dimension $(d+1)$, admitting as base $1, X, \dots, X^d$.

Given a ring homomorphism $\psi : R_1 \rightarrow R_2$, a R_1 -module M_{R_1} , and a R_2 -module, a ψ -module homomorphism between M_{R_1} and M_{R_2} is a map ϕ such that $\phi(\alpha \cdot a + \beta \cdot b) = \psi(\alpha) \cdot \phi(a) + \psi(\beta) \cdot \phi(b)$ for all $\alpha, \beta \in R_1$ and $a, b \in M_{R_1}$.

Given a base $\alpha_1, \dots, \alpha_d$ for M_R , we let $M_{R_{\leq B}}$ be the subset of M_R with elements whose coordinates belong to $R_{\leq B}$ in this base.

A subset $S \subseteq R$ of a commutative ring R is called *exceptional* if for all two distinct $a, b \in R$, the element $a - b$ has a multiplicative inverse in R .

Lemma 3.1 (Generalized Alon–Füredi Theorem [?]). *Let R be a ring and let $S \subseteq R$ be a finite exceptional subset of R . Let $f(\mathbf{Y}) \in R[\mathbf{Y}]$ be a nonzero multilinear polynomial in μ variables. Then $\Pr_{\mathbf{r} \leftarrow S^\mu} [f(\mathbf{r}) = 0] \leq \mu/|S|$.*

3.2 Linear codes and relations

Linear codes Let $\mathcal{C} \subseteq \mathbb{F}^m$ be a linear code over $\mathbb{F}$ of dimension ℓ and blocklength m . For $h \geq 1$, the h -interleaving of $\mathcal{C}$ is the code $\mathcal{C}^{\equiv h} \subseteq (\mathbb{F}^h)^m$ of dimension $\ell \cdot h$ and blocklength $m \cdot h$, whose codewords have the form

$$((c_{i1}, \dots, c_{ih}))_{i \in [m]} \quad \text{for } (c_{1j}, \dots, c_{mj}) \in \mathcal{C} \quad \text{for all } j \in [h].$$

Given two words w_1, w_2 of length n , the relative Hamming distance between w_1 and w_2 is denoted

$$\Delta(w_1, w_2).$$

Relations A relation $\mathcal{R}$ consists of triples $(\mathbf{x}, \mathbf{y}; \mathbf{w})$, where $\mathbf{x}$ is the explicit instance, $\mathbf{y}$ the implicit instance or oracle, and $\mathbf{w}$ the witness. The corresponding language is $\text{LANG}(\mathcal{R}) := \{(\mathbf{x}, \mathbf{y}) \mid \exists \mathbf{w} \text{ such that } (\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}\}$.

Polynomial Equation Satisfiability (PESAT) over finite fields We recall a relation from [?] [Albert: Linear time accumulation paper, or earlier?](#) over finite fields, which captures popular constraints such as R1CS, CCS, AIR (cf. [?]), and lookup and permutation constraints (cf. [?]).

Fix a finite field $\mathbb{F}$, a witness length ℓ , an input length k , a number of rows n , a collection of n polynomials $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ on $k + \ell$ variables, for each $j \in [n]$.

Definition 3.2. The PESAT relation with respect to these parameters is defined as

$$\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \mathbb{F}^k, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in \mathbb{F}^\ell \end{array} \mid \mathbf{p}_i(\mathbf{x}, \mathbf{f}) = 0 \text{ for all } i \in [n] \right\}$$

Let $\mathcal{C} \subseteq \mathbb{F}^m$ be a linear code of dimension ℓ' and blocklength m over $\mathbb{F}$. Let $T : \mathbb{F}^\ell \rightarrow \mathbb{F}^{\ell'}$ be an arbitrary map (looking ahead, for us T will be a packing map that packs elements from $\mathbb{F}_2$ into elements of a larger binary field). We define the *Encoded PESAT relation* as:

$$\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}, \mathcal{C}, T) := \left\{ \left(\begin{array}{l} \mathbf{x} = \mathbf{x} \in \mathbb{F}^k, \\ \mathbf{y} = f \in \mathbb{F}^m, \\ \mathbf{w} = \mathbf{f} \in (\mathbb{F}^\ell) \end{array} \right) \mid \begin{array}{l} \text{Enc}_{\mathcal{C}}(T(\mathbf{f})) = f \\ \wedge \mathbf{p}_i(\mathbf{x}, \mathbf{f}) = 0 \text{ for all } i \in [n] \end{array} \right\}$$

We will drop the reference to T whenever it is understood from the context. When $n = 1$ and $\mathbf{x}$ are such that $\mathbf{x} = (\mathbf{v}, \mu) \in \mathbb{F}^\ell \times \mathbb{F}$; $\mathbf{p}_1(\mathbf{x}, \mathbf{f}) = \langle \mathbf{f}, \mathbf{v} \rangle - \mu$ (and $\ell = k$), the relation $\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}, \mathcal{C}, T)$ is called a *constrained code* [?] [Albert: WHIR?](#). We denote it

$$\mathcal{R}_{\text{Lin}}(\mathbb{F}, \mathcal{C}, T).$$

Definition 3.3 (Relaxation of $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$). Given $\delta \in (0, 1)$, the *relaxation* of $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$, denoted

$$\mathcal{R}_{\text{PESAT}}^\delta, \quad \mathcal{R}_{\text{Lin}}^\delta$$

are the relations defined exactly as $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$, respectively, except that the constraint $\text{Enc}_{\mathcal{C}}(T(\mathbf{f})) = f$ is replaced by

$$\Delta(\text{Enc}_{\mathcal{C}}(T(\mathbf{f})), f) < \delta.$$

3.3 Interactive Oracle Reductions (IOR) of proximity

A public-coin *Interactive Oracle Reduction (IOR) of proximity* $\Pi = (\mathbf{P}, \mathbf{V})$ from a relation $\mathcal{R}$ to a relation $\mathcal{R}'$ is specified as follows. The prover $\mathbf{P}$ receives $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}$. The verifier $\mathbf{V}$ receives $\mathbf{x}$ and query access to $\mathbf{y}$. They interact for k rounds. In round $j \in [k]$, the prover sends an oracle π_j and the verifier sends a uniformly random message $\rho_j \in \{0, 1\}^{r_j}$. After the k rounds, the verifier makes queries to $\mathbf{y}$ and to $\pi_1, \dots, \pi_k$. Then:

- $\mathbf{V}$ either rejects or outputs a new pair $(\mathbf{x}', \mathbf{y}')$, where $\mathbf{y}'$ is specified implicitly in terms of the oracles $\mathbf{y}$ and $\pi_1, \dots, \pi_k$.
- $\mathbf{P}$ outputs a witness $\mathbf{w}'$ such that $(\mathbf{x}', \mathbf{y}', \mathbf{w}') \in \mathcal{R}'$.

We write $\langle \mathbf{P}, \mathbf{V} \rangle(\mathbf{x}, \mathbf{y}, \mathbf{w})$ for the interaction of prover and verifier during the IOR.

Definition 3.4 (Completeness). Π has (*perfect*) *completeness* if for every $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}$,

$$\Pr_{\rho_1, \dots, \rho_k} [(\mathbf{x}', \mathbf{y}'; \mathbf{w}') \in \mathcal{R}' \mid (\mathbf{x}', \mathbf{y}'; \mathbf{w}') \leftarrow \langle \mathbf{P}, \mathbf{V} \rangle(\mathbf{x}, \mathbf{y}, \mathbf{w})] = 1.$$

Remark 3.5. An IOP for a relation $\mathcal{R}$ is an IOR from $\mathcal{R}$ to the trivial relation $\mathcal{R}_{\text{triv}} = \{(\mathbf{x} = \perp, \mathbf{y} = \perp, \mathbf{w} = \perp)\}$.

We proceed to define round-by-round knowledge soundness of an IOR, as presented in [?].

Definition 3.6 (Knowledge state function with relaxation). Let $\Pi = (\mathbf{P}, \mathbf{V})$ be an IOR from a relation $\mathcal{R}$ to $\mathcal{R}'$. A *knowledge state function with relaxation* $(\tilde{\mathcal{R}}, \tilde{\mathcal{R}}')$ for Π is a (possibly inefficient) function KState that, on input $(\mathbf{x}, \mathbf{y})$, interaction transcript tr , and knowledge state witness $\mathbf{w}$, outputs a bit and has the following properties.

- **Empty transcript.** If $\text{tr} = \emptyset$ is the empty transcript, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}; \mathbf{w}) \in \widetilde{\mathcal{R}}$.
- **Prover moves.** If tr is a transcript where the prover is about to move and $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 0$, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \pi, \mathbf{w}) = 0$ for every prover message π .
- **Full transcript.** If $\text{tr} = (\pi_1, \rho_1, \dots, \pi_k, \rho_k)$ is a full transcript, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if V on tr outputs $(\mathbf{x}', \mathbf{y}')$ such that $(\mathbf{x}', \mathbf{y}'; \mathbf{w}) \in \widetilde{\mathcal{R}'}$.

Definition 3.7 (Round-by-round knowledge soundness). A k -round IOR Π has *round-by-round knowledge soundness with relaxation* with errors $(\kappa_1, \dots, \kappa_k)$ if there exists a knowledge state function KState as in ?? and a deterministic extractor $\mathcal{E}_{\text{rbr}}$ such that for every $(\mathbf{x}, \mathbf{y})$, every $j \in [k]$, and every transcript tr where the verifier is about to send ρ_j ,

$$\Pr_{\rho_j} \left[\exists \mathbf{w} \text{ such that } \begin{array}{l} \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \rho_j, \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \rho_j, \mathbf{w}) = 1 \end{array} \right] \leq \kappa_j(\mathbf{x}, \mathbf{y}).$$

If $(\widetilde{\mathcal{R}}, \widetilde{\mathcal{R}'}) = (\mathcal{R}, \mathcal{R}')$ then we say the round-by-round knowledge soundness of Π is *without relaxation*.

Remark 3.8 (Composition of IORs). *Albert: Turn into a theorem/lemma. What can I cite for this? TensorSwitch, Arc, or ZOOK?* IORs can be composed together and inherit completeness and round-by-round knowledge soundness errors. More precisely, let Π_1, Π_2 be perfectly complete IORs from $\mathcal{R}_1$ to $\mathcal{R}_2$ and $\mathcal{R}_3$, respectively, with round-by-round knowledge errors with relaxation $(\widetilde{\mathcal{R}}_1, \widetilde{\mathcal{R}}_2)$ and $(\widetilde{\mathcal{R}}_2, \widetilde{\mathcal{R}}_3)$, respectively; and with errors $\kappa_{11}, \dots, \kappa_{1k_1}$ and $\kappa_{21}, \dots, \kappa_{2k_2}$, respectively. Consider the IOR Π' from $\mathcal{R}_1$ to $\mathcal{R}_3$ consisting of applying IOR Π_1 first, and then Π_2 taking as input for Π_2 the output of Π_1 . Then Π' is a perfectly complete $(k_1 + k_2)$ -round IOR from $\mathcal{R}_1$ to $\mathcal{R}_3$ with round-by-round knowledge soundness with relaxation $(\widetilde{\mathcal{R}}_1, \widetilde{\mathcal{R}}_3)$ and errors $\kappa_{11}, \dots, \kappa_{1k_1}, \kappa_{21}, \dots, \kappa_{2k_2}$.

Grinding *Albert: Review* Grinding [?] is a technique where, given an IOP Π for a relation $\mathcal{R}$ with relaxation $\widetilde{\mathcal{R}}$ with round-by-round knowledge errors $\kappa_1, \dots, \kappa_k$, produces a new IOP Π' for $\mathcal{R}$ with round-by-round knowledge with relaxation $\widetilde{\mathcal{R}}$ with errors

$$2^{-\alpha_1} \cdot \kappa_1, \dots, 2^{-\alpha_k} \cdot \kappa_k.$$

The new IOP Π' works exactly as Π , except that, at each round $i \in [t]$, the verifier samples a random string of α_i bits, called *grinding string*, which sends alongside its message, and rejects unless all bits are 0.

Clearly Π' is not necessarily complete, even if Π is. The goal is to use the BCS compilation of IOP's, and not the IOP's themselves (i.e. the compilation where oracles are replaced by Merkle roots and where interactivensness is removed through the Fiat-Shamir transform) To this end, one chooses $\alpha_1, \dots, \alpha_k$ small enough that, the prover of the BCS transform of Π' can *grind* transcripts until all of the grinding strings in the transcript are the zero string. This results in the prover having to compute per round $i \in [t]$, an average of 2^{α_i} hashes of its i -th message followed by a small string of bits (in practice, one has the prover send a small string of bits next to its message, so that during grinding, it does not need to produce fresh Merkle roots at each grinding iteration). In ??, we provide example grinding times for different values of α_i .

The round-by-round knowledge errors of Π' are obtained by defining a state function which is exactly as the state function of Π , except that the function outputs zero if some of the aforementioned

strings of bits are not the zero string. The extractor for Π' is defined as the extractor for Π , ignoring the additional strings of bits in the transcript.

Table 2: Cost of finding a 64-bit nonce such that $\text{BLAKE3}(\text{msg}, \text{nonce})$ has at least α leading zero bits, with msg of 256 bits. Single and multi-threaded on a MacBook Air M4 16GB. [Albert: As implemented in Flock?](#)

α	18	19	20	21	22	23	24	25
Mean, 1 thread (ms)	12.3	19.5	42.8	82.2	203.8	464.4	808.5	1291.4
Median, 1 thread (ms)	9.2	16.3	27.8	55.2	140.6	358.1	608.5	852.0
Mean, 10 threads (ms)	2.3	3.5	8.9	14.3	35.5	84.6	148.6	240.8
Median, 10 threads (ms)	1.8	2.5	6.8	9.6	26.9	68.0	111.3	161.8

4 BitZ IOP for Lin-BitsDual

4.1 The core Lin-BitsDual relation

In this section we define the main linear relation we work with. We start by fixing notation:

- Let $\mathbb{F}_{2^\nu}$ be a field with 2^ν elements. Let $\beta_1, \dots, \beta_\nu \in \mathbb{F}_{2^\nu}$ be an $\mathbb{F}_2$ -basis of $\mathbb{F}_{2^\nu}$ seen as an $\mathbb{F}_2$ -vector space of dimension ν . Given a vector $\mathbf{u} \in \mathbb{F}_2^\ell$, with ν dividing ℓ , we define

$$\text{Pack}_\nu(\mathbf{u}) \in \mathbb{F}_{2^\nu}^{\ell/\nu}$$

as the vector whose j -th entry is

$$\text{Pack}_\nu(\mathbf{u})_j = \mathbf{u}_{(j-1)\cdot\nu+1} \cdot \beta_1 + \dots + \mathbf{u}_{(j-1)\cdot\nu+\nu-1} \cdot \beta_{\nu-1} + \mathbf{u}_{j\cdot\nu} \cdot \beta_\nu \in \mathbb{F}_{2^\nu}.$$

- Let $\ell \geq 1$ be a dimension parameter divisible by ν , $\mathcal{C}$ a code over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν and blocklength m .
- Let R be a ring (typically R is either $\mathbb{Z}$ or a finite field, not necessarily a prime field) and let $\pi : \mathbb{Z} \rightarrow R$ be a ring homomorphism (not necessarily surjective, and possibly the identity if $R = \mathbb{Z}$).

Definition 4.1. The *Core Lin-BitsDual* relation is defined as

$$\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C}) := \left\{ \left(\begin{array}{l} \mathbf{x} = (\mu, \mathbf{v}) \in R \times R^\ell, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell \end{array} \right) \middle| \begin{array}{l} f = \text{Enc}_\mathcal{C} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f}) \\ \langle \pi(\mathbf{f}), \mathbf{v} \rangle = \mu \end{array} \right\}$$

For $\delta \in (0, 1)$, we let $\mathcal{R}_{\text{Lin-BitsDual}}^\delta$ be the relation $\mathcal{R}_{\text{Lin-BitsDual}}$ after replacing the constraint $f = \text{Enc}_\mathcal{C} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})$ by

$$\Delta(f, \text{Enc}_\mathcal{C} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})) < \delta.$$

4.2 Tensor decomposition and products of constrained codes

We start with some groundwork for our IOP for the Core Lin-BitsDual relation. Let $(\mathbf{x}, \mathbf{y}, \mathbf{w}) = ((\mu, \mathbf{v}), f, \mathbf{f}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. We fix a tensor decomposition of the vector $\mathbf{v} \in R^\ell$ as

$$\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)} = \left(\mathbf{v}_i^{(1)} \cdot \mathbf{v}_j^{(2)} \right)_{i \in [\ell_1], j \in [\ell_2]}$$

for some $\mathbf{v}^{(1)} \in R^{\ell_1}, \mathbf{v}^{(2)} \in R^{\ell_2}$, with $\ell_1 \cdot \ell_2 = \ell$. Note that the trivial tensor decomposition is admitted, i.e. we may effectively not decompose $\mathbf{v}$ by choosing $\ell_1 = \ell, \ell_2 = 1, \mathbf{v}^{(1)} = \mathbf{v}$, and $\mathbf{v}^{(2)} = (1)$.

We index the entries of $\mathbf{f}$ as $\mathbf{f} = (\mathbf{f}_{ij})_{i \in [\ell_1], j \in [\ell_2]}$. Note that

$$\langle \pi(\mathbf{f}), \mathbf{v} \rangle = \sum_{j \in [\ell_2]} \left(\sum_{i \in [\ell_1]} \pi(\mathbf{f}_{ij}) \cdot \mathbf{v}_i^{(1)} \right) \cdot \mathbf{v}_j^{(2)} = \sum_{j \in [\ell_2]} \langle \pi(\mathbf{f}_{*j}), \mathbf{v}^{(1)} \rangle \cdot \mathbf{v}_j^{(2)},$$

where $\mathbf{f}_{*j} = (\mathbf{f}_{ij})_{i \in [\ell_1]}$. Consider the relation (cf. ??) $\mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu)$ for ℓ_2 constraint polynomials Q_{BitZ} so that

$$\begin{aligned} & \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu) \\ &= \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{y}, \boldsymbol{\mu}) \in \mathbb{F}_{2^\nu}^{\ell_1} \times \mathbb{F}_{2^\nu}^{\ell_2}, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in \mathbb{F}_2^{\ell_1 \times \ell_2} \end{array} \right) \mid \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f}) \\ \bigwedge \prod_{i=1}^{\ell_1} ((\mathbf{y}_i - 1) \cdot \mathbf{f}_{ij} + 1) = \boldsymbol{\mu}_j \text{ for all } j \in [\ell_2] \end{array} \right\}. \end{aligned}$$

We omit an explicit description of the polynomials Q_{BitZ} , since the fact that the above relation is a particular instance of the PESAT relation is not relevant in this paper.

We assume prover and verifier have access to an IOR Π_{red} that reduces $\mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu)$ to $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$, where, recall, this latter relation has predicates of the form $\langle \mathbf{g}, \mathbf{u} \rangle = \mu'$ with $\mathbf{g}, \mathbf{u}$ vectors over $\mathbb{F}_{2^\nu}$, and $g = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{g})$. In our instantiation Π_{red} is an optimized GKR protocol for grand products (cf. ??), which reduces the initial grand product claims to MLE evaluation claims of $\mathbf{y}$ and $\mathbf{f}$ over $\mathbb{F}_{2^\nu}$.

We assume Π_{red} works so that the verifier never queries $\mathbf{y}$; and that, given an input

$$(\mathbf{x} = (\mathbf{y}, \boldsymbol{\mu}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu),$$

Π_{red} outputs

$$((\mathbf{v}, \mu'), f, \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$$

for some vector $\mathbf{v}$ and field element μ' . In particular, Π_{red} outputs the same oracle f and witness $\pi_2(\mathbf{f})$ as the oracle and witness it is given as inputs.

This is indeed the case with GKR: GKR reduces the initial circuit evaluation claim with a witness input, into an MLE evaluation claim of the same witness, without querying any oracle in-between.

4.3 The BitZ IOP for the core Lin-BitsDual relation

We provide an IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$ next. We assume the notation from the previous two ??? together with:

- Given a prime p and the morphism $\pi_p : \mathbb{Z} \rightarrow \mathbb{F}_p$ consisting of reduction modulo p , we fix an inverse map of the form $\pi_p^{-1} : \mathbb{F}_p \rightarrow [0 .. p-1] \subseteq \mathbb{Z}$.
- Given a finite field $\mathbb{K}$ which is an extension of a prime field $\mathbb{F}_p$ of degree $\deg(\mathbb{K})$, we let $\pi_{\text{canon}}^{\mathbb{K}} : \mathbb{Z}[X] \rightarrow \mathbb{K}$ be the canonical homomorphism from $\mathbb{Z}[X]$ to $\mathbb{K}$: namely, if $\mathbb{K} = \mathbb{Z}[X]/(p, h(X))$ for a polynomial $h(X)$ irreducible over $\mathbb{F}_p[X]$, then $\pi_{\text{canon}}^{\mathbb{K}}$ consists in modding p out, and taking the quotient by $(h(X))$ (if $\mathbb{K}$ is the prime field $\mathbb{F}_p$, then $\pi_{\text{canon}}^{\mathbb{K}}$ is just π_p). We fix an inverse for $\pi_{\text{canon}}^{\mathbb{K}}$ of the form

$$\pi_{\text{canon}}^{\mathbb{K}}{}^{-1} : \mathbb{K} \rightarrow [0 .. p-1]^{<\deg(\mathbb{K})}[X] \subseteq \mathbb{Z}[X].$$

In words, $\pi_{\text{canon}}^{\mathbb{K}}{}^{-1}$ simply takes an element from $\mathbb{K}$ represented as a degree $< \deg(h(X))$ polynomial over $\mathbb{F}_p$, and applies π_p^{-1} to it. We remark that one could choose any other such “inverse” maps, e.g. for π_p^{-1} one may prefer to use centered representatives.

- Define $\pi_{\text{canon}}^{\mathbb{Z}} : \mathbb{Z} \rightarrow \mathbb{Z}$ as the identity map. We define $\deg(\mathbb{Z}) = 1$.
- Let Π_{iopp} be an IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$. In our instantiation, Π_{iopp} is the WHIR protocol [?] over binary fields, with ring-switching [?] as described in ??, and with proximity parameter set to the Johnson bound.

Construction 4.2 (IOP for the core $\mathcal{R}_{\text{Lin-BitsDual}}$ relation). **Parameters.** $(R, \pi, \mathbb{F}_{2^\nu}, \ell, \ell_1, \ell_2, \mathcal{C})$ comprised of: a ring R and a ring homomorphism $\pi : \mathbb{Z} \rightarrow R$; a binary field $\mathbb{F}_{2^\nu}$; a witness length ℓ divisible by ν ; tensor decomposition dimensions ℓ_1, ℓ_2 satisfying $\ell_1 \cdot \ell_2 = \ell$; a linear code $\mathcal{C}$ over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν and blocklength m .

Let $\mathcal{P}$ be a set of primes with $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$.

Inputs. The prover and verifier receive $\mathbf{x} = (\mu, \mathbf{v}) \in R \times R^\ell$ and $\mathbf{y} = f \in \mathbb{F}_{2^\nu}^m$. The prover receives $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. We assume the tensor split from the previous ??, and adopt its notation.

Let $g \in \mathbb{F}_{2^\nu}$ be a fixed generator of the multiplicative group $\mathbb{F}_{2^\nu}^*$. Let $q \in \mathbb{Z}$ denote the characteristic of R if R has nonzero characteristic, or if R has zero characteristic, then

$$q = \ell_1 \cdot \|\mathbf{v}\|_\infty + 1. \tag{17}$$

1. **Round 1: transforming the constraint over R to an integer constraint with bounded integers.**

- The prover sends $\mu_1, \dots, \mu_{\ell_2} \in [0 .. q-1]^{<\deg(R)}[X]$ such that, over $\mathbb{Z}[X]$,

$$\langle \mathbf{f}_{*j}, \pi_{\text{canon}}^R{}^{-1}(\mathbf{v}^{(1)}) \rangle = \mu_j \quad \text{for all } j \in [\ell_2]. \tag{18}$$

- The verifier checks that

$$\sum_{j \in [\ell_2]} \pi_{\text{canon}}^R(\mu_j) \cdot \mathbf{v}_j^{(2)} = \mu \quad \text{over } R. \tag{19}$$

Next, prover and verifier will focus on certifying that (??) holds for all $j \in [\ell_2]$.

In ?? we describe a variation in which, instead of sending the elements $\mu_1, \dots, \mu_{\ell_2}$, the prover commits to their bit-decomposition over $\mathbb{F}_{2^\nu}$, and proves (??) with ??. This saves on proof size and verifier time when ℓ is large.

- **Optional step: Projecting the integer claim to a prime field, and then lifting it back up to $\mathbb{Z}$.** The following step triggers in the following scenarios:

- $\deg(R) > 1$.
- $\deg(R) = 1$ and $q \geq (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$. [Albert: Review if I need the +1 in the denominator](#)

The verifier uniformly samples and sends a prime q' from $\mathcal{P}$, and, if $\deg(R) > 1$, an element $\alpha \in \mathbb{F}_{q'}$. Now update

$$\begin{aligned} \mathbf{v}^{(1)} &\leftarrow \pi_{q', \alpha} \circ \pi_{\text{canon}}^R{}^{-1} \left(\mathbf{v}^{(1)} \right) \in \mathbb{F}_{q'}^{\ell_1}, \\ \mu_j &\leftarrow \pi_{q'}^{-1} \circ \pi_{q', \alpha}(\mu_j) \in [0 \dots q' - 1] \subseteq \mathbb{Z}, \quad j \in [\ell_2], \\ q &\leftarrow q', \end{aligned}$$

where, recall, $\pi_{q', \alpha} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{q'}$ is defined as coefficient-wise reduction modulo q' and then evaluation of X at α .

Remark 4.3 (Using any module homomorphism instead of $\pi_{q', \alpha}$). Instead of sampling $\pi_{q', \alpha}$, the verifier could sample other $\mathbb{Z}$ -module homomorphisms from $\mathbb{Z}^{<\deg(R)}[X]$ to a prime field. For example it could sample a prime q' and $\deg(R)$ random elements $\alpha_1, \dots, \alpha_{\deg(R)} \in \mathbb{F}_{q'}$, and then prover and verifier use the map $\pi_{q', \alpha_1, \dots, \alpha_{\deg(R)}}(f(X)) = \sum_{i \in [\deg(R)]} \pi_{q'}(f_i) \cdot \alpha_i$, where $f(X) = \sum_{i \in [\deg(R)]} f_i \cdot X^{i-1}$.

For such a class of maps, the error term $(\deg(R) - 1)/q_{\min}$ of ?? becomes $1/q_{\min}$.

2. Phase 2: Transforming the claim into a PESAT claim over $\mathbb{F}_{2^\nu}$, and invoking the IOR Π_{red} .

- Note that, at this point of the protocol, $\mathbf{v}^{(1)} \in \mathbb{F}_q^{\ell_1}$. Prover and verifier define

$$\gamma = \pi_q^{-1}(\mathbf{v}^{(1)}) \in [0 \dots q - 1]^{\ell_1} \subseteq \mathbb{Z}^{\ell_1},$$

and set to prove the following claim:

$$\prod_{i \in [\ell_1]} g^{\gamma_i \cdot \mathbf{f}_{ij}} = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2].$$

This claim is equivalent to

$$\prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2].$$

Notice that

$$(\mathbf{x} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{Bit}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu). \quad (20)$$

- To prove (??), prover and verifier engage in the IOR Π_{red}^4 with input

$$(\mathbf{x} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})).$$

At the end of this IOR, they are left with the following claim to prove:

$$((\mathbf{v}, \mu'), f, \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{u}, \mu) \in \mathbb{F}_{2^\nu}^k, \\ \mathbf{y} = g \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{g} \in (\mathbb{F}_2^\ell) \end{array} \right) \middle| \begin{array}{l} g = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{g}) \\ \langle \mathbf{u}, \mathbf{g} \rangle = \mu \end{array} \right\} \quad (21)$$

for some vector $\mathbf{v}$ and field element μ' output during the IOR Π_{red} .

3. **Phase 3: Using an IOP over $\mathbb{F}_{2^\nu}$.** Prover and verifier engage in an IOP to certify the claims (??).

Remark 4.4 (Reducing proof size by committing to the elements $\mu_1, \dots, \mu_{\ell_2}$). When $\ell = \ell_1 \cdot \ell_2$ is such that ℓ_2 is large, the elements $\mu_1, \dots, \mu_{\ell_2} \in \mathbb{Z}$ sent at Step 1 of ?? may start to dominate the proof size. For example, for $\ell = 2^{13}$ and μ_i of 128 bits, the elements $\mu_1, \dots, \mu_{\ell_2}$ amount to 131 KB. To mitigate this cost, one can have the prover commit to the vector $(\mu_1, \dots, \mu_{\ell_2})$ and then prove (??), i.e.

$$\langle \mathbf{f}_{*j}, \pi_{\text{canon}}^R{}^{-1}(\mathbf{v}^{(1)}) \rangle = \mu_j \quad \text{for all } j \in [\ell_2].$$

using an IOP for constrained codes. Concretely, the prover commits to the vector $\text{bits}(\mu)$ of bits forming the elements $(\mu_1, \dots, \mu_{\ell_2})$ using $\mathbb{B}\text{it}\mathbb{Z}$, and then proves (??) using ?? from ??.

4.4 Soundness and completeness

For readability, we temporarily denote

$$\mathcal{R}_{\text{PESAT}} = \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\mathbb{B}\text{it}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu), \quad \mathcal{R}_{\text{Lin}} = \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu).$$

Theorem 4.5 (Completeness and round-by-round knowledge soundness of ??). *Let $\delta \in (0, 1)$ be a proximity parameter. Assume [Albert](#): can we remove the -1 from $\max \mathcal{P} - 1$?*

$$|\mathbb{F}_{2^\nu}| - 1 > (\ell_1 + 1) \cdot (\max \mathcal{P} - 1).$$

Assume Π_{red} is a perfectly complete t_{red} -round IOR from $\mathcal{R}_{\text{PESAT}}$ to $\mathcal{R}_{\text{Lin}}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{PESAT}}^\delta, \mathcal{R}_{\text{Lin}}^\delta)$, with errors $(\kappa_{\text{red}1}, \dots, \kappa_{\text{red}t_{\text{red}}})$. Assume Π_{iopp} is a perfectly complete t_{iopp} -round IOP for $\mathcal{R}_{\text{Lin}}$ (i.e. an IOR from $\mathcal{R}_{\text{Lin}}$ to the trivial relation $\perp$) with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin}}^\delta, \perp)$, and errors

$$\kappa_{\text{iopp}1}, \dots, \kappa_{\text{iopp}t_{\text{iopp}}}.$$

Then, if $\deg(R) > 1$ or $\deg(R) = 1$ and $q \geq (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$, ?? is a perfectly complete $(1 + t_{\text{red}} + t_{\text{iopp}})$ -round IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^\delta, \perp)$, with errors

$$\kappa_0, (\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]},$$

⁴Recall that, in our instantiation, Π_{red} is the GKR protocol applied to multiple grand products, cf. ??.

where

$$\kappa_0 = \frac{\deg(R) - 1}{q_{\min}} + \frac{\log(\ell_1 + 1) + \log(q - 1)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor}$$

and $q_{\min}$ is the smallest prime in the set of sampleable primes $\mathcal{P}$ from Step 3 of ???. Recall that q is either the characteristic of R if R is a finite field, or $q = \ell_1 \cdot \|\mathbf{v}\|_{\infty} + 1$ if $R = \mathbb{Z}$.

If R is a prime field, then ??? is a $t_{\text{red}} + t_{\text{iopp}}$ -round IOP with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^{\delta}, \perp)$, with errors $(\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]}$.

To prove ??? we begin with the following lemma, which is implicit in [?].

Lemma 4.6. *Let $h(X) \in \mathbb{Z}[X]$ be a nonzero polynomial of degree $\leq d$ and with coefficients in the range $[-2^B + 1 .. 2^B - 1]$. Let $\mathcal{P}$ be a set of primes, and let $q_{\min}$ be the smallest prime in $\mathcal{P}$. Then*

$$\Pr_{\substack{q \leftarrow \mathcal{P}, \\ \alpha \leftarrow \mathbb{F}_q}} [\pi_{q,\alpha}(h(X)) = 0] \leq \frac{B}{\lfloor \log(q_{\min}) \rfloor \cdot |\mathcal{P}|} + \frac{d}{q_{\min}}.$$

Proof. Let $\mathcal{E}_{\text{prime}}$ be the event that $h(X)$ is a multiple of q . Then, by the Schwartz–Zippel lemma,

$$\Pr_{\substack{q \leftarrow \mathcal{P}, \\ \alpha \leftarrow \mathbb{F}_q}} [\pi_{q,\alpha}(h(X)) = 0 \mid \neg \mathcal{E}_{\text{prime}}] \leq \frac{d}{q_{\min}}.$$

Now, every nonzero coefficient of $h(X)$ is divisible by at most $B / \lfloor \log(q_{\min}) \rfloor$ primes from $\mathcal{P}$. Hence, the probability that $\mathcal{E}_{\text{prime}}$ occurs is at most $(B / \lfloor \log(q_{\min}) \rfloor) / |\mathcal{P}|$, and the lemma follows. $\square$

Proof of ???. **Albert: Needs review. Notation may have drifted** The completeness of ??? is clear from its description. To prove its soundness, we decompose ??? into three IORs Π_1, Π_2, Π_3 . The first one, Π_1 , is a 1-round IOR from $\mathcal{R}_{\text{Lin-BitsDual}}$ to $\mathcal{R}_{\text{PESAT}}$, and it consists of applying Round 1 of ???, and outputting

$$(\mathbf{x} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})),$$

as described at the beginning of Phase 1 of ???. The second IOR, Π_2 , is simply Π_{red} , reducing $\mathcal{R}_{\text{PESAT}}$ to $\mathcal{R}_{\text{Lin}}$. Finally, the third IOR, Π_3 , is simply the IOP Π_{iopp} for $\mathcal{R}_{\text{Lin}}$.

By the composition of IORs result (cf. ??), it suffices to show that Π_1 has round-by-round knowledge error κ_0 , and is perfectly complete.

Completeness and soundness of Π_1 We describe a knowledge state function KState . Let $\mathbf{x} = (\mu, \mathbf{v}) \in R \times R^{\ell}$ and $\mathbf{y} = f \in \mathbb{F}_{2^{\nu}}^m$. Let tr be a partial interaction transcript for ???. We start by setting KState so that the first two conditions in ??? are met. Precisely, if $\text{tr} = \emptyset$, we set $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^{\delta}$; and we have KState output zero on inputs $(\mathbf{x}, \mathbf{y}, \text{tr}, m, \mathbf{w})$ where m is a prover message and $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 0$.

After the verifier's first message, tr is as follows:

$$\text{tr} = (\mu_1, \dots, \mu_{\ell_2}, q', \alpha) \in [0 .. q - 1]^{<\deg(R)} [X]^{\ell_2} \times \mathcal{P} \times \mathbb{F}_{q'}.$$

We define $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $\mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell} \subseteq \mathbb{Z}^{\ell}$, $\mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in (\{0, 1\}^{\ell_1})^{\ell_2}$, and

$$\begin{aligned} \langle \mathbf{f}_{*j}, \pi_{q'}^{-1} \circ \pi_{q',\alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle &= \pi_{q'}^{-1} \circ \pi_{q',\alpha}(\mu_j) \quad \text{for all } j \in [\ell_2] \\ \Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\mathbf{f})) &< \delta. \end{aligned} \tag{22}$$

We define $\mathcal{E}_{\text{rbr}}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w}) = \mathbf{w}$. Let $\boldsymbol{\mu} = (\mu_1, \dots, \mu_{\ell_2})$. We next show that

$$\Pr_{q', \alpha} \left[\exists \mathbf{w} \mid \begin{array}{l} \text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, \mathcal{E}_{\text{rbr}}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w}) = 1 \end{array} \right] \leq \kappa_0.$$

Assume

$$\text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w}) = 1 \quad \text{and} \quad \text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, \mathcal{E}_{\text{rbr}}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w})) = \text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, \mathbf{w}) = 0.$$

The latter occurs if and only if $\text{KState}(\mathbb{x}, \mathbb{y}, \mathbf{w}) = 0$ (by Condition 2 in ??), which in turn occurs if and only if $(\mathbb{x}, \mathbb{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$. Since $\text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w}) = 1$, we have $\mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]}$ satisfies (?). Thus we have to bound the probability over q', α that $(\mathbb{x}, \mathbb{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$ and $(\mathbf{f}_{*j})_j$ satisfies (?). To that end, assume $(\mathbb{x}, \mathbb{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$. Then one of the following holds:

$$\langle \pi(\mathbf{f}_{*j}), \mathbf{v}^{(1)} \rangle \neq \mu_j \text{ over } R \text{ for some } j \in [\ell_2] \quad (23)$$

$$\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})) \geq \delta \quad (24)$$

Condition (??) cannot hold, since it would contradict $\text{KState}(\mathbb{x}, \mathbb{y}, \boldsymbol{\mu}, q', \alpha, \mathbf{w}) = 1$. Hence assume (??) holds. Then $\langle \mathbf{f}_{*j}, \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle \neq \mu_j$ over $\mathbb{Z}[X]$. Let

$$p_j(X) := \langle \mathbf{f}_{*j}, \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle - \mu_j \in \mathbb{Z}[X].$$

Then p_j is a nonzero integer polynomial of degree $\leq \deg(R) - 1$ and with coefficients of absolute value at most $\ell_1 \cdot (q - 1) + (q - 1)$ (here we use that all entries in $\mathbf{f}_{*j}$ are bits). Now, since $\mathbf{f}_{*j}$ satisfies (?), applying $\pi_{q'}$ on both sides, reorganizing terms, using that $\pi_{q'}(\mathbf{f}_{*j}) = \pi_{q', \alpha}(\mathbf{f}_{*j})$ and the linearity of $\pi_{q', \alpha}$,

$$0 = \langle \pi_{q'}(\mathbf{f}_{*j}), \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle - \pi_{q', \alpha}(\mu_j) = \pi_{q', \alpha} \left(\langle \mathbf{f}_{*j}, \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle - \mu_j \right) = \pi_{q', \alpha}(p_j(X)).$$

Hence

$$\Pr_{q', \alpha} [(\mathbb{x}, \mathbb{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta, \text{ and } \mathbf{w} = (\mathbf{f}_{*j}) \text{ satisfies } (?)] \leq \Pr_{q', \alpha} [\pi_{q', \alpha}(p_j(X)) = 0]$$

By ??,

$$\Pr_{q', \alpha} [\pi_{q', \alpha}(p_j(X)) = 0] \leq \frac{\log(\ell_1 + 1) + \log(q - 1)}{\lfloor \log(q_{\min}) \rfloor \cdot |\mathcal{P}|} + \frac{\deg(R) - 1}{q_{\min}} = \kappa_0,$$

as needed.

It remains to see that if tr is a full transcript of Π_1 , then $\text{KState}(\mathbb{x}, \mathbb{y}, \text{tr}, \mathbf{w}) = 1$ if and only if the verifier with input $(\mathbb{x}, \mathbb{y}, \text{tr})$ outputs $(\mathbf{x}', \mathbf{y}')$ such that $(\mathbf{x}', \mathbf{y}', \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}^\delta$. The output of the verifier is

$$(\mathbf{x}' = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y}' = f),$$

where

$$\gamma = \pi_{q'}^{-1} \circ \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \in [0..q' - 1]^{\ell_1} \subseteq \mathbb{Z}^{\ell_1}.$$

Now, write as before $\text{tr} = (\boldsymbol{\mu}, q', \alpha)$. By definition, $\text{KState}(\mathbb{x}, \mathbb{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $\mathbf{w} = (\mathbf{f}_{*j})_{j \in [\ell_2]}$ satisfies (?). The equalities in (??) take place in $\mathbb{Z}$. We have that

$$g^{\langle \mathbf{f}_{*j}, \gamma \rangle} = g^{\pi_{q'}^{-1} \circ \pi_{q', \alpha}(\mu_j)}$$

if and only if

$$\langle \mathbf{f}_{*j}, \gamma \rangle = \pi_{q'}^{-1} \circ \pi_{q', \alpha}(\mu_j) \pmod{\text{ord}(g)},$$

where $\text{ord}(g)$ denotes the multiplicative order of g . Now, since $\gamma_i \in [0..q' - 1]$ and $\mathbf{f}_{*j}$ contains bits, we have $|\langle \mathbf{f}_{*j}, \gamma \rangle - \pi_{q'}^{-1} \circ \pi_{q', \alpha}(\mu_j)| \leq (\ell_1 + 1) \cdot (q' - 1)$. Hence if

$$\text{ord}(g) > (\ell_1 + 1) \cdot (q' - 1),$$

then (??) holds if and only if the following hold:

$$\begin{aligned} g^{\langle \mathbf{f}_{*j}, \gamma \rangle} &= g^{\pi_{q'}^{-1} \circ \pi_{q', \alpha}(\mu_j)} \quad \text{for all } j \in [\ell_2], \\ \Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\mathbf{f})) &< \delta. \end{aligned}$$

The above holds if and only if $(\mathbf{x}', \mathbf{y}', (\mathbf{f}_{*j})_{j \in [\ell_2]}) \in \mathcal{R}_{\text{PESAT}}^{\delta}$. □

4.5 The general Lin-BitsDual relation

We now proceed to describe a generalization of $\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$ where the witness $\mathbf{f}$ has entries in an arbitrary module M , instead of in $\{0, 1\} \subseteq \mathbb{Z}$. We assume the notation from the previous sections, together with:

- Let R_M be a ring, possibly different than R , let $\pi : R_M \rightarrow R$ be a ring homomorphism. Let M be a R_M -module and let $\psi : M \rightarrow R$ be a module homomorphism (with underlying ring homomorphism π [Albert: terminology in preliminaries](#)).
- Fix a vector length ℓ_0 . Let $\mathbf{f} \in M_{\leq B}^{\ell_0}$ and write $\mathbf{f} = \sum_{j \in [d]} \mathbf{f}_{*j} \cdot \gamma_j$ with $\mathbf{f}_{*j} \in R_{\leq B}^{\ell_0}$ for all $j \in [d]$. Then write

$$\mathbf{f}_{*j} = \sum_{k=1}^B \mathbf{f}_{*jk} \cdot 2^{k-1}$$

for some vectors

$$\mathbf{f}_{*jk} \in \{0, 1\}^{\ell_0} \subseteq R^{\ell_0},$$

for all j, k . We define $\text{bits}(\mathbf{f}) \in \{0, 1\}^{\ell_0 \cdot d \cdot B} \subseteq \mathbb{Z}^{\ell_0 \cdot d \cdot B}$ as the vector with integer entries obtained by concatenating the vectors $\mathbf{f}_{*jk}$ and, if $R = \mathbb{F}$, lifting their entries to $\{0, 1\} \subseteq \mathbb{Z}$ via π_q^{-1} , i.e.

$$\text{bits}(\mathbf{f}) := (\text{lift}(\mathbf{f}_{*jk}) \mid j \in [d], k \in [B]) \in \{0, 1\}^{\ell_0 \cdot d \cdot B} \subseteq \mathbb{Z}^{\ell_0 \cdot d \cdot B}, \quad (25)$$

where lift is the identity function if $R = \mathbb{Z}$, and $\text{lift} = \pi_q^{-1}$ if $R = \mathbb{F}$.

Remark 4.7 (Non-uniform bit-decomposition sizes). In practical settings, it is possible that not all entries of $\mathbf{f}$ have the same bit size. For example, some chunks of $\mathbf{f}$ may be comprised of vectors, while others may be comprised of generic module elements. However, our definition of $\text{bits}(\mathbf{f})$ allocates $d \cdot B$ bits for every entry of $\mathbf{f}$, no matter its size.

In this case one can define $\text{bits}(\mathbf{f})$ as a vector from $\{0, 1\}^{\ell} \subseteq \mathbb{Z}^{\ell}$ (for a different smaller ℓ) such that

$$L \cdot \pi(\text{bits}(\mathbf{f})) = \mathbf{f}$$

for some public and fixed matrix $L \in M^{\ell_0 \times \ell}$.

Assume $\mathcal{C}$ is a linear code over $\mathbb{F}_{2^\nu}$ of dimension $(\ell_0 \cdot B \cdot d)/\nu$ and blocklength m .

Definition 4.8. The *general Lin-BitsDual* relation is defined as

$$\mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C}) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{v}, \mu) \in R^{\ell_0} \times R, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in M_{\leq B}^{\ell_0} \end{array} \right) \middle| \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\text{bits}(\mathbf{f})), \\ \langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu \end{array} \right\}.$$

When there is no risk of confusion, we denote $\mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C})$ simply by $\mathcal{R}_{\text{Lin-BitsDual}}$. The Core Lin-BitsDual relation is precisely $\mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C})$ after taking $M = \mathbb{Z}$, $B = 1$, and ψ a morphism from $\mathbb{Z}$ to R .

4.6 Bit $\mathbb{Z}$: an IOP for the general Lin-BitsDual relation

Next we describe our IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$. We assume the notation of ???. Namely, M denotes an R_M -module; R is a ring; and $\psi : M \rightarrow R$ is a module homomorphism with underlying ring homomorphism $\pi : R_M \rightarrow R$. Let $\gamma_1, \dots, \gamma_d$ be a R_M -base of M .

We let ℓ_1, ℓ_2 be tensor decomposition parameters (to be used in ???) be such that $\ell_1 \cdot \ell_2 = \ell_0 \cdot d \cdot B$. We assume that $\mathcal{C}$ is a linear code over $\mathbb{F}_{2^\nu}$ of dimension $\ell_0 \cdot d \cdot B/\nu$.

We will need the following result.

Lemma 4.9. Let $\mathbf{f} \in M_{\leq B}^{\ell_0}$, and let $\mathbf{v} \in R^{\ell_0}$. Then

$$\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \langle \pi(\text{bits}(\mathbf{f})), \mathbf{u} \rangle$$

where

$$\mathbf{u} := (\mathbf{v} \otimes (1, \pi(2), \dots, \pi(2^{B-1})) \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d))) \in R^{\ell_0 \cdot d \cdot B} \quad (26)$$

Proof. Rewriting $\langle \psi(\mathbf{f}), \mathbf{v} \rangle$, we obtain

$$\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \left\langle \psi \left(\sum_{\substack{k \in [B], \\ j \in [d]}} \mathbf{f}_{*jk} \cdot 2^{k-1} \cdot \gamma_j \right), \mathbf{v} \right\rangle \quad (27)$$

$$= \left\langle \sum_{\substack{k \in [B], \\ j \in [d]}} \pi(\mathbf{f}_{*jk}) \cdot \pi(2^{k-1}) \cdot \psi(\gamma_j), \mathbf{v} \right\rangle, \quad (28)$$

$$= \sum_{\substack{k \in [B], \\ j \in [d]}} \pi(2^{k-1}) \cdot \psi(\gamma_j) \cdot \langle \pi(\mathbf{f}_{*jk}), \mathbf{v} \rangle$$

$$= \sum_{\substack{k \in [B], \\ j \in [d]}} \sum_{i \in [\ell_0]} \pi(2^{k-1}) \cdot \psi(\gamma_j) \cdot \langle \pi(\mathbf{f}_{ijk}), \mathbf{v}_i \rangle = \langle \pi(\text{bits}(\mathbf{f})), \mathbf{u} \rangle,$$

as needed. □

Construction 4.10 (IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$). **Parameters.**

$$(R_M, R, M, d, \pi, \psi, (\gamma_1, \dots, \gamma_d), \ell_0, B, \ell_1, \ell_2, \mathbb{F}_{2^\nu}, \mathcal{C})$$

comprised of: two rings R_M, R ; a ring morphism $\pi : R_M \rightarrow R$; a R_M -module M of dimension d ; a module homomorphism $\psi : M \rightarrow R$; a R_M -base $\gamma_1, \dots, \gamma_d$ of M_R ; a witness length ℓ_0 ; a bit-size bound B ; tensor decomposition dimensions ℓ_1, ℓ_2 for ?? satisfying $\ell_1 \cdot \ell_2 = \ell_0 \cdot d \cdot B/\nu$; a binary field $\mathbb{F}_{2^\nu}$; a linear code $\mathcal{C}$ over $\mathbb{F}_{2^\nu}$ of dimension $\ell_0 \cdot d \cdot B/\nu$ and blocklength m .

Inputs. The prover and verifier receive $\mathbf{x} = (\mathbf{v}, \mu) \in R^{\ell_0} \times R$ and $y = f \in \mathbb{F}_{2^\nu}^m$. The prover receives $\mathbf{w} = \mathbf{f} \in M_{\leq B}^{\ell_0}$ such that $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C})$, i.e.,

$$f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \quad \langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu.$$

1. By ??, the claim $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C})$ is equivalent to the claim

$$(\mathbf{x}' = (\mathbf{u}, \mu), y' = f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \mathbf{w}' = \text{bits}(\mathbf{f})) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C}). \quad (29)$$

where $\mathbf{u} = (\mathbf{v} \otimes (1, \pi(2), \dots, \pi(2^{B-1})) \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d)))$. Accordingly, prover and verifier use ?? to prove (??). They use parameters $(\mathbb{K}, q, \mathbb{F}_{2^\nu}, \ell_0 \cdot d \cdot B, \ell_1, \ell_2, \mathcal{C})$. See ?? for comments on how to use tensor decompositions of $\mathbf{u}$ in ??.

Remark 4.11 (Tensor decompositions of $\mathbf{u}$). Recall that ?? considers a (possibly trivial) tensor decomposition of its public input vector $\mathbf{u}$ (denoted $\mathbf{v}$ in ??). This decomposition need not be given by some of the tensor factors in ??. For example, in many applications, the input vector $\mathbf{v}$ for $\mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, R, \mathcal{C})$ may admit further tensor decompositions, say $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}$ (e.g. if $\mathbf{v}$ is of the form $(\text{eq}(\mathbf{b}, \mathbf{r}))_{\mathbf{b} \in \{0, 1\}^{\log \ell_0}}$. In that case, we may use a tensor decomposition for $\mathbf{u}$ in ?? of the form, say, $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$, where

$$\mathbf{u}^{(1)} := \mathbf{v}^{(1)}, \quad \mathbf{u}^{(2)} := \mathbf{v}^{(2)} \otimes (1, \pi(2), \dots, \pi(2^{B-1})) \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d)).$$

See ?? for comments on how to choose close-to-optimal tensor split dimensions for $\mathbf{u}^{(1)}$ and $\mathbf{u}^{(2)}$.

Theorem 4.12. *Under the same hypotheses and notation as in ??, if $\deg(R) > 0$, ?? is a perfectly complete $1 + t_{\text{red}} + t_{\text{iopp}}$ IOP for $\mathcal{R}_{\text{Lin-BitsDual}}(M_{\leq B}, \psi, \mathbb{K}, \mathcal{C})$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^{\delta}, \perp)$ and errors*

$$\kappa_0, (\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]},$$

where

$$\kappa_0 = \frac{\deg(R)}{q_{\min}} + \frac{\log(\ell_1 + 1) + \log(q - 1)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor}.$$

If $\deg(R) = 0$, i.e. R is a prime field, then ?? is a $t_{\text{red}} + t_{\text{iopp}}$ -round IOP with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^{\delta}, \perp)$, with errors $\kappa_0, (\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]}$.

Proof. Deferred to ??. In short, the result is a consequence of ??, which establishes an equivalence between triples for $\mathcal{R}_{\text{Lin-BitsDual}}(M_{R \leq B}, \psi, R, \mathcal{C})$ and $\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. ?? further provides a method for efficiently computing one triple from the other. This allows to build a round-by-round knowledge soundness state and an extractor for ?? from those of ??. $\square$

4.7 Instantiation

We instantiate ?? by letting Π_{red} be an optimized GKR protocol for multiple grand product claims of certain form. The IOP Π_{iopp} is taken to be WHIR [?] with the ring switching step from [?] (see ??).

We next provide example instantiations ???? so as to achieve $\lambda = 128$ bits of security, while keeping the protocols as efficient as possible, with the main challenge being to use $\mathbb{F}_{2^\nu}$ of size 2^{128} . Achieving $\lambda = 100$ bits of security is relatively straightforward and we do not discuss it. We use a careful selection of parameters in combination with grinding. In ?? we discuss other instantiations and techniques that do not require grinding.

The contents of this section should be taken as examples of how to instantiate ??, not as limiting strict rules and settings. Other configurations may be perfectly plausible and practical.

In terms of security, we need to ensure the following:

1. $|\mathbb{F}_{2^\nu}| - 1 > (\ell_1 + 1) \cdot (\max \mathcal{P} - 1)$. A requisite in our ???? , to, intuitively, ensure that the inner product in the exponent of $g \in \mathbb{F}_{2^\nu}$ does not overflow.
2. The round-by-round knowledge soundness errors of ?? $\kappa_0^*, (\kappa_{i,\text{red}})_i, (\kappa_{i,\text{iopp}})_i$ are all at most $2^{-\lambda}$.

In terms of efficiency, the most important step of ?? to be aware of is Phase 2, where one uses an IOR Π_{red} to reduce the claim

$$\prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2], \quad (30)$$

into a linear claim on $\pi(\mathbf{f})$. As mentioned, in our instantiation, Π_{red} is the GKR protocol for multiple grand product claims, equipped with the optimizations discuss in ??. This is the most expensive prover part of ???? , constituting between 50% to 66% of prover costs, cf. ??. Because of this, it is important for efficiency that operating in $\mathbb{F}_{2^\nu}$ is as efficient as possible.

With this in mind, we target using $\mathbb{F}_{2^\nu}$ with $\nu \approx 128$. A natural candidate is the 2^{128} sized binary field in GHASH representation

$$\mathbb{F}_{2^\nu} = \mathbb{F}_2[X]/(X^{128} + X^7 + X^2 + X + 1).$$

Before anything else, we discuss the constraint $|\mathbb{F}_{2^\nu}| - 1 > (\ell_1 + 1) \cdot (\max \mathcal{P} - 1)$. First of all, we set ℓ_1, ℓ_2 with $\ell_1 \cdot \ell_2 = \ell$ and

$$\ell_1 \in [\ell^{0.5}, \ell^{0.6}]. \quad (31)$$

In general, choosing $\ell_1 \approx \ell^{0.6}$ can lead to better prover time and proof size for the GKR grand product step, as observed in our experiments. However choosing $\ell_1 \approx \ell^{0.5}$ allows to use slightly smaller soundness errors, which reduces grinding times if targeting 128 bits of security.

We now discuss how to make the errors $\kappa_0^*, (\kappa_{i,\text{red}})_i, (\kappa_{i,\text{iopp}})_i$ be all at most $2^{-\lambda}$.

The errors $\kappa_{i,\text{iopp}}$. Regarding Phase 3, we instantiate our IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \text{Pack}_\nu, \mathcal{C})$ as the WHIR protocol, with the ring switching from [?] (see ??). The ring switching step adds an error term of the form $1/\mathbb{F}_{2^\nu}$. If targetting 128 bits of security, we use grinding [?] of 1 bit in this round. WHIR is configured at any desired $\lambda \leq 128$ bits of security using $|\mathbb{F}_{2^\nu}| = 2^{128}$ following [?, ?]. We do not discuss this further since WHIR is a well-known IOP, and it is well-known that it can achieve 128 bits of security in 128 bit fields when codewords have dimension approximately 2^{30} (assuming an interleaved RS code).

The errors $\kappa_{i,\text{red}}$. As mentioned, our Π_{red} IOR is the GKR protocol for multiple grand product claims from ??, see ??. Each round of the GKR protocol is a round of either a sumcheck for a polynomial of degree 3, or a unification of two evaluation claims of the same multilinear polynomial. Accordingly we have $\varepsilon_{i,\text{red}} \leq 3/|\mathbb{F}_{2^\nu}|$ for all i . This immediately guarantees $\lambda = 100$ bits of security. To achieve $\lambda = 128$ bits of security, one can do two bits of grinding (??) per round.

The error κ_0^* . The error κ_0^* is perhaps the most delicate, as it demands at the same time that $|\mathcal{P}|$ is large, while still satisfying $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$.

Recall that in many scenarios κ_0^* does not exist in the IOP (i.e. sometimes our IOP skips the first verifier message). Concretely, if our initial inner product claim is over $R = \mathbb{F}_q$ a prime field of size $< (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$, then κ_0^* does not appear as an error term, and we do not need to do anything to adjust for it. The term κ_0^* does appear in all other scenarios ($R = \mathbb{Z}$ or R is a prime field of larger size, or R is a proper field extension). We divide our approaches by whether this first step is skipped or not.

For the purposes of this discussion, we assume BitZ is used within a bigger proof system, as described in ?? or ??, so that BitZ is used to prove an inner product claim for a committed witness, as the last phase of a bigger proving system where 1) first the initial constraint is projected to a finite field $\mathbb{F}$, or, alternatively, this step is skipped if the constraint is already over a finite field; and 2) a PIOP is run over $\mathbb{F}$ for the projected constraint, outputting an inner product claim on the projected witness. Now, this inner product claim is the input of ??, and we aim to configure ?? at 128 bits of security for this setting.

Strategy 1: the PIOP achieves 128 bits of security through grinding and we avoid the error term κ_0^* . If allowing, say, 18 – 19 bits of grinding during each round of the PIOP (which adds 1.8 – 3.5ms per round, cf. ??), then, say, a Spartan-like PIOP will achieve 128 bits of security if run on a field $\mathbb{F}$ of size $\approx 2^{112}$ to 2^{113} (since each one of its rounds is either a sumcheck round for a degree ≤ 3 polynomial, or a step that batches some claims with error $1/|\mathbb{F}|$). Hence, if the initial constraint is over $\mathbb{Z}$ or $\mathbb{Z}[X]$, then we may project it to a 112-113 bit random prime field $\mathbb{F}$ and run a PIOP over $\mathbb{F}$. We study this case next. Our analysis also applies if the original constraint was already expressed over a prime field of such size.

Note that the projection of the constraint to $\mathbb{F}$ also requires to be set up to 128 bits of security. We discuss this in ??.

Let $q = |\mathbb{F}|$. Then if

$$q \leq (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1),$$

then κ_0^* is not present as an error term, and there is nothing to adjust. Since $|\mathbb{F}_{2^\nu}| = 2^{128}$, this occurs as long as $\ell_1 \lesssim 2^{15}, 2^{16}$, and using (??), as long as

$$\ell \lesssim 2^{30}, 2^{32} \quad \text{if } \ell_1 \approx \ell^{0.6}$$

and

$$\ell \lesssim 2^{25}, 2^{28} \quad \text{if } \ell_1 \approx \ell^{0.5}$$

with these upper bounds depending on how much grinding one is doing at the PIOP level.

Remark 4.13 (Projecting polynomial constraints to a 112-113 bit prime field with 128 bits of security). **Albert: Need to clean up the UPESAT section first**

Strategy 2: No or essentially no grinding during the PIOP, and κ_0^* is present and grinding is used during the first projection step of ??. We now assume $\mathbb{F}$ is either a prime field with $|\mathbb{F}| > (|\mathbb{F}_{2^\nu}| - 1)/(\ell_1 + 1)$, or a proper extension field. In this case ?? goes through an initial projection step with soundness error κ_0^* . Recall (??)

$$\kappa_0^* = \frac{\deg(\mathbb{F})}{q_{\min}} + \frac{\log(\ell_1 + 1) + \log(q - 1)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor},$$

where q is the characteristic of $\mathbb{F}$. We select $\mathcal{P}$ to be the set of all primes in a range $[2^a, 2^b]$. There are approximately

$$|\mathcal{P}| \approx (2^b/b - 2^a/a)/\log_e(2)$$

such primes. Using $b \leq 128 - \tau \cdot \log(\ell)$, $\ell_1 \leq 2^{\tau \cdot \log(\ell)}$, for $\tau \in [0.5..0.6]$ and plugging these values into κ_0^* we obtain

$$\kappa_0^* \lesssim f(a) := \frac{\deg(R)}{2^a} + \frac{r_1 + \log(q - 1)}{\frac{a}{128 - r_1} \cdot 2^{r_1} - 2^a}.$$

$$r_1 := \tau \cdot \log(\ell)$$

We calculate several minima of $f(a)$ when $\mathbb{F}$ is a field of size $\approx 2^{128}$, for varying values of ℓ and $\tau \in \{0.5, 0.6\}$. We include the corresponding grinding times in order to achieve 128 bits of security for this initial step of ??.

$\log R $	$\deg(R)$	$\log q$	$\log \ell$	τ	a^*	κ_0^*	Bit-deficit to 128	Grinding time (ms)
128	0	128	24	0.5	110	$2^{-109.3}$	19	3.5
128	0	128	24	0.6	107	$2^{-106.9}$	22	35.5
128	0	128	28	0.5	108	$2^{-107.3}$	21	14.3
128	0	128	28	0.6	105	$2^{-104.5}$	24	148.6
128	0	128	30	0.5	107	$2^{-106.3}$	22	35.5
128	0	128	32	0.5	106	$2^{-105.3}$	23	84.6
128	4	32	24	0.5	114	$2^{-110.3}$	18	2.3
128	4	32	24	0.6	111	$2^{-107.8}$	21	14.3
128	4	32	28	0.5	112	$2^{-108.2}$	20	8.9
128	4	32	28	0.6	109	$2^{-105.3}$	23	84.6
128	4	32	30	0.5	111	$2^{-107.2}$	21	14.3
128	4	32	30	0.6	108	$2^{-104.1}$	24	148.6
128	4	32	32	0.5	110	$2^{-106.2}$	22	35.5

Table 3: Minima a^* of $f(a)$ for different configurations of R (of characteristic q and degree $\deg(R) + 1$), witness sizes ℓ , and τ such that $\ell_1 = \ell^\tau$. By choosing primes in the interval $[2^{a^*}, 2^{114}]$, we have that the first error term κ_0^* satisfies $\kappa_0^* = 1/2^{-\log f(a^*)}$. We include the bits left required to achieve $\kappa_0^* < 1/2^{128}$ and the cost of grinding such amount of bits, in a MacBook Air M4 16GB, multi-threaded, using Blake3 as hash function. We omit configurations whose grinding cost exceeds $\approx 150\text{ms}$; in particular $\tau = 0.6$ is omitted for larger ℓ .

Remark 4.14 (128 bits of security for larger witness sizes ℓ , without grinding). Grinding costs scale exponentially in the number of grinding bits. Because of this, we cannot resort to grinding for very large witness sizes ℓ . There are at least two ways to achieve 128 bits of security (or more), in this case.

Method 1: Use a larger binary field $\mathbb{F}_{2^\nu}$, i.e. simply take $\mathbb{F}_{2^\nu}$ with $|\mathbb{F}_{2^\nu}|$ comfortably larger, rather than $|\mathbb{F}_{2^\nu}| = 2^{128}$.

Method 2: We generalize an idea from [?]. In this setting we make the assumption that the entries in γ (cf. Phase 2 of ??) are smaller than $|\mathbb{F}_{2^\nu}| - 1$.

Note that when we attempt to prove the integer equalities $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ ($j \in [\ell_2]$) “in the exponent” during Phase 2 of ??, what we are really doing is proving the equalities modulo $|\mathbb{F}_{2^\nu}| - 1$. So far, we have used $\mathbb{F}_{2^\nu}$ large enough hat this distinction does not matter, in that, because of the total possible magnitudes of $\langle \gamma, \mathbf{f}_{*j} \rangle$ does not exceed $|\mathbb{F}_{2^\nu}| - 2$.

If that is not the case, one can, in addition to proving $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ modulo $|\mathbb{F}_{2^\nu}| - 1$, prove the same equality modulo ℓ_1 . This works because since ℓ_1 and $|\mathbb{F}_{2^\nu}| - 1$ are coprime (because ℓ_1 is a power of two), by the CRT, if both equalities are true, then $\langle \gamma, \mathbf{f}_{*j} \rangle - \mu_j$ is an integer multiple of $\ell_1 \cdot (|\mathbb{F}_{2^\nu}| - 1)$. However, because the entries of $\mathbf{f}$ are bits and the entries of γ are at most $|\mathbb{F}_{2^\nu}| - 2$, this means that $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ as integers.

To prove that $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ modulo ℓ_1 , one can prove the integer equality $\langle \gamma', \mathbf{f}_{*j} \rangle = \mu_j + a_j \cdot \ell_1$ for some $a_j \in \mathbb{Z}$ (provided by the prover), where γ' contains the entries of γ reduced modulo ℓ_1 . As long as $2 \cdot \ell_1 \geq |\mathbb{F}_{2^\nu}| - 1$, this equality can be proved “in the exponent” without causing overflow.

Note that if γ admits a tensor decomposition $\gamma = \gamma^{(1)} \otimes \gamma^{(2)}$, and $3 \cdot \ell_1 < |\mathbb{F}_{2^\nu}| - 1$ then we can take, instead of γ' , the tensor-product $\gamma^{(1)'} \otimes \gamma^{(2)'}$, where $\gamma^{(i)'}$ denotes $\gamma^{(i)}$ reduced modulo ℓ_1 . This allows to use all of our “low-entropy” grand product optimizations from ??.

5 A GKR protocol for low entropy batched grand products via lookup tables

The bulk of the work in $\text{BitZ}(\text{??})$ is to reduce the claim over the binary field $\mathbb{F}_{2^\nu}$

$$\prod_{i \in [\ell_1]} ((g^{\gamma^i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2] \quad (32)$$

to an inner product claim of the form $\langle \mathbf{v}, \pi_2(\mathbf{f}) \rangle = \mu$ over $\mathbb{F}_{2^\nu}$, for some $\mathbf{v} \in \mathbb{F}_{2^\nu}^\ell$, where here $\mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$. Recall that

$$\ell = \ell_1 \cdot \ell_2.$$

To fix intuition, in our instantiation (cf. ??) we set $\ell_1 \approx \ell^{0.6}$, $\ell_2 = \ell^{0.4}$.

Cryptographically speaking, our protocol is simply a standard batched version of the GKR protocol applied to grand products from [?]. However, our implementation uses lookup tables (i.e. preprocessed data) to optimize the prover time of such protocol. In particular, our techniques leverage the fact that the factors in (??) have *low entropy*, in the sense that for every $i \in [\ell_1]$ and $j \in [\ell_2]$, the i -th factor in the j -th grand product of (??) can take only two values, depending on whether $\mathbf{f}_{ij} = 0$ or $\mathbf{f}_{ij} = 1$.

The optimizations we introduce apply to any (batched) grand product with low entropy factors, and not only to the grand product in (??). They also work on any field. Because of this,

We describe a general protocol IOR that, given a finite field $\mathbb{K}$; a subset $S \subseteq \mathbb{K}$; and dimensions $\ell = \ell_1 \cdot \ell_2$, reduces triples from

$$\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{z}, \boldsymbol{\mu}) \in \mathbb{K}^{\ell_1} \times \mathbb{K}^{\ell_2}, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2} \end{array} \right) \left| \prod_{i \in [\ell_1]} p(\mathbf{z}_i, \mathbf{f}_{ij}) = \boldsymbol{\mu}_j \text{ for all } j \in [\ell_2] \right. \right\},$$

to triples in

$$\mathcal{R}_{\text{Lin}}(\mathbb{K}) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{u}, \mu) \in \mathbb{K}^\ell \times \mathbb{K}, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2} \end{array} \right) \left| \langle \mathbf{u}, \mathbf{f} \rangle = \mu \right. \right\}.$$

One can additionally add an encoding $f = \text{Enc}_C \circ T(\mathbf{f})$ as an oracle $\mathbf{y}$ in the above relations (matching the context in which we use the IOR), but our IOR does not make any use of it whatsoever, hence we omit it.

We describe Π_{red} now. We will describe our implementation-level optimizations afterwards. As we mentioned, cryptographically/protocol-wise, our protocol is a standard batched GKR for grand products. We assume ℓ_1, ℓ_2 are powers of two:

$$\ell_1 = 2^{r_1}, \quad \ell_2 = 2^{r_2}.$$

We index $[\ell_1], [\ell_2]$ via elements of the hypercubes $\{0, 1\}^{r_1}, \{0, 1\}^{r_2}$, respectively.

Construction 5.1 (An IOR Π_{red} from $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p)$ to $\mathcal{R}_{\text{Lin}}(\mathbb{K})$). **Input.** Prover and verifier receive $(\mathbf{y}, \boldsymbol{\mu}) \in \mathbb{K}^{\ell_1} \times \mathbb{K}^{\ell_2}$. Prover receives also $\mathbf{f} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2}$.

Let L_{r_1} be a *multilinear* polynomial over $\mathbb{K}$ on $r_1 + r_2$ variables such that

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = p(\mathbf{z}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}.$$

(note that p may not be multilinear). For $i = 0, \dots, r_1 - 1$, let L_i be a multilinear polynomial on $r_2 + i$ variables such that

$$L_i(\mathbf{b}, \mathbf{c}) = L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^i \times \{0, 1\}^{r_2}.$$

1. The verifier sends $\boldsymbol{\alpha}^{(0)} \in \mathbb{K}^{r_2}$. Both prover and verifier compute $C^{(0)} := L_0(\boldsymbol{\alpha}^{(0)})$ (note that L_0 is the multilinear extension of $\boldsymbol{\mu}$ over $\mathbb{K}$).

2. For $i = 0, \dots, r_1 - 1$:

(a) The prover and verifier run the sumcheck protocol for the following claim:

$$C^{(i)} = \sum_{(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^i \times \{0, 1\}^{r_2}} L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)}).$$

At the end of the sumcheck protocol, the prover and verifier are left with a claim of the form

$$L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, 0, \mathbf{r}_{\mathbf{c}}^{(i)}) = a_0^{(i)}, \quad L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, 1, \mathbf{r}_{\mathbf{c}}^{(i)}) = a_1^{(i)} \quad (33)$$

for some $\mathbf{r}_{\mathbf{b}}^{(i)} \in \mathbb{K}^i$, $\mathbf{r}_{\mathbf{c}}^{(i)} \in \mathbb{K}^{r_2}$ and $a_0^{(i)}, a_1^{(i)} \in \mathbb{K}$.

(b) The prover and verifier unify the two claims from (??) into a single one via the following standard procedure:

- i. The verifier sends a random element $\beta^{(i)}$.
- ii. The claim (??) is replaced by

$$L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, \beta^{(i)}, \mathbf{r}_{\mathbf{c}}^{(i)}) = C^{(i+1)} := a_0^{(i)} \cdot (1 - \beta^{(i)}) + a_1^{(i)} \cdot \beta^{(i)}.$$

Let

$$\boldsymbol{\alpha}^{(i+1)} := (\mathbf{r}_{\mathbf{b}}^{(i)}, \beta^{(i)}, \mathbf{r}_{\mathbf{c}}^{(i)}).$$

3. At this stage, the prover and verifier are left with the claim

$$L_{r_1}(\boldsymbol{\alpha}^{(r_1)}) = C^{(r_1)}.$$

This claim is equivalent to

$$\sum_{(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}} p(\mathbf{z}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) \cdot \text{eq}((\mathbf{b}, \mathbf{c}), \boldsymbol{\alpha}^{(r_1)}) = C^{(r_1)}, \quad (34)$$

If p is multilinear, then (??) is a linear claim on $\mathbf{f}$. Namely, it is equivalent to

$$\langle \mathbf{u}, \mathbf{f} \rangle = C^{(r_1)} + \mu$$

for some $\mathbf{u} \in \mathbb{K}^\ell, \mu \in \mathbb{K}$. In this case, prover and verifier output

$$(\mathbf{x}' = (\mathbf{u}, C^{(r_1)} + \mu), \perp, \mathbf{w}' = \mathbf{f}) \in \mathcal{R}_{\text{Lin}}(\mathbb{K}).$$

Otherwise, i.e. if p is not multilinear, prover and verifier run a sumcheck for the claim (??), which results in MLE claims for $\mathbf{y}$ and $\mathbf{f}$. The verifier checks the claim for $\mathbf{y}$ itself, and prover and verifier output the MLE claim for $\mathbf{f}$.

5.1 Optimization via lookup tables

We optimize ?? as follows. For simplicity we assume

$$S = \{0, 1\} \subseteq \mathbb{K},$$

but any small subset of $\mathbb{K}$ is amenable to the techniques discussed here.

Setup We start with the following observation: for every $(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}$, we have $\mathbf{f}_{\mathbf{b}, \mathbf{c}} \in S = \{0, 1\}$, and so

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = p(\mathbf{z}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) = \begin{cases} y_{\mathbf{b}}^{(1)} := p(\mathbf{z}_{\mathbf{b}}, 1) & \text{if } \mathbf{f}_{\mathbf{b}, \mathbf{c}} = 1 \\ y_{\mathbf{b}}^{(0)} := p(\mathbf{z}_{\mathbf{b}}, 0) & \text{if } \mathbf{f}_{\mathbf{b}, \mathbf{c}} = 0. \end{cases}$$

The values $y_{\mathbf{b}}^{(0)}, y_{\mathbf{b}}^{(1)}$ do not depend on $\mathbf{c}$. Hence the prover can set up the lookup table $T_{r_1} \subseteq (\mathbb{K} \times \mathbb{K})^{\{0, 1\}^{r_1}}$ defined as

$$T_{r_1}[\mathbf{b}] = (y_{\mathbf{b}}^{(1)}, y_{\mathbf{b}}^{(0)}) \quad \text{for all } \mathbf{b} \in \{0, 1\}^{r_1}$$

at the beginning of the protocol, and use it to obtain the values $L_{r_1}(\mathbf{b}, \mathbf{c})$. Concretely

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = T_{r_1}[\mathbf{b}][\mathbf{f}_{\mathbf{b}, \mathbf{c}}] \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2} -$$

In our instantiation, $y_{\mathbf{b}}^{(1)} = g^{\gamma_{\mathbf{b}}}$ and $y_{\mathbf{b}}^{(0)} = 1$ (cf. ??), and so the prover precomputes the elements $g^{\gamma_{\mathbf{b}}}$ at the beginning of the protocol.

As we will explain later, the prover will also precompute similar tables for the values $L_{r_1-j}(\mathbf{b}, \mathbf{c})$ of the levels close to the leaves,

$$T_{r_1-j} \in \left(\mathbb{K}^{2^{2^j}} \right)^{\{0, 1\}^{r_1-j}}, \quad (35)$$

as well as other tables relevant for the different sumcheck rounds of ?. Concretely, for $\mathbf{b} \in \{0, 1\}^{r_1-j}$, $T_{r_1-j}[\mathbf{b}]$ consists of all possible products of 2^j consecutive factors of the grand product, namely the factors indexed by $(\mathbf{b}, \mathbf{b}')$ with $\mathbf{b}' \in \{0, 1\}^j$. Each factor can take two values, depending on whether $\mathbf{f}_{\mathbf{b}, \mathbf{b}', \mathbf{c}}$ is 1 or 0. Hence $T_{r_1-j}[\mathbf{b}]$ consists of 2^{2^j} elements. Hence the size of T_{r_1-j} in (??).

Because $|T_{r_1-j}|$ grows doubly-exponentially in j , these tables are viable only for the bottom few levels of the tree ($j = 0, 1, 2$ in our implementation). In higher levels we use standard sumchecks.

Importantly, all these tables have size independent of ℓ_2 . This (among others, cf. ??), is one of the reasons why we use the tensor decomposition structure, i.e. the split given by $\ell = \ell_1 \cdot \ell_2$, in ??

Remark 5.2. All precomputed data in this section depends on the input vector $\mathbf{z}$ of $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p)$, and possibly on further verifier challenges. Since, in a wider context where ?? is a subprocess of a larger proof system, $\mathbf{y}$ typically depends on verifier's challenges, these precomputations must be done at *runtime*.

First prover message in the last sumcheck in Step 2 of ?? We start by analyzing Step 2 with $i = r_1 - 1$. In this step prover and verifier run a sumcheck for the claim

$$C^{(r_1-1)} = \sum_{(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-1} \times \{0,1\}^{r_2}} L_{r_1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)})$$

This is the most expensive step of ?? for the prover (if done naively), since it is the sumcheck over the largest number of variables throughout ??.

For $1 \leq j \leq r_1 - 1$, the j -th prover message in this sumcheck is, if $\boldsymbol{\rho}^{(j)} = (\rho_1, \dots, \rho_{j-1})$ is the vector of verifier challenges so far in this sumcheck,

$$\begin{aligned} h_j(X) &:= \sum_{(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-1-j} \times \{0,1\}^{r_2}} L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)}) \\ &= \sum_{\mathbf{c} \in \{0,1\}^{r_2}} \left(\sum_{\mathbf{b} \in \{0,1\}^{r_1-1-j}} L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\boldsymbol{\rho}^{(j)}, \mathbf{b}; \boldsymbol{\alpha}^{(i,1)}) \right) \cdot \text{eq}(X, \mathbf{c}; \boldsymbol{\alpha}^{(i,2)}) \\ &= \sum_{\mathbf{c} \in \{0,1\}^{r_2}} h_{j,\mathbf{c}}(X) \cdot \text{eq}(X, \mathbf{c}; \boldsymbol{\alpha}^{(i,2)}), \end{aligned}$$

for $\boldsymbol{\alpha}^{(i)} = (\boldsymbol{\alpha}^{(i,1)}, \boldsymbol{\alpha}^{(i,2)}) \in \{0,1\}^{r_1-j-1} \times \{0,1\}^{r_2+1}$. We claim that

$$L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) = \left(L_{r_1}(\boldsymbol{\rho}^{(j)}, 1, \mathbf{b}, 0, \mathbf{c}) - L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 0, \mathbf{c}) \right) \cdot X + L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 0, \mathbf{c}) \quad (36)$$

Indeed, $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c})$ is a linear polynomial, and the right-hand side in the above expression is the line passing through the two evaluations of $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c})$ at $X = 0$ and $X = 1$. An analogous expression holds for $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c})$:

$$L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) = \left(L_{r_1}(\boldsymbol{\rho}^{(j)}, 1, \mathbf{b}, 1, \mathbf{c}) - L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 1, \mathbf{c}) \right) \cdot X + L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 1, \mathbf{c}). \quad (37)$$

Renaming

$$E_{\varepsilon_1 \varepsilon_2}^{(j)}[\mathbf{b}, \mathbf{c}] := L_{r_1}(\boldsymbol{\rho}^{(j)}, \varepsilon_1, \mathbf{b}, \varepsilon_2, \mathbf{c}) \quad \varepsilon_1, \varepsilon_2 \in \{0,1\}, \quad \omega[\mathbf{b}] := \text{eq}(\boldsymbol{\rho}^{(j)}, \mathbf{b}; \boldsymbol{\alpha}^{(i,1)}),$$

we obtain

$$h_{j,\mathbf{c}}(X) = \sum_{\mathbf{b} \in \{0,1\}^{r_1-1-j}} X^2 \cdot A_2^{(j)}[\mathbf{b}, \mathbf{c}] + X \cdot A_1^{(j)}[\mathbf{b}, \mathbf{c}] + A_0^{(j)}[\mathbf{b}, \mathbf{c}]$$

where,

$$\begin{aligned} A_0^{(j)}[\mathbf{b}, \mathbf{c}] &:= E_{00}^{(j)}[\mathbf{b}, \mathbf{c}] \cdot E_{01}^{(j)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}], \\ A_1^{(j)}[\mathbf{b}, \mathbf{c}] &:= \left((E_{10}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{00}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot E_{01}^{(j)}[\mathbf{b}, \mathbf{c}] + (E_{11}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{01}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot E_{00}^{(j)}[\mathbf{b}, \mathbf{c}] \right) \cdot \omega[\mathbf{b}] \\ A_2^{(j)}[\mathbf{b}, \mathbf{c}] &:= (E_{10}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{00}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot (E_{11}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{01}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot \omega[\mathbf{b}]. \end{aligned}$$

Now, for $j = 1$, we have $\boldsymbol{\rho}^{(j)} = \perp$, and hence each value $E_{\varepsilon_1 \varepsilon_2}^{(1)}[\mathbf{b}, \mathbf{c}]$ can be obtained by querying $T_{r_1}[\mathbf{b}]$ and selecting $T_{r_1}[\mathbf{b}][0]$ or $T_{r_1}[\mathbf{b}][1]$ depending on whether $\mathbf{f}_{\mathbf{b}, \mathbf{c}} = 1$ or 0. A consequence is that

the coefficients themselves $A_k^{(1)}[\mathbf{b}, \mathbf{c}]$ ($k \in [3]$) can be obtained through lookup tables, as follows: we have

$$A_0^{(1)}[\mathbf{b}, \mathbf{c}] = T_{r_1}[0, \mathbf{b}, 0][\mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}}] \cdot T_{r_1}[0, \mathbf{b}, 1][\mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}}] \cdot \omega[\mathbf{b}]$$

Hence

$$A_0^{(1)}[\mathbf{b}, \mathbf{c}] = \begin{cases} \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 1, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 1 \\ \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 1, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 0 \\ \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 0, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 1 \\ \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 0, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 0 \end{cases}$$

Thus the prover can precompute the table

$$T_{A_0} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-2}} = \mathbb{K}^{\ell_1}$$

given by

$$T_{A_0}[\mathbf{b}] = (\mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}])$$

and use it to obtain any value $A_0^{(1)}[\mathbf{b}, \mathbf{c}]$ for $(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-2} \times \{0, 1\}^{r_2}$ via a single read into T_{A_0} . The size of T_{A_0} , again, does not depend on $\ell_2 = 2^{r_2}$. Building T_{A_0} requires at most $8 \cdot 2^{r_1-2} = 2^{r_1+1}$ multiplications and no additions.

We now explain how to obtain the values $A_1^{(1)}[\mathbf{b}, \mathbf{c}]$ and $A_2^{(1)}[\mathbf{b}, \mathbf{c}]$, again via the construction of a lookup table. Note that these depend on the *four* bits $\mathbf{f}_{(\varepsilon_1, \mathbf{b}, \varepsilon_2), \mathbf{c}}$, $\varepsilon_1, \varepsilon_2 \in \{0, 1\}$, so our tables will need $2^4 = 16$ entries (as opposed to 4 as in T_{A_0}) per index $\mathbf{b}$. The prover proceeds as follows:

- It precomputes a table $T_{S_1} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-2}}$ analogous to T_{A_0} , which allows to obtain the values

$$S_1^{(1)}[\mathbf{b}, \mathbf{c}] := E_{10}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot E_{11}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}],$$

(observe that T_{A_0} allows to obtain the values $E_{00}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot E_{01}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}]$).

- It precomputes the table $T_{A_2} \in (\mathbb{K}^{16})^{\{0,1\}^{r_1-2}}$ given by

$$T_{A_2}[\mathbf{b}] = \left((\mathbf{y}_{1,\mathbf{b},0}^{(x_1)} - \mathbf{y}_{0,\mathbf{b},0}^{(x_0)}) \cdot (\mathbf{y}_{1,\mathbf{b},1}^{(x'_1)} - \mathbf{y}_{0,\mathbf{b},1}^{(x'_0)}) \cdot \omega[\mathbf{b}] \right)_{(x_0, x_1, x'_0, x'_1) \in \{0,1\}^4}, \quad (38)$$

so that $A_2^{(1)}[\mathbf{b}, \mathbf{c}]$ is the element

$$A_2^{(1)}[\mathbf{b}, \mathbf{c}] = T_{A_2}[\mathbf{b}][\mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(1,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}}, \mathbf{f}_{(1,\mathbf{b},1),\mathbf{c}}].$$

Building T_{A_2} takes at most 20 multiplications and 8 additions per index $\mathbf{b}$ ⁵. I.e. at most $5 \cdot 2^{r_1}$ multiplications and 2^{r_1+1} additions in total.

- It obtains the values $A_1^{(1)}[\mathbf{b}, \mathbf{c}]$ via the identity

$$S_1^{(1)}[\mathbf{b}, \mathbf{c}] = A_0^{(1)}[\mathbf{b}, \mathbf{c}] + A_1^{(1)}[\mathbf{b}, \mathbf{c}] + A_2^{(1)}[\mathbf{b}, \mathbf{c}] \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-2} \times \{0, 1\}^{r_2}.$$

⁵For the multiplications, there are 4 multiplications to compute the 4 elements $t_{x'_0, x'_1, \mathbf{b}} := (\mathbf{y}_{1,\mathbf{b},1}^{(x'_1)} - \mathbf{y}_{0,\mathbf{b},1}^{(x'_0)}) \cdot \omega[\mathbf{b}]$, and then 16 more multiplications to compute $(\mathbf{y}_{1,\mathbf{b},0}^{(x_1)} - \mathbf{y}_{0,\mathbf{b},0}^{(x_0)}) \cdot t_{x'_0, x'_1, \mathbf{b}}$

Once the tables have been computed, obtaining $h_{1,\mathbf{c}}(X)$ for one $\mathbf{c} \in \{0, 1\}^{r_2}$ costs $3 \cdot 2^{r_1-2}$ table reads (one read per table $T_{A_0}, T_{S_1}, T_{A_2}$ per index $\mathbf{b} \in \{0, 1\}^{r_1-2}$) and $3 \cdot 2^{r_1-2} + 2$ additions: this is the cost of adding together the values read at each table, providing $\sum_{\mathbf{b}} A_0[\mathbf{b}, \mathbf{c}]$, $\sum_{\mathbf{b}} S_1[\mathbf{b}, \mathbf{c}]$, $\sum_{\mathbf{b}} A_2[\mathbf{b}, \mathbf{c}]$, and then doing two additions to obtain $\sum_{\mathbf{b}} A_1[\mathbf{b}, \mathbf{c}] = \sum_{\mathbf{b}} S_1[\mathbf{b}, \mathbf{c}] - \sum_{\mathbf{b}} A_0[\mathbf{b}, \mathbf{c}] - \sum_{\mathbf{b}} A_2[\mathbf{b}, \mathbf{c}]$. In total, this amounts to at most $3\ell/4$ table reads and $\approx 3\ell/4 + 2 \cdot \ell_2$ additions to compute $h_{1,\mathbf{c}}$ for all $\mathbf{c}$, and no multiplications (in our instantiation, additions in $\mathbb{K}$ are 128-bit XORs).

Finally, the prover computes $h_1(X)$ from the polynomials $h_{1,\mathbf{c}}(X)$ with at most $5 \cdot 2^{r_2}$ additions and $3 \cdot 2^{r_2}$ multiplications (plus 2^{r_2} multiplications to precompute the values $\text{eq}(\mathbf{c}; \boldsymbol{\alpha}^{(i,2)})$, shared by all rounds of this sumcheck, and 2^{r_1-2} multiplications for the weights $\omega[\mathbf{b}]$).

In ?? we record the total upper bounds for computing $h_1(X)$. In our application we obtain lower numbers since $\mathbf{y}_{\mathbf{b}}^{(0)} = 1$ and we work in characteristic two, hence constant terms cancel inside the differences from ??. Our implementation, in our use-case, builds all three tables with 8 multiplications and roughly 20 additions per index $\mathbf{b} \in \{0, 1\}^{r_1-2}$, i.e. $\approx 2\ell_1$ multiplications in total.

On the other hand, a standard generic approach to compute $h_1(X)$ would require $O(\ell) = O(\ell_1 \cdot \ell_2)$ multiplications and additions, even if using the few sumcheck optimizations that apply in our settings (to our knowledge) **Albert: which ones, exactly?**. Crucially, we note that our sumcheck features summands of full field bit size $\log |\mathbb{K}|$, which voids some known powerful optimizations when the summands are small, e.g. the univariate skip [?] and some optimizations from [?] **Albert: This is a crucial point. It should be highlighted more. And it should be an important point of the introduction and technical overview**

	Our approach	Standard/generic
Multiplications	$\leq 10 \cdot \ell_1 + 4 \cdot \ell_2$	$\Theta(\ell_1 \cdot \ell_2)$
Additions	$\leq 3 \cdot \ell_1 \cdot \ell_2 / 4 + 2 \cdot \ell_1 + 5 \cdot \ell_2$	$\Theta(\ell_1 \cdot \ell_2)$
Table reads	$3 \cdot \ell_1 \cdot \ell_2 / 4$, into tables of $6 \cdot \ell_1$ field elements	–

Table 4: The cost in field operations of the first sumcheck round of the last sumcheck in ?? (Step 2 with $i = r_1 - 1$), where $\ell = \ell_1 \cdot \ell_2 = 2^{r_1} \cdot 2^{r_2}$ is the witness size. In our instantiation $\ell_1 \approx \ell^{0.6} = 2^{r_1}$ and $\ell_2 \approx \ell^{0.4} = 2^{r_2}$ (??).

Subsequent prover messages in the last sumcheck in Step 2 of ?? Next we focus on computing $h_2(X)$. Substituting X by the verifier's first sumcheck challenge ρ_1 in ????, we have, for every $\mathbf{b} \in \{0, 1\}^{r_1-1}$, $\mathbf{c} \in \{0, 1\}^{r_2}$,

$$\begin{aligned}
L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c}) &= (L_{r_1}(1, \mathbf{b}, \mathbf{c}) - L_{r_1}(0, \mathbf{b}, \mathbf{c})) \cdot \rho_1 + L_{r_1}(0, \mathbf{b}, \mathbf{c}) \\
&= \begin{cases} \mathbf{y}_{\mathbf{b}}^{(11)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(1)} - \mathbf{y}_{0,\mathbf{b}}^{(1)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(1)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 1, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 1 \\ \mathbf{y}_{\mathbf{b}}^{(10)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(0)} - \mathbf{y}_{0,\mathbf{b}}^{(1)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(1)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 1, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 0 \\ \mathbf{y}_{\mathbf{b}}^{(01)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(1)} - \mathbf{y}_{0,\mathbf{b}}^{(0)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(0)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 0, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 1 \\ \mathbf{y}_{\mathbf{b}}^{(00)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(0)} - \mathbf{y}_{0,\mathbf{b}}^{(0)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(0)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 0, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 0 \end{cases}
\end{aligned}$$

The idea now is that the prover can compute $h_2(X)$ in essentially the same way as $h_1(X)$, but this time, instead of starting with the table

$$T_{r_1} \in (\mathbb{K}^2)^{\{0,1\}^{r_1}} \quad \text{such that } T_{r_1}[\mathbf{b}] = (\mathbf{y}_{\mathbf{b}}^{(0)}, \mathbf{y}_{\mathbf{b}}^{(1)}) \text{ and } L_{r_1}(\mathbf{b}, \mathbf{c}) = T_{r_1}[\mathbf{b}][\mathbf{f}_{\mathbf{b},\mathbf{c}}], \quad (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2},$$

it starts with the table

$$T_{r_1}^{(2)} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-1}} \quad \text{such that } T_{r_1}^{(2)}[\mathbf{b}] = (\mathbf{y}_{\mathbf{b}}^{(11)}, \mathbf{y}_{\mathbf{b}}^{(10)}, \mathbf{y}_{\mathbf{b}}^{(01)}, \mathbf{y}_{\mathbf{b}}^{(00)}) \quad \text{for } \mathbf{b} \in \{0,1\}^{r_1-1}$$

for which we have, indexing the entries of $T_{r_1}^{(2)}[\mathbf{b}]$ with the 4 values $(1,1), (1,0), (0,1), (0,0)$,

$$L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c}) = T_{r_1}^{(2)}[\mathbf{b}][(\mathbf{f}_{0,\mathbf{b}}, \mathbf{f}_{1,\mathbf{b}})]$$

This allows the prover to obtain the values $L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c})$ for all $(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-1} \times \{0,1\}^{r_2}$ with a simple read in $T_{r_1}^{(2)}$.

We now have, for $(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-3} \times \{0,1\}^{r_2}$,

$$\begin{aligned} A_0^{(2)}[\mathbf{b}, \mathbf{c}] &= L_{r_1}(\rho_1, 0, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\rho_1, 0, \mathbf{b}, 1, \mathbf{c}) \cdot \omega^{(2)}[\mathbf{b}] \\ &= T_{r_1}^{(2)}[0, \mathbf{b}, 0][(\mathbf{f}_{(0,0,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(1,0,\mathbf{b},0),\mathbf{c}})] \cdot T_{r_1}^{(2)}[0, \mathbf{b}, 1][(\mathbf{f}_{(0,0,\mathbf{b},1),\mathbf{c}}, \mathbf{f}_{(1,0,\mathbf{b},1),\mathbf{c}})] \cdot \omega[\mathbf{b}] \end{aligned}$$

for some weights $\omega^{(2)}[\mathbf{b}]$. Hence the elements $A_0^{(2)}[\mathbf{b}, \mathbf{c}]$ can be obtained by querying a table

$$T_{A_0}^{(2)} \in (\mathbb{K}^{16})^{\{0,1\}^{r_1-3}}$$

with

$$T_{A_0}^{(2)}[\mathbf{b}] = (T_{r_1}^{(2)}[0, \mathbf{b}, 0][\mathbf{x}] \cdot T_{r_1}^{(2)}[0, \mathbf{b}, 1][\mathbf{x}'] \cdot \omega[\mathbf{b}])_{\mathbf{x}, \mathbf{x}' \in \{0,1\}^2}$$

The table $T_{S_1}^{(2)}$ is analogous. The coefficient $A_2^{(2)}$, however, now couples *eight* bits — each difference $E_{1\varepsilon}^{(2)} - E_{0\varepsilon}^{(2)}$ is selected by four bits — so tabulating it directly would require a $2^8 = 256$ -case table. Instead of precombining it, our implementation *factors* the term: it gathers the two differences from two 16-case tables and joins them with a single multiplication per slot.

This can be iterated in further sumcheck rounds, for computing $h_3(X), h_4(X)$, etc. The downside is a doubly-exponential growth of the table widths: each further round *squares* the number of cases per index of every table, but it only *halves* the number of indices. Hence, in practice one stops using our optimization after a few rounds of sumcheck, and falls back to continue the sumcheck protocol normally. Concretely, in our implementation we never compute a table with more than 16 elements per index (hence, for example, the table $T_{A_2}^{(2)}$ is not built in our implementation). This is done so that the tables can be kept in cache. ?? displays the different tables built or not built in our implementation, the cost of doing so, and their size at $\ell_1 = 2^{r_1} = 2^{16}$ (corresponding to a total witness size of 2^{27} in our implementation).

Table	Dimension	Adds/mults. in $\mathbb{K}$	Size (MB), $\ell_1 = 2^{16}$	Built?
$T_{r_1}^{(1)}$	$(\mathbb{K}^2)^{2^{r_1}}$	0/0	2.1	yes
$T_{A_0}^{(1)}$	$(\mathbb{K}^{2^2})^{2^{r_1-2}}$	0 / 2^{r_1+1}	1.0	yes
$T_{S_1}^{(1)}$	$(\mathbb{K}^{2^2})^{2^{r_1-2}}$	0 / 2^{r_1+1}	1.0	yes
$T_{A_2}^{(1)}$	$(\mathbb{K}^{2^4})^{2^{r_1-2}}$	2^{r_1+1} / $5 \cdot 2^{r_1}$	4.2	yes
$T_{r_1}^{(2)}$	$(\mathbb{K}^{2^2})^{2^{r_1-1}}$	2^{r_1+2} / 2^{r_1+1}	2.1	yes
$T_{A_0}^{(2)}$	$(\mathbb{K}^{2^4})^{2^{r_1-3}}$	0 / 2^{r_1+2}	2.1	yes
$T_{S_1}^{(2)}$	$(\mathbb{K}^{2^4})^{2^{r_1-3}}$	0 / 2^{r_1+2}	2.1	yes
$T_{A_2}^{(2)}$	$(\mathbb{K}^{2^8})^{2^{r_1-3}}$	—	33.6	no
$T_{r_1}^{(3)}$	$(\mathbb{K}^{2^4})^{2^{r_1-2}}$	2^{r_1+3} / 2^{r_1+2}	4.2	yes
$T_{A_0}^{(3)}$	$(\mathbb{K}^{2^8})^{2^{r_1-4}}$	—	16.8	no
$T_{S_1}^{(3)}$	$(\mathbb{K}^{2^8})^{2^{r_1-4}}$	—	16.8	no
$T_{A_2}^{(3)}$	$(\mathbb{K}^{2^{16}})^{2^{r_1-4}}$	—	4295	no

Table 5: Dimensions, build costs, and sizes of the tables for computing $h_1(X), h_2(X), h_3(X)$, with $\log |\mathbb{K}| \approx 128$. We use the example size $\ell_1 = 2^{r_1} = 2^{16}$, which following our instantiation ($\ell_1 \approx \ell^{0.6}$, ??) this corresponds to a total witness size $\ell \approx 2^{27}$. **Albert:** It seems we do not always build $T_{A_0}^{(2)}$ and $T_{S_1}^{(2)}$?

Instead of building the table $T_{A_2}^{(2)}$ to obtain the values $A_2^{(2)}[\mathbf{b}, \mathbf{c}]$, the prover reads $T_{r_1}^{(2)}$ at appropriate values and computes $A_2^{(2)}[\mathbf{b}, \mathbf{c}]$ using some of the elements in the vectors read this way. In Round 3 we only use the table $T_{r_1}^{(3)}$, and otherwise perform sumcheck normally.

The rest of sumchecks in Step 2 of ?? So far we have discussed the last sumcheck from Step 2 of ??, corresponding to the lower layer of the binary tree of products of our GKR circuit for the grand products.

The sumchecks run for $i = r_1 - 2, r_1 - 3, \dots$ can be optimized in the exact same way as we did for the last sumcheck ($i = r_1 - 1$). In this case, instead of starting with the table $T_{r_1}^{(1)}$, which at each index contains the two values $(\mathbf{y}_{\mathbf{b}}^{(1)}, \mathbf{y}_{\mathbf{b}}^{(0)})$, one starts with the table

$$T_{r_1-j} \in (\mathbb{K}^{2^{2^j}})^{\{0,1\}^{r_1-j}}$$

described in ??. Since T_{r_1-j} has quadratically many more entries per index than T_{r_1-j+1} , all subsequent tables also have quadratically many more entries per index compared to the analogous tables from the $(i+1)$ -th sumcheck. And, further, they require quadratically more operations to comput.

In our implementation we want to keep all tables in cache. Because of this, the smaller i is (i.e. the further we get from the last sumcheck in the GKR protocol), the less number of sumcheck rounds we use tables in. Concretely, for $i = r_1 - 2$ ($j = 1$) we build and use tables $T_{r_1-1}^{(1)}, T_{A_0}^{(1)}, T_{S_1}^{(1)}, T_{r_1-1}^{(2)}$ (note the lack of $T_{A_2}^{(1)}$). For $i = r_1 - 3$ ($j = 3$) we only use the table $T_{r_1-2}^{(1)}$.

5.2 Other optimizations for ??

Using 4 factor multiplication (instead of 2) in the last step of the GKR grand product

Albert: Actually autoresearch ended up undoing this and went back to all 2 factors

In Step 2 we reduce the number of sumchecks by 1 (so now there are $r_1 - 2$ sumchecks) by making the last sumcheck certify the claim

$$C^{(r_1-2)} = \sum_{(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-2} \times \{0,1\}^{r_2}} L_{r_1}(\mathbf{b}, 0, 0, \mathbf{c}) \cdot L_{r_1}(\mathbf{b}, 0, 1, \mathbf{c}) \cdot L_{r_1}(\mathbf{b}, 1, 0, \mathbf{c}) \cdot L_{r_1}(\mathbf{b}, 1, 1, \mathbf{c}) \cdot \text{eq}(\mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)})$$

I.e.

Step 1 of ?? via lookup tables In Step 1 of ??, the prover computes the elements $\mu_1, \dots, \mu_{\ell_2}$ again by means of a lookup table T , which is indexed over $i \in [\ell_1/4]$ and contains the 16-length vectors

$$T[i] = \left(\sum_{k \in [4]} \varepsilon_k \cdot \pi_{\text{canon}}^R \text{ }^{-1}(\mathbf{v}^{(1)})_{4*(i-1)+k} \right)_{\varepsilon_1, \dots, \varepsilon_4 \in \{0,1\}} \in \mathbb{Z}[X]^{16}.$$

6 BitZ for UPESAT

6.1 Universal Polynomial Equation Satisfiability (UPESAT)

Throughout this section R denotes either $\mathbb{Z}$ or a finite field $\mathbb{F}$ with q^e elements. **Albert: Actually it may be good to allow R to be $\mathbb{Q}$, so we can inline constraints more easily (even though we can write the constraints over $\mathbb{Q}$ and then turn them into $\mathbb{Z}$ constraints by multiplying by the LCM, so maybe it's not worth the added complexity**

In this section we describe a variation of the PESAT relation (cf. ??), called *Universal Polynomial Equation Satisfiability (UPESAT)*. It generalizes PESAT by allowing polynomial constraints over a collection of quotient rings of the polynomial ring $R[X]$. UPESAT is a subset of the UCS relation from [?] (cf. ??).

We fix a witness length ℓ_0 which we assume is a power of two; an input length k ; a bit size bound B ; a degree bound d ; a collection of ι rings $R_1, \dots, R_\iota$, where each R_i is a quotient ring $R_i = R[X]/(h_i)$ for some polynomial $h_i \in R[X]$ which is either zero or monic and nonconstant (in the first case, $R_i = R[X]$; when $R = \mathbb{F}$, requiring h_i to be monic is no loss of generality); a number of rows n ; and for each $i \in [\iota]$, a vector of n polynomials

$$\mathbf{p}^{(i)} = \left(\mathbf{p}_j^{(i)} \mid j \in [n] \right) \in ((R[X])[Y_1, \dots, Y_{k+\ell}])^n$$

with coefficients in $R[X]$ and on $k + \ell$ variables. Given $a \in R[X]$ and $i \in [\iota]$, we say that $a = 0$ holds in R_i if the image of a under the canonical projection $R[X] \rightarrow R_i$ is zero. We denote

$$\mathbf{p} = (\mathbf{p}^{(i)})_{i \in [\iota]}, \quad \mathbf{R} = (R_i)_{i \in [\iota]}.$$

The UPESAT relation with respect to these parameters is defined as

$$\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \left(R_{\leq B}^{\leq d}[X] \right)^k, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in \left(R_{\leq B}^{\leq d}[X] \right)^\ell \end{array} \mid \mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{f}) = 0 \text{ in } R_i \text{ for all } j \in [n], i \in [\iota] \right\}$$

For instance, each R_i can be $R[X]$ itself (taking $h_i = 0$), the base ring R (taking $h_i = X$), a cyclotomic ring such as $\mathbb{Z}[X]/(X^{64} + 1)$, or, when $R = \mathbb{F}$, any finite extension field of $\mathbb{F}$ (taking h_i irreducible over $\mathbb{F}$). When $h_i = 0$ for all $i \in [\iota]$, so that the constraints are plain polynomial equations over $R[X]$, we write $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \perp)$.

The next remark explains that all our constructions for $\mathcal{R}_{\text{UPESAT}}$ apply, in particular, to the usual R1CS, CCS [?], AIR [?], lookups, permutation constraints, etc. over finite fields.

Example 6.1 (R1CS constraints over quotients of $R[X]$). Let $R, k, \ell, n, \iota, R_1, \dots, R_\iota, B, d$ be as before, and let $A^{(i)}, B^{(i)}, C^{(i)}$ be $n \times (k + \ell)$ matrices with entries in $R[X]$ (typically, these are sparse matrices with entries in a small subset of R), for all $i \in [\iota]$. Then the R1CS-UPESAT relation is defined as:

$$:= \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \left(R_{\leq B}^{\leq d}[X]\right)^k, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in \left(R_{\leq B}^{\leq d}[X]\right)^\ell \end{array} \middle| \begin{array}{l} \left(\left(A^{(i)} \cdot \mathbf{z}\right) \circ \left(B^{(i)} \cdot \mathbf{z}\right) - \left(C^{(i)} \cdot \mathbf{z}\right) \right)_j = 0 \text{ in } R_i \\ \text{for all } j \in [n], i \in [\iota] \end{array} \right\}, \quad \mathbf{z} = (\mathbf{x}, \mathbf{f})$$

The relation is precisely $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R})$ with

$$\mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{f}) = \left(\left(A^{(i)} \cdot \mathbf{z}\right) \circ \left(B^{(i)} \cdot \mathbf{z}\right) - \left(C^{(i)} \cdot \mathbf{z}\right) \right)_j.$$

Following ?? and [?, ?, ?] one can show that UPESAT allows to express popular constraint systems such as CCS, AIR, and lookup and permutation constraints, over quotients of polynomial rings.

Remark 6.2 (UPESAT vs the UCS relation from Zinc+ [?]). UPESAT is a particular subset of the widely general UCS relation from [?]. In this work, we will build a IOP for UPESAT with witnesses being committed over binary fields. The same ideas can be used to construct an analogous IOP for UCS with witnesses with entries in a bounded subset of $\mathbb{Z}[X]$ (as opposed to $\mathbb{Q}[X]$ [Albert: committing to rational through their bit-decomposition is a bit strange, but it may still be possible?](#)). We choose to focus on UPESAT to favor simplicity of exposition and of our IOP.

The main difference between UPESAT and UCS is that the later allows for witnesses and constraints over arbitrary and multiple domains in the same instance, including rational polynomial rings $\mathbb{Q}[X]$, and it allows to reuse witness elements in $\mathbb{Q}[X]$ or $\mathbb{Z}[X]$ in other domains such as finite fields and polynomial rings over finite fields, by using homomorphic projections.

6.2 An IOP for UPESAT with oracles in binary fields

[Albert: Needs a pass to update notation](#)

We now describe an IOP for $\mathcal{R}_{\text{UPESAT}}$. Given $a \in R[X]$ and $i \in [\iota]$, the equation $a = 0$ holds in $R_i = R[X]/(h_i)$ if and only if a belongs to the ideal (h_i) generated by h_i in $R[X]$. Accordingly, our IOP treats the UPESAT constraints as the ideal membership conditions $\mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{f}) \in (h_i)$ for $j \in [n]$ and $i \in [\iota]$.

We assume $n = 2^{n'}$ is a power of two, and identify $[n]$ with $\{0, 1\}^{n'}$. Let $\mathcal{C}$ be a linear code over $\mathbb{F}_{2^\nu}$ of dimension $(\ell_0 \cdot d \cdot B)/2^\nu$ and blocklength m . Let ℓ_1, ℓ_2 be tensor-decomposition parameters, to be used in ??, so that $\ell_0 \cdot d \cdot B = \ell_1 \cdot \ell_2$. Assume $\mathcal{C} = \mathcal{C}_1^{\equiv \ell_2}$ is the ℓ_2 -interleaving of a linear code $\mathcal{C}_1$ over $\mathbb{F}_{2^\nu}$ of dimension $\ell_1/2^\nu$.

For each prime, recall that $\pi_q : \mathbb{Z} \rightarrow \mathbb{F}_q$ denotes reduction modulo q ; and for every finite field $\mathbb{K}$ of characteristic q , we convene that π_q acts on $\mathbb{K}$ as the identity. For each finite field $\mathbb{K}$ let $\Pi_{\mathbb{K}}$ be an IOR that given $(\mathbf{x}, \perp, \mathbf{w})$ for $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, \mathbf{p})$ (for any collection of polynomials $\mathbf{p}$), outputs $(\mathbf{x}', \perp, \mathbf{w}) \in \mathcal{R}_{\text{Lin}}(\mathbb{K})$. We emphasize that the output of $\Pi_{\mathbb{K}}$ has the same witness $\mathbf{w}$ as the input.

In our instantiations, $\Pi_{\mathbb{K}}$ is either the usual Spartan [?] PIOP over $\mathbb{K}$, or a STARK-like PIOP [?] over $\mathbb{K}$.

Construction 6.3. [IOP for $\mathcal{R}_{\text{UPESAT}}$] **Parameters.** $(R, M_R, d, \ell_0, B, \ell_1, \ell_2, \mathbb{F}_{2^\nu}, \mathcal{C}_1^{\equiv \ell_2})$ comprised of: a ring $R \in \{\mathbb{Z}, \mathbb{F}_{q^e}\}$; an R -module M_R of dimension d ; a witness length ℓ_0 ; a bit-size bound B ; a binary field $\mathbb{F}_{2^\nu}$; tensor decomposition dimensions ℓ_1, ℓ_2 with $\ell_1 \cdot \ell_2 = \ell_0 \cdot d \cdot B$; a ℓ_2 -interleaving of a code $\mathcal{C}_1$ over $\mathbb{F}_{2^\nu}$ of dimension $\ell_1/2^\nu$.

Inputs. The prover and verifier receive $\mathbf{x} = \mathbf{x} \in R_{\leq B}^{\leq d}[X]^k$. The prover additionally receives $\mathbf{w} = \mathbf{f} \in R_{\leq B}^{\leq d}[X]^\ell$ such that $(\mathbf{x}, \perp, \mathbf{w}) \in \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R})$.

1. Round 1: oracles.

- The prover computes

$$f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})) \in \mathbb{F}_{2^\nu}^m$$

(cf. (??)) and sends the verifier oracle access to f .

- The prover forms the multilinear polynomial, for all $i \in [\iota]$,

$$Q^{(i)}(Z_1, \dots, Z_\nu) := \sum_{\mathbf{b} \in \{0,1\}^{n'}} \mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f}) \cdot \text{eq}(\mathbf{b}, Z_1, \dots, Z_\nu) \in (R[X])[Z_1, \dots, Z_\nu].$$

The claim that $\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f}) \in (h_i)$ for all $\mathbf{b} \in \{0,1\}^{n'}$ and $i \in [\iota]$ is equivalent to the claim

$$Q^{(i)}(\mathbf{z}) \in (h_i) \quad \text{for all } \mathbf{z} \in \{0,1\}^{n'}, i \in [\iota]. \quad (39)$$

- If $R = \mathbb{F}$, we let $\mathbb{K}$ be a large enough (to be determined later) extension of $\mathbb{F}$. If $R = \mathbb{Z}$, verifier samples a prime from a set of primes $\mathcal{P}$ (cf. ?? for comments on how to do this in practice). Abusing the notation, we denote the prime by q and denote $\mathbb{F}_q$ by $\mathbb{K}$.

The claim (??) is replaced by

$$\pi_q(Q^{(i)}(\mathbf{z})) \in (\pi_q(h_i)) \quad \text{for all } \mathbf{z} \in \{0,1\}^{n'}, i \in [\iota]. \quad (40)$$

Recall (cf. ??) that π_q is coefficient-wise reduction modulo q when $R = \mathbb{Z}$, and it is the identity map when R is a field $\mathbb{F}$ of characteristic q .

- The verifier samples and sends a random point $\mathbf{r} \in \mathbb{K}^{n'}$.

2. Round 2: Ideal check

- The prover responds with values $u_i \in \mathbb{K}[X]$ for each $i \in [\iota]$. Supposedly, $u_i = \pi_q(Q^{(i)})(\mathbf{r})$. These values can be computed as one would compute the corresponding values in a univariate skip, see ?? [Albert: todo](#). Alternatively, if $Q^{(i)}$ is linear on the multilinear extension of $\mathbf{f}$, then u_i can easily be computed from $\widetilde{\pi_q(\mathbf{f})}(\mathbf{r})$, which in turn can be computed as d MLE evaluations over $\pi_q(R)$ ⁶.

⁶This was already observed at least in [?], when $Q^{(i)}$ is the multilinear extension of the vector $C \cdot \mathbf{z}$ in an R1CS constraint [Albert: check: is that linear on witness?](#).

- The verifier checks that $u_i \in (\pi_q(h_i))$ for all $i \in [\iota]$. Then it samples and sends batching elements $\theta_1, \dots, \theta_\iota \in \mathbb{K}$. Further, if $d > 0$, it samples and sends an element α [Albert: it could send a random \$\mathbb{F}_q\$ -module homomorphism \$\mathbb{K}^{<d}\[X\] \rightarrow \mathbb{K}\$](#) .

3. Using the IOR $\Pi_{\mathbb{K}}$ for the projected claim

- Let $\pi := \pi_{q,\alpha}$ (cf. ??). The prover and verifier replace the claim (??) with the claim

$$\sum_{i \in [\iota]} \sum_{\mathbf{b} \in \{0,1\}^{n'}} \theta_i \cdot \pi(\mathbf{p}_{\mathbf{b}}^{(i)})(\pi(\mathbf{x}), \pi(\mathbf{f})) \cdot \text{eq}(\mathbf{b}, \mathbf{r}) = \sum_{i \in [\iota]} \theta_i \cdot \pi(u_i) \in \mathbb{K}. \quad (41)$$

Observe that $(\pi(\mathbf{x}), \perp, \pi(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{K}, \hat{\mathbf{p}})$ (cf. ??) with $\hat{\mathbf{p}} = (\hat{\mathbf{p}}_1)$ given by

$$\hat{\mathbf{p}}_1 = \sum_{i \in [\iota]} \theta_i \cdot \sum_{\mathbf{b} \in \{0,1\}^{n'}} \pi(\mathbf{p}_{\mathbf{b}}^{(i)})(Y_1, \dots, Y_{k+\ell}) - \sum_{i \in [\iota]} \theta_i \cdot \pi(u_i).$$

Accordingly, prover and verifier use the IOR $\Pi_{\mathbb{K}}$ to reduce (??) to a claim of the form:

$$((\mathbf{v}, \mu), \perp, \pi(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{K}) \quad (42)$$

for some $\mathbf{v} \in \mathbb{K}^{\ell_0}$ and some $\mu \in \mathbb{K}$.

4. Using the BitZ IOR for $\mathcal{R}_{\text{Lin-BitsDual}}$

- The initial claim $(\mathbf{x}, \perp, \mathbf{w}) \in \mathcal{R}_{\text{UPESAT}}$ has been replaced by $((\mathbf{v}, \mu), \perp, \pi(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{K})$. Notice that $\pi : R[X] \rightarrow \mathbb{K}$ induces a π_q -module homomorphism $\psi : R^{<d}[X] \rightarrow \mathbb{K}$ given by $\psi(h(X)) = \pi(h(X))$. Hence

$$((\mathbf{v}, \mu), f, \mathbf{f}) \in \mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{<d}, \psi, \mathbb{K}, \mathcal{C}). \quad (43)$$

Prover and verifier now use ?? to prove (??), with parameters

$$(R, q, R^{<d}[X], d, \mathbb{K}, \psi, (1, X, \dots, X^{d-1}), \ell_0, B, \ell_1, \ell_2, \mathbb{F}_{2^\nu}, \mathcal{C}_1^{\equiv \ell_2}).$$

Remark 6.4 (Randomly sampling a field). [Albert: Add some comment about the recursive cost and the mitigations we have](#)

Theorem 6.5 (Completeness and round-by-round knowledge soundness of ??). *Let $(R, M_R, d, \ell_0, B, \ell_1, \ell_2, \mathbb{F}_{2^\nu}, \mathcal{C}_1^{\equiv \ell_2})$ be parameters for ??. Assume the following:*

- *For every field $\mathbb{K}$ sampleable in Round 1 of ?? (or for the fixed field $\mathbb{K}$ if $R = \mathbb{F}$), $\Pi_{\mathbb{K}}$ is a $t_{\Pi_{\mathbb{K}}}$ -round perfectly complete IOR from $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, \mathbf{p})$ to $\mathcal{R}_{\text{Lin}}(\mathbb{K})$ with round-by-round knowledge soundness (without relaxation) and errors $\kappa_{\Pi_{\mathbb{K}},1}, \dots, \kappa_{\Pi_{\mathbb{K}},t_{\Pi_{\mathbb{K}}}}$.*
- *Let $\delta \in (0,1)$ be a proximity parameter.*
- *Assume $|\mathbb{F}_{2^\nu}| - 1 \geq (\ell_1 + 1) \cdot (q_{\max} - 1)$, where $q_{\max}$ is the largest prime in $\mathcal{P}$ in ??.*
- *Assume the notation and hypotheses of ?????, so that ?? is a perfectly complete $(\tau_{\text{ext}} + t_{\text{red}} + t_{\text{iopp}})$ -round IOP for $\mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{<d}, \psi, \mathbb{K}, \mathcal{C})$, with $\tau_{\text{ext}} = 1$ if the field $\mathbb{K}$ in ?? is a proper*

	$R = \mathbb{Z}$	$R = \mathbb{F}, \mathbb{F} \subsetneq \mathbb{K}$	$R = \mathbb{F} = \mathbb{K}$ is a prime field
Rounds	$2 + t_{\mathbb{K}} + t_{\text{red}} + t_{\text{iopp}}$	$2 + t_{\mathbb{K}} + 1 + t_{\text{red}} + t_{\text{iopp}}$	$2 + t_{\mathbb{K}} + t_{\text{red}} + t_{\text{iopp}}$
$\kappa_{\text{lift-proj}}$	$\perp$	$\frac{\deg(R)}{q_{\min}} + \frac{\log(\ell_1+1)+\log(q-1)}{ \mathcal{P} \cdot \lfloor \log(q_{\min}) \rfloor}$	$\perp$
κ_1	$\frac{B_{\max} \cdot (1 + \max(\nabla(\mathbf{p}) - d_{\min}, 0))}{ \mathcal{P} \cdot \lfloor \log(\mathbb{K}_{\min}) \rfloor} + \frac{1}{ \mathbb{K}_{\min} }$	$\frac{1}{ \mathbb{K}_{\min} }$	$\frac{1}{ \mathbb{K}_{\min} }$

Table 6: Albert: Outdated. Fix. Variations in number of rounds and errors of ?? depending on whether $R = \mathbb{Z}$ or $R = \mathbb{F}$, and on whether $\mathbb{K}$ is a prime field or a proper extension field

extension field and $\tau_{\text{ext}} = 0$ if $\mathbb{K}$ is a prime field, with round-by-round knowledge with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{\leq d}, \psi, \mathbb{K}, \mathcal{C})^\delta, \perp)$ and errors

$$\kappa_{\text{lift-proj}}^*, (\kappa_{\text{red } i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp } i})_{i \in [t_{\text{iopp}}]},$$

where

$$\kappa_{\text{lift-proj}}^* = \begin{cases} \frac{\deg(R)}{q_{\min}} + \frac{\log(\ell_1+1)+\log(q-1)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor} & \text{if } \mathbb{K} \text{ is a proper extension field,} \\ \perp & \text{if } \mathbb{K} \text{ is a prime field.} \end{cases}$$

and $q_{\min}$ is the smallest prime in the set of sampleable primes $\mathcal{P}$.

Then ?? is a perfectly complete $(2 + t_{\mathbb{K}} + \tau_{\text{ext}} + t_{\text{red}} + t_{\text{iopp}})$ -IOP for $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R})$ with round-by-round knowledge soundness without relaxation and errors

$$\kappa_1, \kappa_2, (\kappa_{\Pi_{\mathbb{K}}, i})_{i \in [t_{\Pi_{\mathbb{K}}}]}, \kappa_{\text{lift-proj}}^*, (\kappa_{\text{red } i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp } i})_{i \in [t_{\text{iopp}}]},$$

with

$$\begin{aligned} \kappa_1 &:= \left(\frac{B_{\max} \cdot (1 + \max(\nabla(\mathbf{p}) - d_{\min}, 0))}{|\mathcal{P}| \cdot \lfloor \log(|\mathbb{K}_{\min}|) \rfloor} \right) \cdot \tau_{\mathbb{Z}} + \frac{1}{|\mathbb{K}_{\min}|} \\ \kappa_2 &:= \frac{\nabla(\mathbf{p})}{|\mathbb{K}_{\min}|} + \frac{1}{|\mathbb{K}_{\min}|}, \end{aligned}$$

where:

- $\tau_{\mathbb{Z}} = 1$ if $R = \mathbb{Z}$ and $\tau_{\mathbb{Z}} = 0$ otherwise,
- $\mathbb{K}_{\min}$ is either $\mathbb{K}$ if $R = \mathbb{F}$, or the field $\mathbb{F}_q$ with $q \in \mathcal{P}$ with the smallest size if $R = \mathbb{Z}$,
- $d_{\min} := \min \{ \deg(h_i) \mid i \in [\ell], h_i \neq 0 \}$, with the convention $d_{\min} = +\infty$ when $h_i = 0$ for all $i \in [\ell]$
- $\nabla(\mathbf{p}) := \max \left\{ \max_{\mathbf{b} \in \{0,1\}^{n'}, i \in [\ell]} \{ \deg_X(\mathbf{p}_j^{(i)}) + (d-1) \cdot \max_{t \in [k+\ell]} \deg_{Y_t}(\mathbf{p}_j^i) \cdot (k+\ell) \}, d \right\}$
- $B_{\max} := \max_{\mathbf{b} \in \{0,1\}^{n'}, i \in [\ell]} \lceil \log(\|\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})\|_{\infty}) \rceil$.

In ?? we describe the variations in number of rounds and errors of ?? depending on whether $R = \mathbb{Z}$ or $R = \mathbb{F}$, and on whether $\mathbb{K}$ is a prime field or a proper extension field.

We defer the proof of ?? to ?? **Albert: Errors after composition of IOR's cannot be the same, since they receive different inputs?**. We state and prove a key lemma here:

Lemma 6.6 (Ideal membership is preserved under modular reduction). *Let $d, B \geq 1$ and $f, h \in \mathbb{Z}_{\leq B}^{\leq d}[X]$, with h either 0 or a nonzero monic polynomial. Let $\mathcal{P}$ be a set of primes. Then*

$$\Pr_{p \leftarrow \mathcal{P}} \left[\pi_p(f) \in (\pi_p(h)) \wedge f \notin (h) \right] < \frac{B \cdot (1 + \max(\deg(f) - \deg(h) + 1, 0))}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor}.$$

where $q_{\min} := \min \mathcal{P}$.

Proof. Assume $f \notin (h)$. In particular $f \neq 0$. If $h = 0$, then $(\pi_p(h)) = \{0\}$, and so $\pi_p(f) = 0$. Since $\|f\|_{\infty} \leq 2^B - 1$, there are at most $B / \lfloor \log(q_{\min}) \rfloor$ primes p such that $\pi_p(f) = 0$, as needed.

If $h \neq 0$ and $\deg(h) = 0$, then $h = 1$ (since h is monic) and so $f \in (h) = \mathbb{Z}[X]$, a contradiction. Assume thus that $\deg(h) \geq 1$. Let $Q, R \in \mathbb{Z}[X]$ be such that $f = h \cdot Q + R$ with $\deg(R) < \deg(h)$ (such integer polynomials exist because h is monic). Since $f \notin (h)$, we have $R \neq 0$. By ??,

$$\|R\|_{\infty} \leq \|f\|_{\infty} \cdot (1 + \|h\|_{\infty})^{\max(\deg(f) - \deg(h) + 1, 0)}.$$

Now since $\deg(\pi_p(h)) = \deg(h)$ (because h is monic) we have $\deg(\pi_p(h)) > \deg(R) \geq \deg(\pi_p(R))$. The condition $\pi_p(f) \in (\pi_p(h))$ thus implies that $\pi_p(R) = 0$. There are at most $\log_2(\|R\|_{\infty}) / \lfloor \log q_{\min} \rfloor$ primes dividing the coefficients of R . Hence

$$\Pr_{p \leftarrow \mathcal{P}} [f \notin (h) \wedge \pi_p(f) \in (\pi_p(h))] \leq \frac{\log_2(\|R\|_{\infty})}{|\mathcal{P}| \cdot \lfloor \log q_{\min} \rfloor} \leq \frac{B + \max(\deg(f) - \deg(h) + 1, 0) \cdot B}{|\mathcal{P}| \cdot \lfloor \log q_{\min} \rfloor},$$

as needed. $\square$

Albert: Ideal membership as a generalization of univariate skip? Albert: Repeated computations through ideal membership?

6.3 Instantiation

Albert: TODO: what prime ranges to choose to achieve 128 bits of security?

7 Experiments

7.1 Instantiation

Remark 7.1 (The intermediate IOR $\Pi_{\mathbb{K}}$ as a batched sumcheck). **Albert: Scratch. Maybe remove** In many reasonable instances of $\mathcal{R}_{\text{UPESAT}}$ (e.g. R1CS, CCS, or AIR-like instantiations), we have that

$$\mathbf{p}_{\mathbf{b}}^{(i)}(Y_1, \dots, Y_{k+\ell}) = \mathbf{p}'^{(i)}(\mathbf{b}, Y_1, \dots, Y_{k+\ell}) \quad \text{for all } i \in [\iota],$$

for some polynomials $\mathbf{p}'^{(i)}$ on $n' + k + \ell$ **Albert: ℓ_0 ?** variables.

In this case, one can take the IOR $\Pi_{\mathbb{K}}$ from $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, \hat{\mathbf{p}})$ to $\mathcal{R}_{\text{Lin}}(\mathbb{K})$ from at Step 3 of ?? to consist simply in applying the batched sumcheck protocol over $\mathbb{K}$ for the claims

$$\sum_{\mathbf{b} \in \{0,1\}^{n'}} \pi(\mathbf{p}'^{(i)})(\mathbf{b}, \pi(\mathbf{x}), \pi(\mathbf{f})) \cdot \text{eq}(\mathbf{b}, \mathbf{r}) = \pi(u_i) \in \mathbb{K} \quad \text{for all } i \in [\iota].$$

At the end of batched sumcheck, they are left with a claim of the form

$$\pi(\mathbf{p}'^{(i)})(\mathbf{r}^{(1)}, \pi(\mathbf{x}), \pi(\mathbf{f})) = u'_i \in \mathbb{K} \quad \text{for all } i \in [\iota] \tag{44}$$

for some $\mathbf{r}^{(1)} \in \mathbb{K}^\nu$, $u'_i \in \mathbb{K}$.

....
...

The claim (??) forms a valid instance-witness for the PESAT relation (cf. ??) over $\mathbb{K}$ with polynomial constraints given by $\pi(\mathbf{p}'^{(i)})(\mathbf{r}^{(1)}, Y_1, \dots, Y_{k+\ell}) - u'_i$, i.e. for such constraints and corresponding parameters

$$(\pi(\mathbf{x}), \perp, \pi(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}.$$

Accordingly, prover and verifier use the IOR $\Pi_{\mathbb{K}}$ to reduce (??) to a claim of the form

$$\widetilde{\pi(\mathbf{f})}(\mathbf{r}^{(2)}) = \mu \tag{45}$$

for some $\mathbf{r}^{(2)} \in \mathbb{K}^{\log \ell}$ and some $\mu \in \mathbb{K}$. In other words, $((\mathbf{r}^{(2)}, \mu), \perp, \pi(\mathbf{f})) \in \mathcal{R}_{\text{mle}}$ over $\mathbb{K}$.

7.2 Experimental results

7.2.1 Raw performance of BitZ

We measure the performance of ?? for committing to witnesses $\mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$, and proving claims of the form $\langle \mathbf{v}, \pi_q(\mathbf{f}) \rangle = \mu$ over a finite field $\mathbb{F}_{q^d}$, where π_q denotes reduction modulo a prime q , and $\mathbf{v} \in \mathbb{F}_{q^d}^\ell$. We instantiate ?? following the guideline in ?. For WHIR we use a Reed-Solomon code of rate $1/8$ over $\mathbb{F}_{2^{128}}$.

In ?? we present performance for the case $q \leq 2^{105}$, $d = 1$, in which case so the optional step of Round 1 of ?? is omitted (cf. ??). In ?? we make the same measurement for the case $d = 5$ and $q \leq 2^{32}$.

Albert: Measure Binius64 proving the same inner product directly in Bin fields?

7.2.2 Integer and modular multiplication

In ?? we measure the cost of proving 2^i integer multiplications of the form $x \cdot y = z$, for x, y random 32-bit integers, and z a 64-bit integer. We use our IOP for the UPESAT relation ?. The specific form of the UPESAT relation is as follows: the witness and the constraints are over $\mathbb{Z}$

7.2.3 SHA-256 hashing

SHA-256 compression overview We begin with a brief overview of SHA-256 [?] to establish notation, then formalize the relation we seek to arithmetize.

For the purposes of SHA-256, a *word* is a ≤ 32 string of bits, which we represent as a binary integer polynomial, i.e. as an element of $\{0, 1\}^{<32}[X]$. The SHA-256 compression function receives as input a 512-bit string, represented as sixteen *message words* $(M_t)_{t \in [16]} \in \{0, 1\}^{<32}[X]^{16}$. The algorithm also uses 64 fixed *round constants* $(K_t)_{t \in [64]} \in \{0, 1\}^{<32}[X]^{64}$.

We let

$$\begin{aligned} \psi_2 : \mathbb{Z}[X] &\rightarrow \mathbb{Z} \\ a(X) &\mapsto a(2) \end{aligned} \tag{46}$$

be the ring homomorphism sending integer polynomials to their evaluation at $X = 2$. Notice that, given $a \in \{0, 1\}^{<32}[X]$, the integer $\psi_2(a) \in \mathbb{Z}$ is the integer whose bit-representation coincides with the coefficients of a . The restriction

$$\psi_2 : \{0, 1\}^{<32}[X] \longrightarrow \{0, \dots, 2^{32} - 1\}$$

is a bijection; throughout this subsection, ψ_2^{-1} denotes the inverse of this restricted map, not an inverse on all of $\mathbb{Z}[X]$.

SHA-256 is built from the following six auxiliary functions. For $x, y, z \in \{0, 1\}^{<32}[X]$, define:

$$\begin{aligned}
\Sigma_0(x) &:= \text{ROTR}^2(x) \text{ XOR } \text{ROTR}^{13}(x) \text{ XOR } \text{ROTR}^{22}(x) && \in \{0, 1\}^{<32}[X] \\
\Sigma_1(x) &:= \text{ROTR}^6(x) \text{ XOR } \text{ROTR}^{11}(x) \text{ XOR } \text{ROTR}^{25}(x) && \in \{0, 1\}^{<32}[X] \\
\text{Ch}(x, y, z) &:= (x \text{ AND } y) \text{ XOR } (\neg x \text{ AND } z) && \in \{0, 1\}^{<32}[X] \\
\text{Maj}(x, y, z) &:= (x \text{ AND } y) \text{ XOR } (x \text{ AND } z) \text{ XOR } (y \text{ AND } z) && \in \{0, 1\}^{<32}[X] \\
\sigma_0(x) &:= \text{ROTR}^7(x) \text{ XOR } \text{ROTR}^{18}(x) \text{ XOR } \text{SHR}^3(x) && \in \{0, 1\}^{<32}[X] \\
\sigma_1(x) &:= \text{ROTR}^{17}(x) \text{ XOR } \text{ROTR}^{19}(x) \text{ XOR } \text{SHR}^{10}(x) && \in \{0, 1\}^{<32}[X]
\end{aligned} \tag{47}$$

Here ROTR^r rotates a word right by r positions, SHR^r shifts it right by r positions and fills the high positions with zero, and XOR , AND , $\neg$ act coefficient-wise on bits.

Message schedule. At the beginning of the SHA-256 compression function, the sixteen input message words $(M_t)_{t \in [16]}$ are expanded into a so-called *message schedule*, which consists of 64 words $(W_t)_{t \in [64]} \in (\{0, 1\}^{<32}[X])^{64}$ defined as follows:

$$W_t := \begin{cases} M_t & t \in [16], \\ \psi_2^{-1}(\psi_2(W_{t-16}) + \psi_2(\sigma_0(W_{t-15})) + \psi_2(W_{t-7}) + \psi_2(\sigma_1(W_{t-2})) \bmod 2^{32}) & t \in \{17, \dots, 64\}. \end{cases} \tag{48}$$

Compression loop. After the message schedule is computed, SHA-256 performs the so-called *compression loop*. Throughout it, the function maintains an 8-word internal state, conventionally labeled as

$$(a, b, c, d, e, f, g, h) \in \{0, 1\}^{<32}[X]^8.$$

The compression loop iterates 64 rounds. At each round $t \in [64]$, the state entering that round is denoted $(a_t, b_t, c_t, d_t, e_t, f_t, g_t, h_t) \in \{0, 1\}^{<32}[X]^8$. The per-round update is given by, for $t \in [64]$,

$$\begin{aligned}
T_{1,t} &:= \psi_2^{-1}(\psi_2(h_t) + \psi_2(\Sigma_1(e_t)) + \psi_2(\text{Ch}(e_t, f_t, g_t)) + \psi_2(K_t) + \psi_2(W_t) \bmod 2^{32}), \\
T_{2,t} &:= \psi_2^{-1}(\psi_2(\Sigma_0(a_t)) + \psi_2(\text{Maj}(a_t, b_t, c_t)) \bmod 2^{32}), \\
a_{t+1} &:= \psi_2^{-1}(\psi_2(T_{1,t}) + \psi_2(T_{2,t}) \bmod 2^{32}), \\
e_{t+1} &:= \psi_2^{-1}(\psi_2(d_t) + \psi_2(T_{1,t}) \bmod 2^{32}), \\
(b_{t+1}, c_{t+1}, d_{t+1}, f_{t+1}, g_{t+1}, h_{t+1}) &:= (a_t, b_t, c_t, e_t, f_t, g_t).
\end{aligned}$$

For the feed-forward step, let

$$\begin{aligned}
S^{\text{in}} &= (a_1, b_1, c_1, d_1, e_1, f_1, g_1, h_1), \\
S^{\text{fin}} &= (a_{65}, b_{65}, c_{65}, d_{65}, e_{65}, f_{65}, g_{65}, h_{65}).
\end{aligned}$$

The compression output is the word vector H^{out} defined by

$$H_j^{\text{out}} = \psi_2^{-1}(\psi_2(S_j^{\text{in}}) + \psi_2(S_j^{\text{fin}}) \bmod 2^{32}) \quad (j \in \{0, \dots, 7\}).$$

7.2.4 Falcon signature verification

7.2.5 RSA and Poseidon

Albert: Let's compare to with the circuit of Limber

7.2.6 Hashing followed by Falcon signature verification

7.2.7 Comparison with Hyrax

Albert: Comparing with Hyrax may be more interesting than it sounds initially. Hyrax is pay-per-bit and opens on large prime fields: a profile that F2Z can adopt. Hence (maybe) projects using Hyrax could use F2Z as a drop-in replacement which is PQ secure and more efficient

7.3 Open questions

Albert: We may want to keep these to ourselves

Albert: This section is just scratch/notes/thoughts

- When having $\mathbb{F}_2$ -virtualization in the grand product, there's potential optimisations. So far I've only found a way to cheapen the following settings: say first and two vectors in the grand product are $\mathbf{g}_1, \mathbf{g}_2$, and third is $\mathbf{g}_1 \text{ XOR } \mathbf{g}_2$. This is a tricky problem. We should look more into it
- Should we allow for $W > 1$? Currently just setting $W = 1$ and thinking of all bits as unrelated leads to better performance, but that sounds suboptimal.
- Folding instances of $\mathcal{R}_{\text{Lin-BitsDual}}$

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A Deferred proofs

Albert: Notation changed significantly recently. REVIEW

A.1 Completeness and soundness of ??

Albert: I wrote this quite quickly. It may need polishing

The result follows from ?? and the fact that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{R \leq B}, \psi, \mathbb{K}, \mathcal{C})$ if and only if (??) holds (due to ??). More concretely: perfect completeness is immediate. For round-by-round knowledge soundness, let KState' be a state function for ??. Then we define a state function KState for ?? as, when tr is neither an empty transcript or a transcript containing only a prover message (and no verifier messages),

$$\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = \text{KState}'(\mathbf{x}', \mathbf{y}, \text{tr}, \mathbf{w}),$$

where if $\mathbf{x} = (\mathbf{v}, \mu)$ then $\mathbf{x}' = (\mathbf{u}, \mu)$ is defined as in ??. For empty transcripts we let $\text{KState}(\mathbf{x}, \mathbf{y}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{R \leq B}, \psi, \mathbb{K}, \mathcal{C}_1^{\equiv \ell_2})^\delta$. If tr is a transcript containing uniquely the first prover message, then we let $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = \text{KState}(\mathbf{x}, \mathbf{y}, \mathbf{w})$.

Further, let $\mathcal{E}'$ be an extractor for ??. We define an extractor $\mathcal{E}$ for ?? as follows: if tr is neither empty nor consisting in just one prover message, $\mathcal{E}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) := \mathcal{E}'(\mathbf{x}', \mathbf{y}, \text{tr}, \mathbf{w})$. For tr empty, we let $\mathcal{E}(\mathbf{x}, \mathbf{y}, \mathbf{w})$ proceeds as follows: first it computes

$$\mathbf{g} = \mathcal{E}'(\mathbf{x}', \mathbf{y}, \mathbf{w}).$$

If $\mathbf{g} \in \{0, 1\}^{\ell_0 \cdot B \cdot d} \subseteq \mathbb{Z}^{\ell_0 \cdot B \cdot d}$, then let $\mathbf{f} \in M_{R \leq B}^{\ell_0}$ such that $\text{bits}(\mathbf{f}) = \mathbf{g}$, concretely, index $\mathbf{f}$ as $\mathbf{g} = (\mathbf{g}_{ijk})_{i \in [\ell_0], j \in [d], k \in [B]}$. Let $\phi = \text{id}$ be the identity map if $R = \mathbb{Z}$, and $\phi = \pi_q$ if R is a field of characteristic q . Then output

$$\mathbf{f} := \sum_{j \in [d], k \in [B]} \gamma_j \cdot 2^{k-1} \cdot \phi(\mathbf{g}_{*jk}),$$

where $\mathbf{g}_{*jk} = (\mathbf{g}_{ijk})_{i \in [\ell_0]} \in \{0, 1\}^{\ell_0} \subseteq \mathbb{Z}^{\ell_0}$. Then $\text{bits}(\mathbf{f}) = \mathbf{g}$ [Albert: more detail?](#).

If tr is a transcript consisting of only one prover message, we let $\mathcal{E}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = \mathcal{E}(\mathbf{x}, \mathbf{y}, \mathbf{w})$.

For the last two cases, we claim that

$$((\mathbf{u}, \mu), \mathbf{y}, \mathbf{g}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi_q, \mathbb{K}, \mathcal{C}_1^{\ell_2}) \Leftrightarrow ((\mathbf{v}, \mu), \mathbf{y}, \mathbf{f}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{R \leq B}, \psi, \mathbb{K}, \mathcal{C}_1^{\equiv \ell_2}).$$

Indeed, this follows from ??, which yields

$$\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \langle \pi_q(\text{bits}(\mathbf{f})), \mathbf{u} \rangle = \langle \pi_q(\mathbf{g}), \mathbf{u} \rangle.$$

[Albert: increase detail?](#)

We now prove that, for any transcript tr where verifier is about to send its j -th message

$$P_j := \Pr_{\rho} \left[\exists \mathbf{w} \left| \begin{array}{l} \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \text{tr}, \rho, \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \rho, \mathbf{w}) = 1 \end{array} \right. \right] \leq \kappa'_j.$$

where κ'_j is the j -th error in ??.

Indeed, the inequality is immediate if tr does not consist of a single prover message, since then KState and $\mathcal{E}$ coincide with the state KState' and extractor $\mathcal{E}'$ of $??$. For P_1 , namely if tr consists of a single prover message, then

$$P_1 = \Pr_{\rho} \left[\exists \mathbf{w} \mid \begin{array}{l} \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \text{tr}, \rho, \mathbf{w})) = 0 \\ \wedge \text{KState}'(\mathbf{x}', \mathbf{y}, \text{tr}, \rho, \mathbf{w}) = 1 \end{array} \right]$$

If $\text{KState}'(\mathbf{x}', \mathbf{y}, \text{tr}, \rho, \mathbf{w}) = 1$ then with probability $1 - \kappa_0$, $\mathcal{E}'(\mathbf{x}', \mathbf{y}, \text{tr}, \mathbf{w})$ outputs $\mathbf{g}$ such that $\text{KState}'(\mathbf{x}', \mathbf{y}, \text{tr}, \mathbf{g}) = 1$, hence, in this case, $\text{KState}'(\mathbf{x}', \mathbf{y}, \mathbf{g}) = 1$, which holds if and only if $(\mathbf{x}', \mathbf{y}, \mathbf{g}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi_q, \mathbb{K}, \mathcal{C}_1^{\equiv \ell_2})$. Hence, with probability at least $1 - \kappa_0$, $\mathcal{E}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w})$ outputs $\mathbf{f}$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{f}) \in \mathcal{R}_{\text{Lin-BitsDual}}(M_{R_{\leq B}}, \psi, \mathbb{K}, \mathcal{C}_1^{\equiv \ell_2})$. It follows that $P_1 \leq \kappa_0$.

A.2 Completeness and soundness of $??$

To prove $??$, we will need to introduce an extension of the definition of $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, \mathbf{p})$ and $\mathcal{R}_{\text{Lin}}(\mathbb{K})$ so that these relations can hold witnesses in $R_{\leq B}^{\leq d}$ and their encodings over a binary field. Namely, given a field of characteristic q and a π_q -module homomorphism (cf. $??$) $\psi_{\mathbb{K}} : R^{\leq d}[X] \rightarrow \mathbb{K}$, we define:

$$\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \mathbf{p}, \psi_{\mathbb{K}}, \mathcal{C}) := \left\{ \left(\begin{array}{l} \mathbf{x} = \mathbf{x} \in \mathbb{K}^k, \\ \mathbf{y} = f \in (\mathbb{F}_{2^\nu})^m, \\ \mathbf{w} = \mathbf{f} \in (R_{\leq B}^{\leq d})^{\ell_0} \end{array} \right) \mid \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})) \\ \wedge \mathbf{p}_i(\mathbf{x}, \psi_{\mathbb{K}}(\mathbf{f})) = 0 \text{ for all } i \in [n] \end{array} \right\}$$

Note that when $\mathbf{p}$ contains a single linear polynomial, then

$$\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \mathbf{p}, \psi, \mathcal{C}) = \mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{\leq d}, \psi, \mathbb{K}, \mathcal{C}).$$

We will also need an encoded version of UPESAT: we define $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ as

$$\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in (R_{\leq B}^{\leq d}[X])^k, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in (R_{\leq B}^{\leq d}[X])^\ell \end{array} \mid \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \\ \mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{f}) = 0 \text{ in } R_i \text{ for all } j \in [n], i \in [\iota] \end{array} \right\}$$

As usual, for the relations $\mathcal{R}$ that contain the constraint $f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f}))$, we also consider their *relaxed* version $\mathcal{R}^\delta$, where such constraint is replaced by $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f}))) < \delta$.

Lemma A.1. *Let $\mathbb{K}$ be a finite field of characteristic q and let $\psi_{\mathbb{K}} : R^{\leq d}[X] \rightarrow \mathbb{K}$ be a π_q -module homomorphism. **Albert: correct terminology? And above?.** Let $\Pi_{\mathbb{K}}$ be a $t_{\mathbb{K}}$ -round perfectly complete IOR from $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, \mathbf{p})$ to $\mathcal{R}_{\text{Lin}}(\mathbb{K})$ with round-by-round knowledge soundness (with no relaxation) with errors $(\kappa_{\Pi_{\mathbb{K}}, i})_{i \in [t_{\mathbb{K}}]}$.*

Let $\delta \in (0, 1)$. Then there is an IOR $\Pi'_{\mathbb{K}}$ from $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \mathbf{p}, \psi_{\mathbb{K}}, \mathcal{C})$ to $\mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{\leq d}, \psi_{\mathbb{K}}, \mathbb{K}, \mathcal{C})$ with $t_{\mathbb{K}}$ rounds, perfectly complete, and with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \mathbf{p}, \psi_{\mathbb{K}}, \mathcal{C}))^\delta, \mathcal{R}_{\text{Lin-BitsDual}}(R_{\leq B}^{\leq d}, \psi_{\mathbb{K}}, \mathbb{K}, \mathcal{C})^\delta$ and with errors $(\kappa_{\Pi_{\mathbb{K}}, i})_{i \in [t_{\mathbb{K}}]}$.

Albert: Could there be an issue with extracting the correct bit size B ? No because prover outputs $\mathbf{f}$, not $\mathbf{f}'$ This IOR $\Pi'_{\mathbb{K}}$ is exactly $\Pi_{\mathbb{K}}$, except that prover and verifier receive an oracle f , which

they do not use throughout the IOR; and additionally the prover receives a witness $\mathbf{f}$ with entries in $R_{\leq B}^{\leq d}$ instead of in $\mathbb{K}$. Then the prover of $\Pi'_{\mathbb{K}}$ proceeds exactly as the prover of $\Pi_{\mathbb{K}}$ after replacing $\mathbf{f}$ by $\psi_{\mathbb{K}}(\mathbf{f})$. At the end of the IOR, both prover and verifier of $\Pi'_{\mathbb{K}}$ output the same $\mathbf{x}$ output by $\Pi_{\mathbb{K}}$. Additionally, they output $\mathbf{y} = f$, and the prover of $\Pi'_{\mathbb{K}}$ outputs any vector in the preimage of $\psi_{\mathbb{K}}^{-1}(\mathbf{f}')$, where $\mathbf{f}'$ is the witness output by the prover of $\Pi_{\mathbb{K}}$.

Proof. **Albert:** TODO. Anywhere in the literature? □

We build three IORs Π_1, Π_2, Π_3 whose composition closely mimics ??.

Description of Π_1 . The IOR Π_1 is a two-round IOR from $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ to $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})$, for some polynomials $\hat{\mathbf{p}}$ and π_q -module homomorphism $\pi : R_{\leq B}^{\leq d} \rightarrow \mathbb{K}$. In Π_1 , verifier and prover receive $(\mathbf{x}, f)$ and $(\mathbf{x}, f, \mathbf{f}) \in \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ as inputs, respectively. In the first round, prover and verifier simulate Round 1 of ??, skipping the first item (computing and sending the encoding f of $\mathbf{f}$), since f is already in the input of the prover and verifier. The prover instead sends an empty message $\perp$. For the second round of Π_1 , prover and verifier simulate Round 2 of ??. At the end of Π_1 , prover and verifier output

$$(\pi(\mathbf{x}), f, \mathbf{f}) \in \mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C}),$$

Description of Π_1 . The IOR Π_1 is a two-round IOR from $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ to $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})$, for some polynomials $\hat{\mathbf{p}}$ and π_q -module homomorphism $\pi : R_{\leq B}^{\leq d} \rightarrow \mathbb{K}$. In Π_1 , verifier and prover receive $(\mathbf{x}, f)$ and $(\mathbf{x}, f, \mathbf{f}) \in \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ as inputs, respectively. In the first round, prover and verifier simulate Round 1 of ??, skipping the first item (computing and sending the encoding f of $\mathbf{f}$), since f is already in the input of the prover and verifier. The prover instead sends an empty message $\perp$. For the second round of Π_1 , prover and verifier simulate Round 2 of ??. At the end of Π_1 , prover and verifier output

$$(\pi(\mathbf{x}), f, \mathbf{f}) \in \mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C}),$$

where $\hat{\mathbf{p}}$ and π are determined during the execution of Π_1 **Albert:** This may be formally incorrect: I don't think current difinitions support relations in an IOR changing depending on the run of the IOR, so we'd need to add the parameters in $\mathbf{x}$.

Description of Π_2 . The IOR Π_2 is an IOR from $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})$ to $\mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \mathcal{C})$. Prover and verifier receive $(\pi(\mathbf{x}), f)$ and $(\pi(\mathbf{x}), f, \mathbf{f})$ as inputs. Then they apply the IOR $\bar{\Pi}'_{\mathbb{K}}$ given by ??, where $\Pi_{\mathbb{K}}$ is the IOR applied in Item 3 of ??. They use as inputs $(\pi(\mathbf{x}), f)$ and $(\pi(\mathbf{x}), f, \mathbf{f})$, respectively. By ??, $\Pi'_{\mathbb{K}}$ is an IOR from $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})$ to $\mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \pi, \mathcal{C})$, perfectly complete, with $t_{\mathbb{K}}$ rounds, and with round-by-round knowledge with relaxation $(\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})^\delta, \mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \pi, \mathcal{C})^\delta)$ and with errors $(\kappa_{\Pi_{\mathbb{K}}, i})_{i \in t_{\mathbb{K}}}$

Description of Π_3 The IOR Π_3 is an IOP for $\mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \pi, \mathcal{C})$. It consists in applying Phase 4 of ??, i.e. simulating ?? with inputs $((\mathbf{v}, \mu), f, \mathbf{f})$. By ?? and the hypothesis of ??, ?? is a $(\tau_{\text{ext}} + t_{\text{red}} + t_{\text{iopp}})$ -round perfectly complete IOP for $\mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \pi, \mathcal{C})$ with round-by-round knowledge with relaxation $(\mathcal{R}_{\text{Lin}}(R_{\leq B}^{\leq d}, \mathbb{K}, \pi, \mathcal{C})^\delta, \perp)$, with errors $\kappa_{\text{lift-proj}}^*, (\kappa_{\text{red } i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp } i})_{i \in [t_{\text{iopp}}]}$.

The next lemma states that the composition of Π_1, Π_2 and Π_3 is an IOR whose completeness and soundness implies completeness and soundness of ??.

Lemma A.2. *Let Π_{123} be the composition of Π_1, Π_2 and Π_3 . Then Π_{123} is a $(2 + t_{\mathbb{K}} + \tau_{\text{ext}} + t_{\text{red}} + t_{\text{iopp}})$ -round IOP for $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$. Assume Π_{123} is perfectly complete and, for any $\delta \in (0, 1)$, has round-by-round knowledge with relaxation $(\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta, \perp)$ with errors $(\kappa'_j)_{j \in [2+t_{\mathbb{K}}+\tau_{\text{ext}}+t_{\text{red}}+t_{\text{iopp}}]}$.*

Then ?? is a $(2 + t_{\mathbb{K}} + \tau_{\text{ext}} + t_{\text{red}} + t_{\text{iopp}})$ -round perfectly complete IOP for $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R})$, and has round-by-round knowledge without relaxation with errors $(\kappa_j)_{j \in [2+t_{\mathbb{K}}+\tau_{\text{ext}}+t_{\text{red}}+t_{\text{iopp}}]}$ defined as

$$\kappa_j(\mathbb{x}) := \max_{f \text{ s.t. } (\mathbb{x}, f) \notin \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta} \kappa_j(\mathbb{x}, f)'.$$

Albert: Need to argue somewhere that these errors are negligible if κ'_j are – I'd prefer to just use the errors from κ'_j directly. This may be related with the issue about composition of IORs and having ill-defined errors

Proof. Deferred to ??. □

Due to ??, it suffices to show that Π_{123} is perfectly complete and has the desired soundness properties. We prove this by using the composition of IOR result ??. Both Π_2 and Π_3 already have the desired properties by definition and hypotheses of ??. Thus it suffices to show the following:

Lemma A.3 (Completeness and soundness of Π_1). *Let $\delta \in (0, 1)$. The IOR Π_1 is a perfectly complete 2-round IOR from $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$ to $\mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta, \mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})^\delta)$ and with errors*

$$\begin{aligned} \kappa_1 &= \frac{B_{\max} \cdot (1 + \max(\nabla(\mathbf{p}) - d_{\min}, 0))}{|\mathcal{P}| \cdot \lfloor \log(|\mathbb{K}_{\min}|) \rfloor} + \frac{1}{|\mathbb{K}_{\min}|} \\ \kappa_2 &= \frac{\nabla(\mathbf{p})}{|\mathbb{K}_{\min}|} + \frac{1}{|\mathbb{K}_{\min}|}, \end{aligned}$$

Proof. **Albert: Probably needs some polishing and review** Indeed, let $\mathbb{x} = \mathbf{x}, \mathbb{y} = f$ be an input and oracle for $\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})$. For any $\mathbf{w}$ define $\text{KState}(\mathbb{x}, f, \mathbf{w}) = 1$ if and only if $(\mathbb{x}, f, \mathbf{w}) \in \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta$. Given a full transcript $\text{tr} = (\perp, (\mathbb{K}, \mathbf{r}), (u_1, \dots, u_\ell), (\theta_1, \dots, \theta_\ell, \alpha))$ for Π_1 , define $\text{KState}(\mathbb{x}, f, \text{tr}, \mathbf{w}) = 1$ if and only if $(\pi(\mathbf{x}), f, \pi(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \pi, \mathcal{C})^\delta$, where $\pi = \pi_{q, \alpha}$, q being the characteristic of $\mathbb{K}$ and α one of the verifier's messages in tr . For partial transcripts where the prover tr is about to send a message m , let $\text{KState}(\mathbb{x}, f, \text{tr}, m, \mathbf{w}) = \text{KState}(\mathbb{x}, f, \text{tr}, \mathbf{w})$.

We now define $\text{KState}(\mathbb{x}, f, \text{tr}, \mathbf{w})$ where tr is nonempty and such that prover is about to move. The only undefined setting is when $\text{tr} = (\perp, (\mathbb{K}, \mathbf{r}))$. In this case we define $\text{KState}(\mathbb{x}, f, \text{tr}, \mathbf{w}) = 1$ if and only if

$$\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{w}))) < \delta$$

and $\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \in (\pi_q(h_i))$ for all $i \in [\ell]$, where

$$Q_{\mathbf{w}}^{(i)}(\mathbf{Z}) := \sum_{\mathbf{b} \in \{0,1\}^{n'}} \mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{w}) \cdot \text{eq}(\mathbf{b}, \mathbf{Z}) \in R[X].$$

Observe that KState satisfies the three conditions of ??.

We define the extractor $\mathcal{E}_{\text{rbr}}$ as $\mathcal{E}_{\text{rbr}}(\mathbb{x}, f, \text{tr}, \mathbf{w}) = \mathbf{w}$ for all $\mathbb{x}, f, \text{tr}, \mathbf{w}$. Let tr be a transcript for Π_1 where the verifier is about to move, so that either $\text{tr} = (\perp)$ or $\text{tr} = (\perp, (\mathbb{K}, \mathbf{r}), (u_i)_{i \in [\ell]})$. For a transcript $\text{tr} = (\perp)$, we need to bound the following probability, corresponding to the round-by-round knowledge soundness of Round 1 of Π_1 :

$$P_1 := \Pr_{\mathbb{K}, \mathbf{r}} \left[\exists \mathbf{w} \mid \begin{array}{l} \wedge \text{KState}(\mathbb{x}, f, \perp, \mathbf{w}) = 0 \\ \wedge \text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), \mathbf{w}) = 1 \end{array} \right]$$

By our definition of KState , P_1 is the probability that

$$\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \in (\pi_q(h_i)) \text{ for all } i \in [\ell] \text{ and } \Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{w}))) < \delta \quad (49)$$

while at the same time

$$\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{w}) \notin (h_i) \text{ for some } \mathbf{b} \in \{0, 1\}^{n'} \text{ and } i \in [\ell] \quad (50)$$

or $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{w}))) \geq \delta$. Clearly, the latter condition cannot hold if (??) holds; hence we will study the probability that (??) and (??) hold simultaneously. We ultimately want to prove that $P_1 \leq \kappa_1$.

Let E_1 be the event that (??) holds and $\pi_q(\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{w})) \in (\pi_q(h_i))$ for all $\mathbf{b}, i$. By ??,

$$\Pr_{\mathbb{K}, \mathbf{r}}[E_1] \leq \max_{\mathbf{b} \in \{0, 1\}^{n'}, i \in [\ell]} \left\{ \frac{B' \cdot (1 + \max(\deg(\mathbf{p}_{\mathbf{b}}^i(\mathbf{x}, \mathbf{f})) - \deg(h_i) + 1, 0))}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor} \right\},$$

where $\deg(h_i) := +\infty$ if $h_i = 0$; and B' is $\lceil \log(\|\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})\|_{\infty}) \rceil$. We have, with $d_{\min}$ as in ??, [Albert: more detail?](#)

$$\begin{aligned} & \max_{\mathbf{b} \in \{0, 1\}^{n'}, i \in [\ell]} \{ \deg(\mathbf{p}_{\mathbf{b}}^i(\mathbf{x}, \mathbf{f})) - \deg(h_i) \} \\ & \leq \nabla(\mathbf{p}) - d_{\min} = \max_{\mathbf{b} \in \{0, 1\}^{n'}, i \in [\ell]} \{ \deg_X(\mathbf{p}_{\mathbf{b}}^i) + (d - 1) \cdot \max_{t \in [k + \ell]} \deg_{Y_t}(\mathbf{p}_{\mathbf{b}}^i) \} + d - d_{\min}, \\ & \max_{\mathbf{b} \in \{0, 1\}^{n'}, i \in [\ell]} B' = \max_{\mathbf{b} \in \{0, 1\}^{n'}, i \in [\ell]} \lceil \log(\|\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})\|_{\infty}) \rceil = B_{\max}. \end{aligned}$$

Now, let E be the event that both (??) and (??) hold. Then

$$P_1 = \Pr_{\mathbb{K}, \mathbf{r}}[E] \leq \Pr_{\mathbb{K}, \mathbf{r}}[E \mid \neg E_1] + \Pr_{\mathbb{K}, \mathbf{r}}[E_1].$$

We next analyze $\Pr_{\mathbb{K}, \mathbf{r}}[E \mid \neg E_1]$. If E_1 does not occur, then the multilinear polynomial $\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{Z})$ is not the zero polynomial in $\mathbb{K}[X]$. On the other hand, for E to occur, we need $\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) = 0$ in $\mathbb{K}$. Hence, by Schwartz-Zippel lemma, $\Pr_{\mathbb{K}, \mathbf{r}}[E \mid \neg E_1] \leq 1/|\mathbb{K}_{\min}|$. It follows that $P_1 \leq \kappa_1$, as needed.

Finally, we bound the following probability, corresponding to the round-by-round knowledge soundness of the second round of Π_1 .

$$P_2 := \Pr_{(\theta_i)_{i \in [\ell], \alpha}} \left[\exists \mathbf{w} \mid \begin{array}{l} \wedge \text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), (u_i)_{i \in [\ell]}, \mathbf{w}) = 0 \\ \wedge \text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), (u_i)_{i \in [\ell]}, ((\theta_i)_{i \in [\ell]}, \alpha), \mathbf{w}) = 1 \end{array} \right]$$

By definition,

$$\text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), (u_i)_{i \in [\iota]}, ((\theta_i)_{i \in [\iota]}, \alpha), \mathbf{w}) = 1 \Leftrightarrow (\pi(\mathbb{x}), f, \pi(f)) \in \mathcal{R}_{\text{PESAT}}(R_{\leq B}^{\leq d}, \mathbb{K}, \hat{\mathbf{p}}, \mathcal{C})^\delta \quad (51)$$

and verifier does not abort [Albert: It's not completely transparent how verifier accepts or aborts in the definition of RBR KS](#); while

$$\text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), (u_i)_{i \in [\iota]}, \mathbf{w}) = \text{KState}(\mathbb{x}, f, \perp, (\mathbb{K}, \mathbf{r}), \mathbf{w}) = 0$$

if and only if $\Delta(f, \text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\text{bits}(\mathbf{w}))) \geq \delta$ or

$$\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \notin (\pi_q(h_i)) \text{ for some } i \in [\iota]. \quad (52)$$

Let E be the event that (??) and (??) hold. Again, $P_2 = \Pr_{(\theta_i), \alpha}[E]$, since (??) forces $\Delta(f, \text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\text{bits}(\mathbf{w}))) < \delta$.

Now, if E occurs then

$$\sum_{i \in [\iota]} \sum_{\mathbf{b} \in \{0,1\}^{n'}} \theta_i \cdot \pi(\mathbf{p}_{\mathbf{b}}^{(i)})(\pi(\mathbf{x}), \pi(\mathbf{w})) \cdot \text{eq}(\mathbf{b}, \mathbf{r}) = \sum_{i \in [\iota]} \theta_i \cdot \pi(u_i) \in \mathbb{K},$$

and at the same time

$$\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) = \sum_{\mathbf{b} \in \{0,1\}^{n'}} \pi_q(\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})) \cdot \text{eq}(\mathbf{b}, \mathbf{r}) \notin (\pi_q(h_i)) \subseteq \mathbb{K}[X]$$

for some $i \in [\iota]$. Since the verifier does not reject, $u_i \in (\pi_q(h_i))$ for all $i \in [\iota]$. Hence, if E holds, then for some i , we have $\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \neq u_i$. In particular $u'_i := \pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) - u_i \in \mathbb{K}[X]$ is a nonzero polynomial with coefficients in $\mathbb{K}$.

Let E_1 be the event that E holds and $\pi(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) = \pi(u_i)$ for all $i \in [\iota]$ (recall that $\pi = \pi_{q, \alpha}$ acts on $\mathbb{K}[X]$ by replacing X by α). Now, since $u'_i(X)$ is a nonzero polynomial from $\mathbb{K}[X]$, we have that $\pi(u'_i(X)) = u'_i(\alpha) = 0$ with probability at most $\deg(u'_i)/|\mathbb{K}_{\min}|$ by Schwartz-Zippel lemma. It follows that [Albert: Again, I'm not missing a union bound right?](#)

$$\begin{aligned} \Pr_{(\theta_i)_{i \in [\iota]}, \alpha}[E_1] &\leq \max_{i \in [\iota]} \frac{\deg(\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) - u_i)}{|\mathbb{K}_{\min}|} \leq \max_{i \in [\iota]} \frac{\max\{\deg(\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r})), d\}}{|\mathbb{K}_{\min}|} \\ &\leq \max_{\mathbf{b} \in \{0,1\}^{n'}, i \in [\iota]} \frac{\max\{\deg(\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})), d\}}{|\mathbb{K}_{\min}|} \leq \frac{\nabla(\mathbf{p})}{|\mathbb{K}_{\min}|}, \end{aligned}$$

where we used that $\deg_X(u_i) < d$; that $\pi_q(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \in \mathbb{K}[X]$ is a $\mathbb{K}$ -linear combination of the polynomials $\pi_q(\mathbf{p}_{\mathbf{b}}^{(i)}(\mathbf{x}, \mathbf{f})) \in \mathbb{K}[X]$ [Albert: More details about the last inequality?](#). Now,

$$\Pr[E] \leq \Pr[E \mid \neg E_1] + \Pr[E_1].$$

If E_1 does not hold, but E holds, then $\pi(Q_{\mathbf{w}}^{(i)})(\mathbf{r}) \neq \pi(u_i)$ for some $i \in [\iota]$. Then the polynomial on variables $U_1, \dots, U_\iota$,

$$Q'(U_1, \dots, U_\ell) := \sum_{i \in [\ell]} U_i \cdot \pi(Q_{\mathbf{w}}^{(i)}) - \sum_{i \in [\ell]} U_i \cdot \pi(u_i)$$

is a nonzero multilinear polynomial with coefficients in $\mathbb{K}$. Now, $[E \mid \neg E_1]$ holds if and only if $Q'(\theta_1, \dots, \theta_\ell) = 0 \in \mathbb{K}$. By Schwartz-Zippel lemma, this probability is at most $1/|\mathbb{K}_{\min}|$. It follows that $P_2 \leq \kappa_2$, as required. $\square$

This completes the proof of ??.

A.2.1 Proof of ??

We prove ??. Observe that the prover and verifier of Π_{123} proceed in exactly the same way as the prover and verifier of ?? after the prover sends f in Item 1 of Round 1 (except that in Π_{123} the prover sends an initial empty message).

If Π_{123} is perfectly complete, then clearly ?? is as well [Albert: more details?](#). Further, transcripts between Π_{123} and ?? are the same, except that the prover in Π_{123} sends an empty first message when the prover in ?? sends the encoding f . More precisely, a (partial) nonempty transcript of ?? has the form (f, tr') for some tr' and some first prover message f , and the “mirror” transcript of Π_{123} is $(\perp, \text{tr}')$, with f being now part of the instance oracle.

From any knowledge state function with relaxation $\text{KState}_{\Pi_{123}}$ for Π_{123} , we define a state function KState for ?? as follows: let tr be a partial or full transcript of ??. If tr is not empty, write $\text{tr} = (f, \text{tr}')$ as above. Then we define

$$\text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \mathbf{w}) = \text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \mathbf{w}).$$

If tr is empty, we define

$$\text{KState}(\mathbb{x}, \perp, \mathbf{w}) = 1 \Leftrightarrow (\mathbb{x}, \perp, \mathbf{w}) \in \mathcal{R}_{\text{UPESAT}}.$$

Hence KState satisfies Condition 1 of ?? (Empty transcript). We now show it satisfies Conditions 2 (Prover moves) and 3 (Full transcript), assuming $\text{KState}_{\Pi_{123}}$ satisfies ??.

Let tr be a transcript of ?? where the prover is about to move and $\text{KState}(\mathbb{x}, \perp, \text{tr}, \mathbf{w}) = 0$. If tr is nonempty, then $\text{tr} = (f, \text{tr}')$ as above, and for any next prover message m of ??, $\text{KState}(\mathbb{x}, \perp, (\text{tr}, m), \mathbf{w}) = \text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}', m), \mathbf{w}) = 0$ because, since $\text{KState}_{\Pi_{123}}$ satisfies ??, $\text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \mathbf{w}) = \text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \mathbf{w}) = 0$. If tr is empty, then the next prover's message is f . If $\text{KState}(\mathbb{x}, \perp, \mathbf{w}) = 0$, then $(\mathbb{x}, \perp, \mathbf{w}) \notin \mathcal{R}_{\text{UPESAT}}$. Now $\text{KState}(\mathbb{x}, \perp, f, \mathbf{w}) = \text{KState}_{\Pi_{123}}(\mathbb{x}, f, \perp, \mathbf{w})$. By the “Prover moves” condition of ??, if $\text{KState}_{\Pi_{123}}(\mathbb{x}, f, \mathbf{w}) = 0$ (empty transcript where the prover is about to move), then $\text{KState}_{\Pi_{123}}(\mathbb{x}, f, \perp, \mathbf{w}) = 0$ ($\perp$ is always the first message of the prover in Π_{123}). Now $\text{KState}_{\Pi_{123}}(\mathbb{x}, f, \mathbf{w}) = 0$ indeed because by the condition “Empty transcript” of ??, $\text{KState}_{\Pi_{123}}(\mathbb{x}, f, \mathbf{w}) = 0$ if and only if $(\mathbb{x}, f, \mathbf{w}) \notin \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta$, and since $(\mathbb{x}, \perp, \mathbf{w}) \notin \mathcal{R}_{\text{UPESAT}}$, it follows that $(\mathbb{x}, f, \mathbf{w}) \notin \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta$. Hence $\text{KState}(\mathbb{x}, \perp, f, \mathbf{w}) = 0$ and KState satisfies the condition “Prover moves” of ??.

Regarding the third condition, “Full transcript”, let (f, tr') be a full transcript for ??. Then for all $\mathbb{x}, \mathbf{w}$,

$$\text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \mathbf{w}) = \text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \mathbf{w}) = 1 \Leftrightarrow V_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}')) = 1,$$

where $V_{\Pi_{123}}$ is the verifier of Π_{123} .

Notice that the verifier of ?? given $(\mathbb{x}, \perp, (f, \text{tr}'))$ makes exactly the same checks as $V_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'))$. Hence the verifier of ?? accepts $(\mathbb{x}, \perp, (f, \text{tr}'))$ if and only if $\text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \mathbf{w}) = 1$, as needed. This shows that KState satisfies the three conditions of ??.

Now assume Π_{123} has round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta, \perp)$ with errors $(\kappa'_j)_{j \in [2+t_{\mathbb{K}}, 1+t_{\text{red}}+t_{\text{iopp}}]}$, with respect to the state function $\text{KState}_{\Pi_{123}}$. We claim that then ?? has round-by-round knowledge soundness without relaxation, with errors $(\kappa'_j)_{j \in [2+t_{\mathbb{K}}, 1+t_{\text{red}}+t_{\text{iopp}}]}$, with respect to the state function KState .

We start by defining an extractor $\mathcal{E}$ for ?? simply as

$$\mathcal{E}(\mathbb{x}, \perp, (f, \text{tr}'), \mathbf{w}) = \mathcal{E}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \mathbf{w})$$

for every (partial) transcript (f, tr') where the verifier is about to move. We need not define an extractor for, say $(\mathbb{x}, \perp, \mathbf{w})$, since the verifier is not about to move after an empty transcript $\perp$ (the prover is about to send f).

Now, let $\text{tr} = (f, \text{tr}')$ be a (partial) transcript of ?? where the verifier is about to send its j -th message ρ . Then we have, using the definitions of KState , $\mathcal{E}$ and our assumption on the round-by-round knowledge soundness of Π_{123} ,

$$\begin{aligned} & \Pr_{\rho} \left[\exists \mathbf{w} \left| \begin{array}{l} \wedge \text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \mathcal{E}(\mathbb{x}, \perp, (f, \text{tr}'), \rho, \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbb{x}, \perp, (f, \text{tr}'), \rho, \mathbf{w}) = 1 \end{array} \right. \right] \\ &= \Pr_{\rho} \left[\exists \mathbf{w} \left| \begin{array}{l} \wedge \text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \mathcal{E}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \rho, \mathbf{w})) = 0 \\ \wedge \text{KState}_{\Pi_{123}}(\mathbb{x}, f, (\perp, \text{tr}'), \rho, \mathbf{w}) = 1 \end{array} \right. \right] \\ &\leq \kappa_j(\mathbb{x}, f)' \leq \max_{f \text{ s.t. } (\mathbb{x}, f) \notin \mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}, \mathbf{p}, \mathbf{R}, \mathcal{C})^\delta} \kappa_j(\mathbb{x}) = \kappa_j(\mathbb{x}). \end{aligned}$$

This concludes the proof of ??.

B Ring switching via Galois orbits

Albert: Context: how does it relate to other references?

Let $b_1, \dots, b_m \in \mathbb{F}_2^{2^n}$ **Albert:** Use boldface. Using an $\mathbb{F}_2$ -basis $\beta_1, \dots, \beta_m$ of $\mathbb{F}_{2^m}$ we pack m inner products

$$\langle b_i, \mathbf{v} \rangle = \sum_{j \in \{0,1\}^n} b_{i,j} \cdot \mathbf{v}_j, \quad i = 1, \dots, m,$$

coefficient-wise into a single inner product

$$\begin{aligned} b &:= b_1 \cdot \beta_1 + \dots b_m \cdot \beta_m \in \mathbb{F}_{2^m}^{2^n} \\ \langle b, \mathbf{v} \rangle &= \langle b_1, \mathbf{v} \rangle \cdot \beta_1 + \dots + \langle b_m, \mathbf{v} \rangle \cdot \beta_m \\ &= \sum_{j \in \{0,1\}^n} (b_{1,j} \cdot \beta_1 + \dots + b_{m,j} \cdot \beta_m) \cdot \mathbf{v}_j. \end{aligned}$$

The Frobenius automorphism $\phi(z) = z^2$ keeps $\mathbb{F}_2 \subset \mathbb{F}_{2^m}$ fixed, and has no other fixed points. We extract the value of $\langle b_i, \mathbf{v} \rangle$ from the values of $\langle b, \mathbf{v} \rangle$ along the Galois orbit

$$\mathbf{v}, \mathbf{v}^2, \mathbf{v}^{2^2}, \dots, \mathbf{v}^{2^{m-1}},$$

as follows, where by $\mathbf{v}^t$ we mean that each component of $\mathbf{v}$ is raised to the power t . The Moore matrix

$$M = \begin{pmatrix} \beta_1 & \beta_2 & \dots & \beta_m \\ \beta_1^2 & \beta_2^2 & \dots & \beta_m^2 \\ \beta_1^{2^2} & \beta_2^{2^2} & \dots & \beta_m^{2^2} \\ \vdots & & & \\ \beta_1^{2^{m-1}} & \beta_2^{2^{m-1}} & \dots & \beta_m^{2^{m-1}} \end{pmatrix}$$

consists of the Galois orbits of the basis vectors, and is invertible. (See below.) Using that the component polynomials are over $\mathbb{F}_2$, we see that

$$\phi^k \left(\langle b, \phi^{-k}(\mathbf{v}) \rangle \right) = \langle b_1, \mathbf{v} \rangle \cdot \phi^k \beta_1 + \dots + \langle b_m, \mathbf{v} \rangle \cdot \phi^k \beta_m,$$

for any k . To this end, observe that

$$M \cdot \begin{pmatrix} \langle b_1, \mathbf{v} \rangle \\ \langle b_2, \mathbf{v} \rangle \\ \langle b_3, \mathbf{v} \rangle \\ \vdots \\ \langle b_m, \mathbf{v} \rangle \end{pmatrix} = \begin{pmatrix} \langle b, \mathbf{v} \rangle \\ \phi \left(\langle b, \phi^{-1}(\mathbf{v}) \rangle \right) \\ \phi^2 \left(\langle b, \phi^{-2}(\mathbf{v}) \rangle \right) \\ \vdots \\ \phi^{m-1} \left(\langle b, \phi^{-(m-1)}(\mathbf{v}) \rangle \right) \end{pmatrix}.$$

Multiplying both sides by M^{-1} (M is a Moore matrix and hence invertible because the $\beta_1, \dots, \beta_m$ are invertible) we obtain an expression of the values $\langle b_i, \mathbf{v} \rangle$ as a linear expression of the values $\phi^k \left(\langle b, \phi^{-k}(\mathbf{v}) \rangle \right)$.

Now one can prove the inner product claims $\langle b, \phi^{-i}(\mathbf{v}) \rangle$ for all i by first batching them all into one single claim on the value of $\langle b, \mathbf{u} \rangle$, and then proving this later claim normally, since we had committed to the vector b initially. The batching step introduces the error term

$$1/|\mathbb{F}_{2^\nu}|.$$

Ulrich: Invertibility of M follows from that of M^t : any linear functional $(c_1, \dots, c_m)$ in the kernel of M^t gives rise to the linearizable polynomial

$$p(X) = \sum_{i=0}^{m-1} c_i \cdot X^{2^i}$$

which has $\beta_1, \dots, \beta_m$, and hence the whole space F as roots. By dimension, $p(X)$ must be trivial, meaning that all its coefficients are zero.

C Bilinear Embeddings

Given a field $\mathbb{F}$ and a degree- d extension $\mathbb{E}$. Our goal is to commit to a vector $\mathbf{w} \in \mathbb{F}^{nd}$ so that we can later use it to prove an inner product claim of the form $\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} = h$ for some $\mathbf{a} \in \mathbb{F}^{nd}$ and $h \in \mathbb{F}$. However, we only have an efficient inner product commitment scheme to commit to vectors $\tilde{\mathbf{w}} \in \mathbb{E}^n$ and open as $\langle \tilde{\mathbf{w}}, \tilde{\mathbf{a}} \rangle_{\mathbb{E}} = \tilde{h}$ for some $\tilde{\mathbf{a}} \in \mathbb{E}^n$ and $\tilde{h} \in \mathbb{E}$.

To do this, we seek a triple of maps (W, A, H) that satisfy:

$$\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} = H \left(\langle W(\mathbf{w}), A(\mathbf{a}) \rangle_{\mathbb{E}} \right).$$

C.1 Background

C.1.1 Perfect pairings

In the following let $\mathbb{E}/\mathbb{F}$ be a finite field extension of degree d . If we disregard multiplication in $\mathbb{E}$, then $\mathbb{E}$ is a d -dimensional vector space over $\mathbb{F}$, hence isomorphic to $\mathbb{F}^d$. Let $\mathbf{e}_1, \dots, \mathbf{e}_d$ denote the standard basis of $\mathbb{F}^d$. Recall that a function $\mu : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{F}$ is a *bilinear form* if it is linear in each argument separately. It is *symmetric* if $\mu(x, y) = \mu(y, x)$ for all $x, y \in \mathbb{E}$. For each $x \in \mathbb{E}$, the partial application in the first argument defines a linear form

$$\mu_x : \mathbb{E} \rightarrow \mathbb{F} \quad y \mapsto \mu(x, y).$$

Thus $\mu_x \in \mathbb{E}^*$ where $\mathbb{E}^* = \text{Hom}_{\mathbb{F}}(\mathbb{E}, \mathbb{F})$ denotes the dual vector space. The bilinear form therefore induces an $\mathbb{F}$ -linear map

$$\Phi_\mu : \mathbb{E} \rightarrow \mathbb{E}^* \quad x \mapsto \mu_x.$$

We call μ *nondegenerate*, or equivalently a *perfect pairing*, if Φ_μ is an isomorphism. Or equivalently for finite spaces, [Albert: add a comment that injectivity implies surjectivity when the vector spaces have finite dimension?](#)

$$\mu(x, y) = 0 \text{ for all } y \in \mathbb{E} \implies x = 0.$$

Since μ is a perfect pairing, every basis $(\beta_1, \dots, \beta_d)$ of $\mathbb{E}$ has a unique dual basis $(\beta_1^\vee, \dots, \beta_d^\vee)$ satisfying $\mu(\beta_i, \beta_j^\vee) = \delta_{ij}$.

John: It would be good to explain how you would typically derive $\beta_j^\vee$

C.1.2 Succinct vectors

Given a field $\mathbb{F}$, a vector $\mathbf{a} \in \mathbb{F}^n$ for $n = 2^k$ is called *succinct* if it admits a tensor decomposition

$$\begin{bmatrix} m_{00} \\ m_{01} \end{bmatrix} \otimes \begin{bmatrix} m_{10} \\ m_{11} \end{bmatrix} \otimes \dots \otimes \begin{bmatrix} m_{(k-1)0} \\ m_{(k-1)1} \end{bmatrix} \quad (53)$$

where typically $m_{ij} \in \mathbb{F}$. The multilinear extension of $\mathbf{a}$, viewed as a function on $\{0, 1\}^k$, then becomes

$$\tilde{a}(\mathbf{r}) = \prod_{i=0}^{k-1} ((1 - r_i)m_{i0} + r_i m_{i1}). \quad (54)$$

which can be evaluated in $O(k)$ field operations.

More generally, the coefficients m_{ij} need not themselves be field elements. It suffices that they belong to an associative $\mathbb{F}$ -algebra in which every complete product $m_{0b_0}m_{1b_1} \dots m_{(k-1)b_{k-1}}$ evaluates to an element of $\mathbb{F}$. Consequently, every vector entry and every value of the multilinear extension remains an element of $\mathbb{F}$. As the m_{ij} no longer need to be commutative we must therefore also fix a permutation of the variables $\mathbf{r}$.

Since every finite-dimensional associative algebra admits a matrix representation, we may, without loss of generality, assume that the objects m_{ij} are matrices. In fact, it suffices to take $m_{ij} \in \mathbb{F}^{w_i \times w_{i+1}}$ with boundary dimensions $w_0 = w_k = 1$. We call $\max w_i$ the *width* of the representation, which is known as a *matrix branching program* and is closely related to matrix product states and weighted finite automata.

The canonical example of a succinct vector is the multilinear evaluation vector $a_i = \text{eq}(\mathbf{r}, \text{bits}(i))$ obtained by taking $m_{i0} = 1 - r_i$ and $m_{i1} = r_i$. Another classical example is the geometric progression $(1, x, x^2, \dots)$ obtained by taking $m_{i0} = 1$ and $m_{i1} = x^{2^i}$. An illustrative non-scalar example is the counting vector $(0, 1, 2, \dots)$ represented by

$$m_{0b} = \begin{bmatrix} 2 & b \end{bmatrix} \quad m_{ib} = \begin{bmatrix} 2 & b \\ 0 & 1 \end{bmatrix} \quad m_{(k-1)b} = \begin{bmatrix} b \\ 1 \end{bmatrix}.$$

C.2 Structure

We are now equipped to prove a theorem on the structure of embeddings. Under the assumption that the output map is $\mathbb{F}$ -linear, we show that for $n = 1$ the space of possible embeddings is very structured.

Theorem C.1 (Structure of bilinear embeddings). *Let $W, A : \mathbb{F}^d \rightarrow \mathbb{E}, H : \mathbb{E} \rightarrow \mathbb{F}$ be maps such that for all $\mathbf{w}, \mathbf{a} \in \mathbb{F}^d$,*

$$\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} = H(W(\mathbf{w})A(\mathbf{a})).$$

Suppose that H is $\mathbb{F}$ -linear. Then:

1. *W and A are $\mathbb{F}$ -linear isomorphisms,*
2. *$\mu_H(x, y) = H(xy)$ is a perfect pairing,*
3. *$(W(\mathbf{e}_1), \dots, W(\mathbf{e}_d))$ and $(A(\mathbf{e}_1), \dots, A(\mathbf{e}_d))$ are dual bases with respect to μ_H .*

Conversely, if W and A , and H are $\mathbb{F}$ -linear, μ_H is a perfect pairing and the images of the standard basis under W and A are dual bases with respect to μ_H , then the identity

$$\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} = H(W(\mathbf{w})A(\mathbf{a})).$$

holds for all $\mathbf{w}, \mathbf{a} \in \mathbb{F}^d$.

Proof of the forward direction. First, observe that $\mu_H(W(\mathbf{e}_i), A(\mathbf{e}_j)) = \langle \mathbf{e}_i, \mathbf{e}_j \rangle_{\mathbb{F}} = \delta_{ij}$. Taking $i = j$ shows H is nonzero. Since H is $\mathbb{F}$ -linear, μ_H is a symmetric bilinear form, we show that it is nondegenerate. Let $x \in \mathbb{E}$ be nonzero. Since multiplication by x is a bijection of $\mathbb{E}$ and H is nonzero, there exists $y \in \mathbb{E}$ such that $\mu_H(x, y) = H(xy) \neq 0$. Thus μ_H is nondegenerate, and hence a perfect pairing.

Next, we show that W and A are bijective. Suppose $W(\mathbf{w}_1) = W(\mathbf{w}_2)$. Then for every $\mathbf{a} \in \mathbb{F}^d$,

$$\langle \mathbf{w}_1 - \mathbf{w}_2, \mathbf{a} \rangle_{\mathbb{F}} = \mu_H(W(\mathbf{w}_1) - W(\mathbf{w}_2), A(\mathbf{a})) = 0.$$

By nondegeneracy of the standard inner product on $\mathbb{F}^d$, this implies $\mathbf{w}_1 = \mathbf{w}_2$. Hence W is injective. Since $\mathbb{F}^d$ and $\mathbb{E}$ have the same finite cardinality, W is bijective. The same argument shows that A is bijective.

We now prove that W is $\mathbb{F}$ -linear. For $\mathbf{w}_1, \mathbf{w}_2, \mathbf{a} \in \mathbb{F}^d$, linearity of H gives

$$\begin{aligned} & \mu_H(W(\mathbf{w}_1 + \mathbf{w}_2) - W(\mathbf{w}_1) - W(\mathbf{w}_2), A(\mathbf{a})) \\ &= \langle \mathbf{w}_1 + \mathbf{w}_2, \mathbf{a} \rangle_{\mathbb{F}} - \langle \mathbf{w}_1, \mathbf{a} \rangle_{\mathbb{F}} - \langle \mathbf{w}_2, \mathbf{a} \rangle_{\mathbb{F}} \\ &= 0. \end{aligned}$$

Since A is surjective, the second argument ranges over all of $\mathbb{E}$. Nondegeneracy of μ_H therefore implies $W(\mathbf{w}_1 + \mathbf{w}_2) = W(\mathbf{w}_1) + W(\mathbf{w}_2)$. Similarly, for $\lambda \in \mathbb{F}$,

$$\begin{aligned} & \mu_H(W(\lambda \mathbf{w}) - \lambda W(\mathbf{w}), A(\mathbf{a})) \\ &= \langle \lambda \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} - \lambda \langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}} \\ &= 0. \end{aligned}$$

Again using surjectivity of A and nondegeneracy of μ_H , we obtain $W(\lambda \mathbf{w}) = \lambda W(\mathbf{w})$. Hence W is $\mathbb{F}$ -linear. By symmetry, A is also $\mathbb{F}$ -linear. Since they are also bijective, W and A are linear isomorphisms. From this it follows that $(W(\mathbf{e}_1), \dots, W(\mathbf{e}_d))$ and $(A(\mathbf{e}_1), \dots, A(\mathbf{e}_d))$ are bases of $\mathbb{E}$. Finally, from $\mu_H(W(\mathbf{e}_i), A(\mathbf{e}_j)) = \delta_{ij}$ it follows they are dual with respect to μ_H . $\square$

Proof of the converse direction. Can be directly shown using a basis expansion. Write $\mathbf{w} = \sum_{i=1}^d w_i \mathbf{e}_i$ and $\mathbf{a} = \sum_{j=1}^d a_j \mathbf{e}_j$. Since W and A are $\mathbb{F}$ -linear, $W(\mathbf{w}) = \sum_{i=1}^d w_i W(\mathbf{e}_i)$ and $A(\mathbf{a}) = \sum_{j=1}^d a_j A(\mathbf{e}_j)$. Therefore,

$$\begin{aligned} H(W(\mathbf{w})A(\mathbf{a})) &= \mu_H(W(\mathbf{w}), A(\mathbf{a})) \\ &= \sum_{i,j} w_i a_j \mu_H(W(\mathbf{e}_i), A(\mathbf{e}_j)) \\ &= \sum_{i,j} w_i a_j \delta_{ij} \\ &= \sum_i w_i a_i \\ &= \langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}}. \end{aligned}$$

$\square$

The $n = 1$ case extends naturally to $n > 1$ by doing blockwise packing, i.e. we embed using $W' = I_n \otimes W$ and $A' = I_n \otimes A$ and keep the same H . Let $\mathbf{w} = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ and $\mathbf{a} = (\mathbf{a}_1, \dots, \mathbf{a}_n)$ then

$$\begin{aligned} H(\langle W'(\mathbf{w}), A'(\mathbf{a}) \rangle_{\mathbb{E}}) &= H\left(\sum_{i=1}^n W(\mathbf{w}_i)A(\mathbf{a}_i)\right) \\ &= \sum_{i=1}^n H(W(\mathbf{w}_i)A(\mathbf{a}_i)) \\ &= \sum_{i=1}^n \langle \mathbf{w}_i, \mathbf{a}_i \rangle_{\mathbb{F}} \\ &= \langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{F}}. \end{aligned}$$

Going forward, we use W and A also for their blockwise extensions when the meaning is clear. We do not consider embeddings that mix coordinates across different blocks of size d , as such embeddings are unlikely to be useful in practice.

We have shown that every $n = 1$ embedding with a linear output map is determined by a choice of a nonzero linear map H and linear isomorphism W . The map W then determines the first basis $\beta_i = W(\mathbf{e}_i)$ and the map A is then forced to map to the μ_H -dual basis.

The linearity assumption on the output map is necessary. Indeed, composing a linear embedding with a multiplicative bijection $g : \mathbb{E} \rightarrow \mathbb{E}$ generally produces an embedding that is nonlinear.

Whether every nonlinear embedding arises from such a multiplicative reparameterization is a separate question, which we do not need here.

An important property of bilinear embeddings is that they preserve succinctness if the block size aligns with the tensor decomposition:

Theorem C.2 (Preservation of succinctness). *Given $d = 2^k$, $n = 2^l$ and a $\mathbb{F}$ -linear blockwise map $A = I_n \otimes A_0$ for some isomorphism $A_0 : \mathbb{F}^d \rightarrow \mathbb{E}$, then if $\mathbf{a}$ is succinct then so is $A(\mathbf{a})$. In particular if*

$$\mathbf{a} = \bigotimes_{i=0}^{k+l-1} \begin{bmatrix} m_{i0} \\ m_{i1} \end{bmatrix}$$

where the m_{ij} can be scalars or $w_i \times w_{i+1}$ matrices, then

$$A(\mathbf{a}) = \left(\bigotimes_{i=0}^{l-1} \begin{bmatrix} m_{i0} \\ m_{i1} \end{bmatrix} \right) \cdot (I_{w_l} \otimes A_0) \left(\bigotimes_{i=l}^{l+k-1} \begin{bmatrix} m_{i0} \\ m_{i1} \end{bmatrix} \right).$$

where the second factor is column vector in $\mathbb{E}^{w_l}$ which can be absorbed into the matrices $m_{(l-1)b}$, bringing the representation back into canonical form.

C.3 Constructions

The set of nonzero linear output maps H is $\mathbb{E}^* \setminus \{0\}$ which is isomorphic to $\mathbb{E}^\times$. The set of linear isomorphism W is isomorphic to $\text{GL}_d(\mathbb{F})$, hence the total number of bilinear embeddings is $|\mathbb{E}^\times| \cdot |\text{GL}_d(\mathbb{F})|$. We'll now look at specific choices and their properties.

C.3.1 Trace embedding

Let $q = |\mathbb{F}|$, the q -power Frobenius map is $\varphi : \mathbb{E} \rightarrow \mathbb{E}$ given by $x \mapsto x^q$. It is an $\mathbb{F}$ -linear field automorphism of order d , and its fixed field is precisely $\mathbb{F}$:

$$\mathbb{E}^\varphi = \{x \in \mathbb{E} : \varphi(x) = x\} = \mathbb{F}.$$

The field trace is the $\mathbb{F}$ -linear map $\text{Tr}_{\mathbb{E}/\mathbb{F}} : \mathbb{E} \rightarrow \mathbb{F}$ defined by

$$\text{Tr}_{\mathbb{E}/\mathbb{F}}(x) = \sum_{i=0}^{d-1} \varphi^i(x) = \sum_{i=0}^{d-1} x^{q^i}.$$

To see that this sum is indeed in $\mathbb{F} \subseteq \mathbb{E}$ observe the sum is fixed by φ , since

$$\varphi \left(\sum_{i=0}^{d-1} \varphi^i(x) \right) = \sum_{i=1}^d \varphi^i(x) = \sum_{i=0}^{d-1} \varphi^i(x),$$

where $\varphi^d(x) = \varphi^0(x)$. Hence the trace lies in the fixed field $\mathbb{F}$.

This makes $\text{Tr}_{\mathbb{E}/\mathbb{F}}$ a natural choice of output map H . More generally, every $\mathbb{F}$ -linear map $H : \mathbb{E} \rightarrow \mathbb{F}$ is uniquely of the form $H(x) = \text{Tr}_{\mathbb{E}/\mathbb{F}}(\alpha x)$ for some $\alpha \in \mathbb{E}$. Consequently, the nonzero output maps are parameterized by $\mathbb{E}^\times$.

The Galois orbit embedding chooses the basis $\beta_i = \varphi^i(\gamma)$ which has dual $\beta_i^\vee = \varphi^i(\gamma^\vee)$.

Ulrich: This sounds to me very much related to Galois Fröbenius on an inner product with a general covector. You would end up with the multi-claim of the inner products of the packed poly with the Galois orbit of the covector. This is how I view it: Given $\mathbf{p}_1, \dots, \mathbf{p}_d \in \mathbb{F}^n$, and $\mathbf{q}_1, \dots, \mathbf{q}_d \in \mathbb{F}^n$, we are interested in

$$\sum_{k=1}^d \langle \mathbf{p}_k, \mathbf{q}_k \rangle,$$

right?

The packing map sends the coeff vectors $\mathbf{p}_1, \dots, \mathbf{p}_d$ to $\mathbf{p} = \beta_1 \cdot \mathbf{p}_1 + \dots + \beta_d \cdot \mathbf{p}_d$, where $\beta_1, \dots, \beta_d$ is an $\mathbb{F}$ -basis of $\mathbb{E}$. Choose the element a_k in $\mathbb{E}$ which extracts the k -th coordinate

$$\mathbf{p}_k = \Phi_k(\mathbf{p}) = \text{Tr}_{\mathbb{E}/\mathbb{F}}(a_k \cdot \mathbf{p}).$$

Then

$$\begin{aligned} \langle \mathbf{p}_k, \mathbf{q}_k \rangle &= \left\langle \sum_{i=0}^d \varphi^i(a_k \cdot \mathbf{p}), \mathbf{q}_k \right\rangle \\ &= \sum_{i=0}^d \varphi^i(a_k) \cdot \varphi^i(\langle \mathbf{p}, \mathbf{q}_k \rangle), \end{aligned}$$

since $\mathbf{q}_k$ is over $\mathbb{F}$. This reduces our inner product claim to that of the packed poly $\mathbf{p}$ with the unpacked polys $\mathbf{q}_1, \dots, \mathbf{q}_d$.

But does that match our use case? We are rather interested in $\langle \mathbf{p}_k, \mathbf{v} \rangle$ for some $\mathbf{v} \in \mathbb{E}^n$. In this case,

$$\begin{aligned} \langle \mathbf{p}_k, \mathbf{v} \rangle &= \left\langle \sum_{i=0}^d \varphi^i(a_k \cdot \mathbf{p}), \mathbf{q}_k \right\rangle \\ &= \sum_{i=0}^d \varphi^i(a_k) \cdot \varphi^i(\langle \mathbf{p}, \varphi^{-i}(\mathbf{v}) \rangle), \end{aligned}$$

and we end up with inner products of the packed poly with the Galois orbit of $\mathbf{v}$.

C.3.2 Coefficient projection

In the coefficient projection we assume a concrete representation of $\mathbb{E}$ as $\mathbb{F}[X]/(f(X))$ for some monic degree d polynomial f irreducible over $\mathbb{F}$. We assume f is chosen for implementation efficiency: it is sparse with mostly low order terms. Further we represent members of $\mathbb{E}$ in the monomial basis $(1, X, X^2, \dots, X^{d-1})$. We can then represent W and A by invertible matrices in $\mathbb{F}^{d \times d}$ and H by a vector in $\mathbb{F}^d$. For implementation efficiency we seek an embedding with sparse W and A . It turns out taking the most efficient choices for H and W also gives efficient A :

Theorem C.3. *Write $f(X) = f_d X^d + f_{d-1} X^{d-1} + \dots + f_1 X + f_0$ so that $f_d = 1$ and $f_0 \neq 0$. Take $W = I_d$ and $H = (1, 0, \dots, 0)$. Then the dual-basis embedding A is given by*

$$A_{ij} = \begin{cases} 1, & i = j = 0, \\ -f_{i+j}/f_0, & i, j \geq 1 \text{ and } i + j \leq d, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, A is a bordered upper anti-triangular Hankel matrix. For $d = 5$ it looks like

$$A = \frac{1}{f_0} \begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ 0 & -f_2 & -f_3 & -f_4 & -1 \\ 0 & -f_3 & -f_4 & -1 & 0 \\ 0 & -f_4 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

Notably f_1 does not appear and for $k > 1$ the f_k coefficient appears $k - 1$ times. This perfectly matches the usual implementation preference for sparse reduction polynomials with mostly low-order terms.

When doing inner product openings, the $1/f_0$ factor can be moved out of the inner product.

For the 127-bit field $\mathbb{F}_2[X]/(X^{127} + X + 1)$ the A map is simply a reversal of the $i > 0$ coordinates. For the GHASH field $\mathbb{F}_2[X]/(X^{128} + X^7 + X^2 + X + 1)$ the map additionally needs seven XOR corrections.

C.4 Extension openings

Assume now that we wish to open a base-field witness $\mathbf{w} \in \mathbb{F}^{nd}$ against extension-field weights $\mathbf{a} \in \mathbb{F}^{nd}$, proving the claim $\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{E}} = h$ for some $h \in \mathbb{E}$. We present a one-round batching protocol and a $\log d$ -round folding protocol.

C.4.1 Batching protocol

Given a bilinear embedding (W, A, H) and a basis $(\beta_1, \dots, \beta_d)$ for $\mathbb{E}/\mathbb{F}$, we reduce the extension-field claim by decomposing into basis coordinates and batching the resulting claims with a random linear combination.

Construction C.4 (Batching protocol for extension opening). Given the claim $\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{E}} = h$, define $\mathbf{a}_i \in \mathbb{F}^{nd}$ such that

$$\mathbf{a} = \sum_{i=1}^d \mathbf{a}_i \beta_i.$$

The parties proceed as follows.

1. The prover sends $h_i = \langle W(\mathbf{w}), A(\mathbf{a}_i) \rangle_{\mathbb{E}}$ for $i = 1, \dots, d$.
2. The verifier checks $h = \sum_{i=1}^d H(h_i) \beta_i$.
3. The verifier samples an ϵ -zero-evader $\boldsymbol{\rho} \leftarrow_{\$} \mathbb{E}^d$.
4. The parties reduce the claim to $\langle W(\mathbf{w}), \mathbf{a}' \rangle_{\mathbb{E}} = h'$ where

$$h' = \sum_{i=1}^d \rho_i h_i, \quad \mathbf{a}' = \sum_{i=1}^d \rho_i A(\mathbf{a}_i).$$

This interactive oracle reduction is always complete, has round-by-round soundness error ϵ , and proof size $d \log |\mathbb{E}| = d^2 \log |\mathbb{F}|$.

Theorem C.5 (Succinctness of coordinate decomposition). *Given a succinct vector $\mathbf{a} \in \mathbb{E}^{nd}$ for $d = 2^k$ and a basis (β_i) of $\mathbb{E}/\mathbb{F}$ then the β -coordinate decomposed vectors are also succinct. In particular if*

$$\mathbf{a} = \bigotimes_{i=0}^{k+l-1} \begin{bmatrix} m_{i0} \\ m_{i1} \end{bmatrix}$$

where the m_{ij} can be scalars or $w_i \times w_{i+1}$ matrices, then

$$\mathbf{a}_i = \mathbf{u}_i \cdot \left(\bigotimes_{j=0}^{k+l-1} \begin{bmatrix} \Phi(m_{j0}) \\ \Phi(m_{j1}) \end{bmatrix} \right) \cdot \mathbf{v}$$

where $\Phi : \mathbb{E} \rightarrow \mathbb{F}^{d \times d}$ is the map that represents an $\mathbb{E}$ -element m by the matrix representing multiplication by m in basis (β_i) , $\mathbf{u}_i$ is the one-hot vector with $u_{ij} = \delta_{ij}$, and $\mathbf{v}$ is the base (β_i) representation of $1 \in \mathbb{E}$. Φ is applied elementwise to $w_i \times w_{i+1}$ matrices over $\mathbb{E}$ to produce $dw_i \times dw_{i+1}$ block matrices over $\mathbb{F}$. Absorbing the outer vectors in the matrices brings the representation back into canonical form, with the width increased by at most a factor d .

For practical implementations it makes sense to evaluate the branching program over $\mathbb{E}$ directly and then extract the coordinates in the basis (β_i) . This is equivalent to omitting the output selector $\mathbf{t}_i$ to keep the output dimension $w_0 = d$.

If $A = I_n \otimes A_0$ is blockwise we can also apply theorem ?? and compose it with the combination weights $\boldsymbol{\rho} \in \mathbb{E}^d$ to get a succinct representation for $\mathbf{a}'$:

$$\mathbf{a}' = \boldsymbol{\rho} \cdot \left(\bigotimes_{i=0}^{l-1} \begin{bmatrix} \Phi(m_{i0}) \\ \Phi(m_{i1}) \end{bmatrix} \right) \cdot (I_{dw_l} \otimes A_0) \left(\bigotimes_{i=l}^{l+k-1} \begin{bmatrix} \Phi(m_{i0}) \\ \Phi(m_{i1}) \end{bmatrix} \right) \cdot \mathbf{v}. \quad (55)$$

The map Φ is an algebra embedding, so replacing each scalar $m_{ij} \in \mathbb{E}$ by $\Phi(m_{ij})$ leaves the branching-program computation unchanged, except that the values are now represented in the basis (β_i) representation of $\mathbb{E}$. The only operations outside the embedded $\mathbb{E}$ computation are the blockwise embedding A_0 and the random linear combination $\boldsymbol{\rho}$. Thus, apart from the applications of A_0 and $\boldsymbol{\rho}$, evaluating $\tilde{\mathbf{a}}'$ is the same computation as evaluating $\tilde{\mathbf{a}}$, performed in the regular representation of $\mathbb{E}$. **Remco: Something is off with the complexity. Should really check with a Sage implementation. Probably needs a factor d .**

The batch protocol is quadratic in d in terms of $\mathbb{F}$ -element operations and proof size. This is acceptable when d is small, but binary fields where $d = 128$ the communication would be 2,048 bytes. Avoiding this quadratic dependence on d appears to require further rounds of interaction and more sophisticated decomposition.

C.4.2 Folding protocol

In the folding protocol we interactively reduce the claim $\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{E}} = h$ through a sequence of embeddings W_i where at round i the claim is

$$\langle W_i(\mathbf{w}), A_i(\mathbf{a}) \rangle_{\mathbb{E}} = h_i \quad W_i : \mathbb{F}^{nd} \rightarrow \mathbb{E}^{nd_i} \quad A_i : \mathbb{E}^{nd} \rightarrow \mathbb{E}^{nd_i}$$

Here $d = d_0 \geq \dots \geq d_k = 1$ is a series of folding dimensions. We start with W_0 and A_0 the identity and $h_0 = h$ so that the initial claim is indeed $\langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{E}} = h$. We also stipulate that the embeddings

are blockwise such that $W_i = I_n \otimes W'_i$ and $A_i = I_n \otimes A'_i$. For simplicity we will only consider the $n = 1$ case from here on as $n > 1$ follows by blockwise repetition.

The final map is $W_k : \mathbb{F}^d \rightarrow \mathbb{E}$ which is an F -linear isomorphism. Let (β_i) be the implied basis such that

$$W_k(\mathbf{x}) = \sum_{i=1}^d x_i \beta_i.$$

Working backwards from the final embedding we create a sequence

$$R_i \circ W_{i+1} = W_i$$

where each map R_i has only $\left\lceil \frac{d_i}{d_{i+1}} \right\rceil$ Frobenius components.

Every $\mathbb{F}$ -linear map $T : \mathbb{E}^m \rightarrow \mathbb{E}^n$ can be viewed as being built from $\mathbb{E}$ -linear maps and $\mathbb{F}$ -linear scalar operations on $\mathbb{E}$. To recursively fold such maps, the scalar operations must be compatible with arbitrary $\mathbb{E}$ -linear coordinate transformations. Concretely, we require an intertwining

$$\beta(R\mathbf{x}) = \alpha(R)\beta(\mathbf{x}),$$

where α acts entrywise on R . This property singles out semilinear scalar operators. Over finite fields, these are precisely the Frobenius maps σ^i up to scalar multiples, and the Frobenius powers therefore provide the canonical scalar decomposition.

Theorem C.6 (Frobenius decomposition). *Let $\mathbb{E}/\mathbb{F}$ be a finite extension of degree d , and let $\sigma : \mathbb{E} \rightarrow \mathbb{E}$ be the Frobenius map $x \mapsto x^{|\mathbb{F}|}$. Then every $\mathbb{F}$ -linear map $T : \mathbb{E}^m \rightarrow \mathbb{E}^n$ admits a unique decomposition*

$$T(\mathbf{x}) = \sum_{i=0}^{d-1} M_i \cdot (\mathbf{I}_m \otimes \sigma^i)(\mathbf{x}), \quad M_i \in \mathbb{E}^{n \times m}.$$

We define $\text{supp}_\sigma(T) := \{i : M_i \neq 0\}$.

Single folding round. Assume the current claim is

$$\langle W_i(\mathbf{w}), A_i(\mathbf{a}) \rangle_{\mathbb{E}} = h_i.$$

Let

$$R_i : \mathbb{E}^{nd_{i+1}} \rightarrow \mathbb{E}^{nd_i}$$

be a fixed $\mathbb{F}$ -linear map satisfying

$$R_i \circ W_{i+1} = W_i.$$

By the Frobenius decomposition there exist unique matrices

$$M_{i,0}, \dots, M_{i,d-1} \in \mathbb{E}^{nd_i \times nd_{i+1}}$$

such that

$$R_i = \sum_{j=0}^{d-1} M_{i,j} \circ (\mathbf{I}_{nd_{i+1}} \otimes \sigma^j).$$

For every j , define the $\mathbb{F}$ -linear map

$$C_{i,j} = (\mathbf{I}_{nd_{i+1}} \otimes \sigma^{-j}) \circ M_{i,j}^\top.$$

Then, for every $\mathbf{x} \in \mathbb{E}^{nd_{i+1}}$ and $\mathbf{b} \in \mathbb{E}^{nd_i}$,

$$\langle R_i(\mathbf{x}), \mathbf{b} \rangle_{\mathbb{E}} = \sum_{j=0}^{d-1} \sigma^j(\langle \mathbf{x}, C_{i,j}(\mathbf{b}) \rangle_{\mathbb{E}}).$$

Applying this identity to

$$\mathbf{x} = W_{i+1}(\mathbf{w}), \quad \mathbf{b} = A_i(\mathbf{a}),$$

gives

$$\langle W_i(\mathbf{w}), A_i(\mathbf{a}) \rangle_{\mathbb{E}} = \sum_{j=0}^{d-1} \sigma^j(\langle W_{i+1}(\mathbf{w}), C_{i,j}(A_i(\mathbf{a})) \rangle_{\mathbb{E}}).$$

The prover sends the values

$$g_{i,j} = \langle W_{i+1}(\mathbf{w}), C_{i,j}(A_i(\mathbf{a})) \rangle_{\mathbb{E}}, \quad j = 0, \dots, d-1.$$

The verifier checks

$$h_i = \sum_{j=0}^{d-1} \sigma^j(g_{i,j}),$$

samples a random challenge $\rho_i \in \mathbb{E}$, and defines

$$A_{i+1} = \left(\sum_{j=0}^{d-1} \rho_i^j C_{i,j} \right) \circ A_i.$$

Finally, letting

$$h_{i+1} = \sum_{j=0}^{d-1} \rho_i^j g_{i,j},$$

the next claim is

$$\langle W_{i+1}(\mathbf{w}), A_{i+1}(\mathbf{a}) \rangle_{\mathbb{E}} = h_{i+1}.$$

We therefore seek an interactive reduction from a mixed claim to a packed claim

$$\begin{array}{ccc} \langle \mathbf{w}, \mathbf{a} \rangle_{\mathbb{E}} = h & \mathbf{w} \in \mathbb{F}^{nd} & \mathbf{a} \in \mathbb{E}^{nd} \\ \Downarrow & & \\ \langle W(\mathbf{w}), \mathbf{a}' \rangle_{\mathbb{E}} = h' & W(\mathbf{w}) \in \mathbb{E}^n & \mathbf{a}' \in \mathbb{E}^n \end{array}$$

where $W : \mathbb{F}^{nd} \rightarrow \mathbb{E}^n$ is a fixed $\mathbb{F}$ -linear embedding and $\mathbf{a}' \in \mathbb{E}^n$, $h' \in \mathbb{E}$ are determined by $\mathbf{a}$, h and the transcript. Ideally $\mathbf{a} \mapsto \mathbf{a}'$ should also preserve useful properties like efficient computation of its multilinear extension.

Above we decomposed $\mathbf{a}$ with respect to a basis of $\mathbb{E}/\mathbb{F}$. Here we instead decompose the induced $\mathbb{F}$ -linear functional on packed witnesses. Define

$$T_{\mathbf{a}} : \mathbb{E}^n \rightarrow \mathbb{E} \quad \mathbf{x} \mapsto \langle W^{-1}(\mathbf{x}), \mathbf{a} \rangle_{\mathbb{E}}.$$

This is an $\mathbb{F}$ -linear functional on $\mathbb{E}^n$, but is not generally $\mathbb{E}$ -linear. We'll first prove a representation lemma for $\mathbb{F}$ -linear $\mathbb{E} \rightarrow \mathbb{E}$ maps and then a decomposition theorem.

Theorem C.7. *Let $\mathbb{F} \subseteq \mathbb{E}$ be finite fields with $q = |\mathbb{F}|$ and $[\mathbb{E} : \mathbb{F}] = d$, then every $\mathbb{F}$ -linear map $T : \mathbb{E} \rightarrow \mathbb{E}$ is uniquely represented by a q -linearized polynomial*

$$P(X) = \sum_{j=0}^{d-1} c_j X^{q^j} \quad c_j \in \mathbb{E}.$$

Proof. By interpolation every map $T : \mathbb{E} \rightarrow \mathbb{E}$ can be represented uniquely by a polynomial $P \in \mathbb{E}[X]$ of degree less than $|\mathbb{E}| = q^d$

$$P(X) = \sum_{j=0}^{q^d-1} a_j X^j \quad a_j \in \mathbb{E}.$$

From $\mathbb{F}$ -linearity follows $P(0) = 0$ and hence $a_0 = 0$, further we have $P(X + Y) = P(X) + P(Y)$ and hence

$$\begin{aligned} 0 &= P(X + Y) - P(X) - P(Y) \\ &= \sum_{j=1}^{q^d-1} a_j (X + Y)^j - \sum_{j=1}^{q^d-1} a_j (X^j + Y^j) \\ &= \sum_{j=1}^{q^d-1} a_j \sum_{k=0}^j \binom{j}{k} X^k Y^{j-k} - \sum_{j=1}^{q^d-1} a_j (X^j + Y^j) \\ &= \sum_{j=1}^{q^d-1} a_j \sum_{k=1}^{j-1} \binom{j}{k} X^k Y^{j-k} \end{aligned}$$

Write $q = p^r$ with p the characteristic of $\mathbb{F}$. For every j with $a_j \neq 0$ we must have $\binom{j}{k} = 0 \pmod p$ for all $0 < k < j$, by Lucas's theorem this holds iff $j = p^s$ for some s . Hence we can reindex P keeping only the non-zero coefficients

$$P(X) = \sum_{j=0}^{rd-1} b_j X^{p^j} \quad b_j \in \mathbb{E}.$$

From $\mathbb{F}$ -linearity also follows $P(\lambda x) = \lambda P(x)$ for all $x \in \mathbb{E}$ and $\lambda \in \mathbb{F}$. For every $b_j \neq 0$ this requires $\lambda^{p^j} = \lambda$ for every $\lambda \in \mathbb{F}$. This holds exactly when $j = rs$ for some s . Reindexing P again we get our claimed result

$$P(X) = \sum_{j=0}^{d-1} c_j X^{q^j} \quad c_j \in \mathbb{E}.$$

□

Theorem C.8 (Frobenius decomposition). *Let $\mathbb{F} \subseteq \mathbb{K} \subseteq \mathbb{E}$ be finite fields with $[\mathbb{K} : \mathbb{F}] = m$, and let $\sigma(x) = x^{|\mathbb{F}|}$ denote the $\mathbb{F}$ -Frobenius automorphism. Then every $\mathbb{F}$ -linear map $T : \mathbb{E} \rightarrow \mathbb{E}$ admits a unique decomposition*

$$T(x) = \sum_{i=0}^{m-1} \sigma^i \circ T_i$$

where each T_i is $\mathbb{K}$ -linear.

Proof. Write $q = |\mathbb{F}|$ and $d = [\mathbb{E} : \mathbb{F}]$, by the preceding theorem T has a unique representation

$$T(X) = \sum_{j=0}^{d-1} c_j X^{q^j} \quad c_j \in \mathbb{E}.$$

Since $\mathbb{K} \subseteq \mathbb{E}$, we have $m \mid d$. Write each index uniquely as $j = ms + i$, with $0 \leq i < m$ and $0 \leq s < d/m$. Grouping the terms by their residue modulo m gives

$$T(X) = \sum_{i=0}^{m-1} \sum_{s=0}^{d/m-1} c_{ms+i} X^{q^{ms+i}}.$$

For each residue class i , define

$$T_i(X) = \sum_{s=0}^{d/m-1} c_{ms+i}^{q^{-i}} X^{q^{ms}},$$

where $z \mapsto z^{q^{-i}}$ denotes the inverse of the automorphism $z \mapsto z^{q^i}$ of $\mathbb{E}$. Each T_i is now a q^m -linearized polynomial hence applying the preceding theorem to the extension $\mathbb{K} \subseteq \mathbb{E}$, each T_i is $\mathbb{K}$ -linear. Then

$$(\sigma^i \circ T_i)(X) = T_i(X)^{q^i} = \sum_{s=0}^{d/m-1} c_{ms+i} X^{q^{ms+i}}.$$

Therefore T represented as

$$T = \sum_{i=0}^{m-1} \sigma^i \circ T_i.$$

Uniqueness follows immediately from the uniqueness of the q -linear polynomial representation. $\square$

The decomposition theorem extends coordinatewise to maps $\mathbb{E}^n \rightarrow \mathbb{E}$. In particular, the functional $T_{\mathbf{a}}(\mathbf{x}) = \langle W^{-1}(\mathbf{x}), \mathbf{a} \rangle_{\mathbb{E}}$ admits a unique decomposition

$$T_{\mathbf{a}} = \sum_{i=0}^{m-1} \sigma^i \circ T_i, \tag{56}$$

where each $T_i : \mathbb{E}^n \rightarrow \mathbb{E}$ is $\mathbb{K}$ -linear. Thus the initial claim may be written as

$$h = T_{\mathbf{a}}(W(\mathbf{w})) = \sum_{i=0}^{m-1} \sigma^i(T_i(W(\mathbf{w}))).$$

Construction C.9 (Frobenius fold). Given the claim $T_{\mathbf{a}}(W(\mathbf{w})) = h$, the parties proceed as follows.

1. The prover sends $v_i = T_i(W(\mathbf{w}))$ for $i = 1, \dots, m-1$.
2. The verifier derives $v_0 = h - \sum_{i=1}^{m-1} \sigma^i(v_i)$.
3. The verifier samples independent challenges $\rho_1, \dots, \rho_{m-1} \leftarrow \$ \mathbb{E}$.
4. The parties reduce to the claim $T'(W(\mathbf{w})) = h'$, where

$$T' = T_0 + \sum_{i=1}^{m-1} \rho_i T_i \quad \text{and} \quad h' = v_0 + \sum_{i=1}^{m-1} \rho_i v_i.$$

Construction C.10 (Folding protocol for extension opening). Choose a tower of intermediate fields $\mathbb{F} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \dots \subseteq \mathbb{K}_t = \mathbb{E}$, and write $m_i = [\mathbb{K}_{i+1} : \mathbb{K}_i]$. Initially, $T_{\mathbf{a}}$ is $\mathbb{K}_0$ -linear. At round i , apply the Frobenius fold with $\mathbb{F} = \mathbb{K}_i$, and $\mathbb{K} = \mathbb{K}_{i+1}$. This reduces the current $\mathbb{K}_i$ -linear functional to a $\mathbb{K}_{i+1}$ -linear functional.

After t rounds, the residual functional $T' : \mathbb{E}^n \rightarrow \mathbb{E}$ is $\mathbb{E}$ -linear. Consequently, there exists $\mathbf{a}' \in \mathbb{E}^n$ such that $T'(\mathbf{x}) = \langle \mathbf{x}, \mathbf{a}' \rangle_{\mathbb{E}}$. The final claim is therefore $\langle W(\mathbf{w}), \mathbf{a}' \rangle_{\mathbb{E}} = h'$, as required.

Each round requires sending $m_i - 1$ elements of $\mathbb{E}$. For a binary tower where $m_i = 2$, $\mathbb{F} = \mathbb{F}_2$, and $\mathbb{E} = \mathbb{F}_{2^{128}}$ we have 7 rounds of one $\mathbb{E}$ element message each, for a total of 112 bytes. Compare to the non-interactive protocol this reduces the communication complexity from $O(d^2)$ to $O(d \log d)$ elements of $\mathbb{F}_2$.

Towerless folding. Say $\mathbb{F} \subseteq \mathbb{E}$ admits no tower of subfields, for example $\mathbb{E} = \mathbb{F}_{2^{127}}$. Take again $d = [\mathbb{E} : \mathbb{F}]$ and decompose $T_{\mathbf{a}}$ as

$$T_{\mathbf{a}} = \sum_{i=0}^{d-1} \sigma^i \circ T_i,$$

We can still recursively fold using the protocol above with $m = 2$ by grouping even and odd exponents of σ^i (zero-padding as necessary), as there is no actual dependence on the subfield.

This folding progressively halves d (rounding up) in the $\mathbb{E}^{nd}$ inner product. This induces a sequence of perfect pairings

$$(\mathbb{F}^{nd}, \mathbb{E}^{nd}) \rightsquigarrow (\mathbb{E}^{nd/m}, \mathbb{E}^{nd/m}) \rightsquigarrow (\mathbb{K}_i^{nd/m}, \mathbb{E}^{nd/m})$$

$$\langle W_i(\mathbf{w}), A_i(\mathbf{a}) \rangle = \langle \mathbf{w}, \mathbf{a} \rangle.$$

The final result is a transcript dependent map $A : \mathbb{E}^{dn} \rightarrow \mathbb{E}^n$. If W is blockwise with $I_n \otimes W_0$ for some $W_0 : \mathbb{F}^d \rightarrow \mathbb{E}$ then so is A with $A_0 : \mathbb{E}^d \rightarrow \mathbb{E}$.

D [Remove?] Virtualization of $\mathbb{F}_2$ -linear combinations lifted to integers

Albert: Do we really need this section? Isn't the technical overview enough? Having contents only in the TO is a bit weird academically speaking, but this section is really a straightforward rewriting of the corresponding TO section, with some more formal symbols and language

In this section we expand the core relation $\mathcal{R}_{\text{Lin-BitsDual}}$ to impose, instead of constraints of the form $f = \text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})$ and $\langle \pi_q(\mathbf{f}), \mathbf{v} \rangle = \mu$, where $\mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$, constraints as $\text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f}) = f$ and

$$\langle \pi_q \circ \pi_2^{-1}(A \cdot \pi_2(\mathbf{f})) \rangle = \mu$$

for a matrix $A \in \mathbb{F}_2^{\ell' \times \ell}$. Precisely, given a new parameter ℓ' , we define

$$\mathcal{R}_{\text{Lin-BitsDual}}^{(\{0, 1\}, \pi_q, R, C)}$$

$$:= \left\{ \left(\begin{array}{l} \mathbf{x} = (\mu, \mathbf{v}, A) \in R \times R^{\ell'} \times \mathbb{F}_2^{\ell' \times \ell}, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell \end{array} \right) \middle| \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f}) \\ \wedge \langle \pi_q \circ \pi_2^{-1}(A \cdot \pi_2(\mathbf{f})), \mathbf{v} \rangle = \mu \end{array} \right\}$$

Albert: We could replace $M \cdot \pi_2(\mathbf{f})$ by the output of a GKR circuit on $\pi_2(\mathbf{f})$. The relation has the tag “virtual” to reference the intuition that $\mathcal{R}_{\text{Lin-BitsDual}}^{\langle \{0, 1\}, \pi_q, R, \mathcal{C} \rangle}$ expresses a linear constraint on the vector $\mathbf{h} = \pi_q \circ \pi_2^{-1}(A \cdot \pi_2(\mathbf{f})) \in R^{\ell'}$, which is not committed directly: only $\pi_2(\mathbf{f})$ is. Note that $\mathbf{h}$ is a linear expression over $\mathbb{F}_{2^\nu}$ of the elements in $\pi_2(\mathbf{f})$, lifted to the integers, and then projected to R .

Construction D.1 (IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$). **Parameters.** $(R, \pi, \mathbb{F}_{2^\nu}, \ell', \ell_1, \ell_2, \ell, \mathcal{C})$ comprised of: a ring R (either $\mathbb{Z}$ or a finite field); a ring homomorphism $\pi : \mathbb{Z} \rightarrow R$; a binary field $\mathbb{F}_{2^\nu}$; a virtual witness length ℓ' ; tensor decomposition dimensions ℓ_1, ℓ_2 satisfying $\ell_1 \cdot \ell_2 = \ell'$; a non-virtual witness length ℓ ; a linear code $\mathcal{C}$ over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν and blocklength m .

Inputs. The prover and verifier receive $\mathbf{x} = (\mu, \mathbf{v}, A) \in R \times R^{\ell'} \times \mathbb{F}_2^{\ell' \times \ell}$ and $\mathbf{y} = f \in \mathbb{F}_{2^\nu}^m$. The prover receives $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^{\langle \{0, 1\}, \pi_q, R, \mathcal{C} \rangle}$.

1. Define $\mathbf{h} = \pi_2^{-1}(A \cdot \pi_2(\mathbf{f})) \in \{0, 1\}^{\ell'} \subseteq \mathbb{Z}^{\ell'}$. Prover and verifier execute ?? until before Phase 3, with parameters⁷

$$(R, \pi, \mathbb{F}_{2^\nu}, \ell', \ell_1, \ell_2, \mathcal{C})$$

and inputs

$$\mathbf{x}' = (\mu, \mathbf{v}), \quad \mathbf{y}' = f, \quad \mathbf{w}' = \mathbf{h} \in \{0, 1\}^{\ell'} \subseteq \mathbb{Z}^{\ell'}.$$

Note that until that point, the verifier has not made any queries to $\mathbf{y}'$. So, in terms of completeness, it does not matter, yet, that f is not the encoding of $\pi_2(\mathbf{h}) = M \cdot \pi_2(\mathbf{f})$.

2. At the end of Phase 2 of ??, prover and verifier are left with the claims

$$\begin{aligned} & ((\mathbf{v}, \mu'), f, \pi_2(\mathbf{h})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu) \\ &= \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{u}, \mu) \in \mathbb{F}_{2^\nu}^k, \\ \mathbf{y} = g \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{g} \in (\mathbb{F}_{2^\nu}^\ell) \end{array} \right) \middle| \begin{array}{l} g = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{g}) \\ \langle \mathbf{u}, \mathbf{g} \rangle = \mu \end{array} \right\} \end{aligned}$$

for some vector $\mathbf{v} \in \mathbb{F}_{2^\nu}^{\ell'}$. **Albert:** $\mathbf{v}$ was already used in the input and field element $\mu' \in \mathbb{F}_{2^\nu}$ output during the IOR Π_{red} .

As we mentioned, this claim as is is false, because $f \neq \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{h})$ –indeed, the dimension of $\mathcal{C}$ does not match that of $\text{Pack}_\nu \circ \pi_2(\mathbf{h})$. Instead of, as done in ??, using an IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$ to attempt to prove the above claim, prover and verifier proceed as follows.

Observe that, if the prover is honest,

$$\mu' = \langle \mathbf{v}, \pi_2(\mathbf{h}) \rangle = \langle \mathbf{v}, A \cdot \pi_2(\mathbf{f}) \rangle = \langle A^\top \cdot \mathbf{v}, \pi_2(\mathbf{f}) \rangle,$$

⁷?? would require a code $\mathcal{C}$ as parameter whose dimension matches ℓ' . This is not the case in our current setting, but since ?? is run only up to before Phase 3 (and so the code $\mathcal{C}$ is not used in this partial run), this is not an issue.

where $\top$ denotes transposition. Hence the following claim is true:

$$((A^\top \cdot \mathbf{v}, \mu'), f, \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu). \quad (57)$$

Now prover and verifier use an IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$ to prove (??) [Albert: comments on verifier?](#).

Remark D.2. [Albert: Comments on how the verifier handles \$A^\top \cdot \mathbf{v}\$](#)

Theorem D.3 (Completeness and round-by-round knowledge soundness of ??). *Albert: Should we be more formal with this theorem? Albert: Completeness missing* Under the same hypothesis as ?? for $\delta, g, \ell', \ell_1, \ell_2, \Pi_{\text{red}}, \Pi_{\text{iopp}}$, ?? is a perfectly complete IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$ and has round-by-round knowledge soundness with relaxation $((\mathcal{R}_{\text{Lin-BitsDual}})^\delta, \perp)$, with errors

$$\kappa_0^*, (\kappa_{\text{red},i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp},i})_{i \in [t_{\text{iopp}}]}$$

as defined in ??. The errors $\kappa_{\text{red},i}$ correspond to the errors of the IOR Π_{red} when used on triples of the form

$$(\mathbb{x}_{\text{red}} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbb{y}_{\text{red}} = f, \mathbb{w}_{\text{red}} = \pi_2(\mathbf{h})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \text{Pack}_\nu)$$

for certain γ_i and μ_j (so, the witness has dimension ℓ' and we use the decomposition $\ell' = \ell_1 \cdot \ell_2$). The errors $\kappa_{\text{iopp},i}$ correspond to the errors of the IOP Π_{iopp} when used on triples of the form

$$(\mathbb{x}_{\text{iopp}} = (A^\top \cdot \mathbf{v}, \mu'), \mathbb{y}_{\text{iopp}} = f, \mathbb{w}_{\text{iopp}} = \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu),$$

so that the witness has dimension ℓ .

The error κ_0^* is only present if $\deg(R) > 0$. Otherwise, ?? (and thus ?? has one round less) and κ_0^* is not part of the soundness errors.

Proof. [Albert: Too vague?](#) The proof results from observing that ?? is the composition of an IOR Π_1 from $\mathcal{R}_{\text{Lin-BitsDual}}^{\text{virtual}}(\{0, 1\}, \pi, R, \mathcal{C})$ to $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$, with an IOP Π_2 for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$. In Π_1 the input is $(\mathbb{x} = (\mu, \mathbf{v}, A), \mathbb{y} = f, \mathbb{w} = \mathbf{f})$. The output is $((A^\top \cdot \mathbf{v}, \mu'), f, \pi_2(\mathbf{f}))$. The IOR Π_1 is exactly the same as ?? with input $((\mu, \mathbf{v}), f, \mathbf{h})$, until Phase 3. Hence by the same arguments as in the proof of ??, it is perfectly complete and has errors $\kappa_0^*, (\kappa_{\text{red},i})_i$ with parameters set as $(R, \pi, \mathbb{F}_{2^\nu}, \ell', \ell_1, \ell_2, \mathcal{C})$. The IOP Π_2 is perfectly complete and has round-by-round knowledge errors $\kappa_{\text{iopp},i}$ by assumption. $\square$

Remark D.4 (Instantiating ??). ?? works in the exact same way as ?? until Phase 3, where an IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$ is invoked on a slightly different instance-oracle-triple. The latter IOP is a standard combination of ring switching ?? and WHIR ??, plus perhaps additional processes to make verification time cheaper ??. The whole ?? can be instantiated as discussed in ??, assuming this latter IOP is instantiated correctly (it is a standard IOP, so we do not discuss it further).